Predictive regressions: a machine learning perspective

Guillaume Coqueret* and Romain Deguest

November 9, 2020

Abstract

In this article, we quantify the forecasting efficiency of the OLS estimator in univariate predictive regressions. We link the prediction accuracy to three key quantities: the persistence of the underlying series, the forecasting horizon, and the sample size. We find that high autocorrelation in the dependent variable is required to reach reasonably low levels of mean squared errors. In this case, we identify two configurations which generate positive out-of-sample R^2: short term forecasting with small samples and long horizon predictions with very deep samples. Two examples of such configurations can easily be found in financial economics: the short term volatility and the long term equity premium. We confirm our results via an empirical study on the S&P 500 with a series of 15 popular predictors used in the literature.

1. Introduction

Predictive regressions (PRs) have a special place in the technical toolkit of social scientists in general and financial economists in particular. Nevertheless, the apparent simplicity of their formulation masks the complexity of the underlying statistical machinery. Numerous studies have warned scholars and practitioners to be careful when relying on simple (sample) estimates and tests stemming from PRs.[1] Since the seminal paper of Stambaugh (1999), it is for instance known that when regressors are autocorrelated, the OLS estimators of PRs are biased and should be corrected. In response to these technical hurdles, researchers have proposed a large palette of solutions (e.g., Campbell and Yogo (2006), Jansson and Moreira (2006), Pástor and Stambaugh (2009) and Johnson (2019)) and developed extensions to the ideas of Stambaugh (1999).[2]

One additional challenge arises when the dependent variable is also autocorrelated, which occurs in finance when returns are computed over long horizons (Campbell (2001), Lanne (2002), Hjalmarsson (2012)). Indeed, since Fama and French (1988a,b), it has widely been documented that statistical significance is empirically more pronounced when return horizons are large, though this phenomenon vanishes beyond 12 to 15 years (Bandi et al. (2019)).[3] However, these seemingly robust results may in fact be spurious because long term returns yield larger t-statistics *by construction* (Valkanov (2003), Boudoukh et al. (2008), Hjalmarsson (2008a)). This has led to the

*EMLYON Business School, 23 avenue Guy de Collongue, 69130 Ecully, FRANCE. E-mail: coqueret@em-lyon.com

[1] A selective list is: Ferson et al. (2003), Deng (2014) and Phillips (2015).

[2] Two such directions are panel regressions (Hjalmarsson (2008b)) and multivariate regressions (Amihud and Hurvich (2004), Andersen and Varneskov (2020)). Heavy-tailed errors are also an issue, and we refer to Bao (2007) and Shephard (2020) for a treatment on this subject. For a general approach to errors, we point to Ibragimov et al. (2020) and the references therein.

[3] One notable exception is Ang and Bekaert (2007), where the authors argue that predictability is a short-term phenomenon.

Electronic copy available at: https://ssrn.com/abstract=3709412

blossoming of an abundant literature that proposes alternative approaches designed to mitigate this issue (see, e.g., Hjalmarsson (2011), Phillips and Lee (2013), Sizova (2014), Chevillon (2017), Xu (2020) and Boudoukh et al. (2020)). In the asset pricing literature, present value models, via latent variables, may also circumvent technical impediments (Van Binsbergen and Koijen (2010)), but may over-reject the null of no predictability (Piatti and Trojani (2020)).

One common feature of the aforementioned articles is that they are all intended for *inference*. Their aim is to evaluate the significance of a linear relationship between two (or more) possibly lagged variables. This is useful to determine, via a null hypothesis, if one variable can be predicted by another one with a given degree of confidence. Most often, this is achieved by making assumptions on model errors and deriving properties of estimators. In the present paper, our premise is different. We depart from the traditional analyses by considering that the processes are given and our goal is then to characterize the errors stemming from PRs. The paper focuses on quantifying the pure predictive ability which evaluates *how well* one variable will predict the other, based on the OLS estimator (see Table 1).[4] Simply put, our research question is: given two autoregressive processes, how efficient is the OLS estimator in predictive regressions? There is no perfect yardstick for efficiency, but we set a heuristic limit and consider that a model (or, equivalently, an estimator) is only worthwhile if it is able to deliver strictly positive R^2 out-of-sample.

	Assumptions on	Results on
traditional inference	X, ϵ, and $\hat{b}$	$\hat{b} \rightarrow$ tests
this paper	X, y, and $\hat{b}$ (OLS)	$\epsilon \rightarrow$ MSE, R^2

Table 1: **Relative positioning of the paper**. y is the dependent variable, X represents predictors and ϵ stands for model errors. $\hat{b}$ defines the form of the estimator and MSE is the acronym for Mean Squared Error.

Our approach is motivated by current developments in asset pricing studies that rely on machine learning methods (see, e.g., Bryzgalova et al. (2020), Chen et al. (2020), Gu et al. (2020a,b)). In machine learning, the focal performance indicator is out-of-sample fit; therefore, it is valuable to formally characterize this measure of accuracy for the most elementary model of all, the simple regression. The most common way to assess out-of-sample fit is to compute the mean squared error (MSE) between model predictions and realized values, from which the R-squared is usually derived.

While several papers have already addressed the issue of forecasting accuracy for one autoregressive process (see Hoque et al. (1988), Magnus and Pesaran (1989, 1991) or, more recently, Bao (2007) when error terms are non-Gaussian), there are, to the best of our knowledge no such results when two series are involved, which is the case in predictive regressions. This article is meant to fill this void. As is often the case when analyzing regression coefficients, our results rely heavily on moments of ratios of quadratic forms of multivariate Gaussian distributions.[5] To compute such moments, we rely on Magnus (1990) who proposes expressions based on simple integral formulas for nontrivial exponents.[6]

[4]Statistically, it is of course possible to devise tests that characterize forecasting accuracy (see, among others, West (1996), Hansen (2005), Rossi (2005) and Rossi and Inoue (2012)). Comparison tests are in fact notoriously complex (Li et al. (2020), Zhu and Timmermann (2020)). These issues are beyond the scope of this paper.

[5]This particular topic is reviewed in Appendix B of Paolella (2018) and in Bao and Kan (2013). In a similar vein, Kan and Wang (2010) use an elegant characterization of eigenvalues to derive moments for the sample estimator of the autocorrelation.

[6]The contribution of Magnus (1990) generalizes previous work on the subject (Sawa (1972) and Magnus (1986)

Electronic copy available at: https://ssrn.com/abstract=3709412

Our results are undoubtedly linked to the abundant empirical literature on the predictability of asset returns. While some contributions argue in favor of predictability (Lewellen (2004), Cochrane (2008)), or against (Bossaerts and Hillion (1999), Ferson et al. (2003), Goyal and Welch (2003), Welch and Goyal (2008)), some studies take a more unifying stance by concluding that the relationship is likely time-varying (Dangl and Halling (2012), Zhu (2015), Farmer et al. (2019)). Similarly, our conclusions are not clear cut, as we do not arbitrate for or against predictability. Rather, we detail the econometric configurations under which predictive regressions are likely to deliver satisfactory out-of-sample accuracy.

In particular, we document that accuracy results are naturally very sensitive to the horizons of forecasts. When the horizon is short, the best fit is obtained when the dependent variable is highly persistent and when sample sizes are small. This description is for instance well suited to volatility forecasting. On the other hand, when the horizon is longer (e.g., for annual returns), the R^2 is highest when the predictors are not persistent, and, most importantly, when training samples are very deep (hundreds, or, ideally, thousands of points).

Our contributions are threefold. First, we derive previously unknown analytical identities and properties for the mean quadratic error of PRs when estimates are based on finite sample OLS. Second, we numerically evaluate the sensitivity of these errors with respect to autocorrelations (of both processes), sample sizes and forecasting horizons. Surprisingly, our results reveal that choosing predictors with high autocorrelation often leads to lower performance, especially when the autocorrelation of the predicted variable is high. Finally, we contextualize our findings in an empirically study of return predictability where the dependent variable is the S&P500 index and the predictors are taken from the studies from Welch and Goyal (2008) and Novy-Marx (2014).

Our empirical findings confirm the stylized results obtained in the numerical section. Notably, it corroborates the impact of the persistence of the dependent variable on the mean quadratic error. We show that the predictive performance improves when economic constraints suggest a post-hoc treatment of predictions. This substantiate similar analyses on the benefits of enforcing such constraints (Campbell and Thompson (2008) and Pettenuzzo et al. (2014)). We also propose applications of return predictability to dynamic portfolio construction. In particular, we design a smooth signal based on the wisdom of the crowds (predictor ensemble) and show that it leads to higher risk-adjusted performances compared to a buy-and-hold investment in the S&P500. Finally, we study the short-term predictability of the S&P500 variance and corroborate that for persistent dependent variables, short training samples yield larger R-squared. It is important to underline that while our empirical study is focused on asset pricing, the theoretical results of the paper can be valuable beyond financial disciplines.

The remainder of the paper is structured as follows. In Section 2, we formulate our research problem, provide all the required notations and lay out our analytical results. Section 3 is dedicated to numerical analyses and stylized properties. In Section 4, we confirm our theoretical conclusions on empirical data, while Section 5 concludes. All the technical details and proofs are located in the Appendix.

notably). Alternative solutions involve infinite series expansions (see, e.g., Smith (1989), Hillier et al. (2009) or Hillier et al. (2014)), but the integral approach is arguably more tractable and better suited to numerical analyses.

Electronic copy available at: https://ssrn.com/abstract=3709412

2. Problem and theoretical results

2.1. Notations

We are given two strictly stationary first order auto-regressive (AR(1)) processes x^* and y^* defined by

$$x^*_{t+1} = \alpha_x + \rho_x x^*_t + e_{x^*,t+1}, \tag{1}$$
$$y^*_{t+1} = \alpha_y + \rho_y y^*_t + e_{y^*,t+1}, \tag{2}$$

with constants α_x and α_y, autocorrelations ρ_x and ρ_y satisfying $|\rho_x| < 1$ and $|\rho_y| < 1$, and correlated Gaussian white noise processes e_{x^*} and e_{y^*} with variances σ^2_x and σ^2_y and correlation ρ satisfying $|\rho| < 1$. AR(1) processes are commonplace in the financial literature when modelling both dependent variables and predictors (see, e.g., Stambaugh (1999), Campbell and Yogo (2006), and Van Binsbergen and Koijen (2010), to cite but a few).

The econometrician seeks to build a forecasting model that aims to predict y^*_{t+k} based on the current level of x^*_t and resorts to a linear regression:[7]

$$y^*_{t+k} = a + b\, x^*_t + e_{t+k}, \tag{3}$$

where, plainly, a is the intercept and b the slope. The above predictive model takes into consideration a lag of k time steps between the regressor and the predicted quantity to avoid any look-ahead bias in the training sample. A canonical example of this specification takes y to be the annual return of a portfolio (or index) with value P_t, monitored on a monthly basis. In this case, $y_{t+12} = \frac{P_{t+12}}{P_t} - 1$, and x represents some meaningful macroeconomic predictor such as the aggregate dividend yield, or proxies for term or credit spreads.

In practice, the two parameters a and b are unknown and must be estimated from a dataset, which we will call the training sample, as is customary in the machine learning (ML) jargon. If we denote with T the size of this training sample $\mathcal{D}^*_t = \{(x^*_{t-k-T+1}, y^*_{t-T+1}), ..., (x^*_{t-k}, y^*_t)\}$, then the canonical expressions for the Ordinary Least Squares (OLS) estimator are

$$\hat{b}(\mathcal{D}^*_t) = \frac{\sum_{s=0}^{T-1}(x^*_{t-k-s} - \bar{x}^*)(y^*_{t-s} - \bar{y}^*)}{\sum_{s=0}^{T-1}(x^*_{t-k-s} - \bar{x}^*)^2} \tag{4}$$
$$\hat{a}(\mathcal{D}^*_t) = \bar{y}^* - \hat{b}(\mathcal{D}^*_t)\bar{x}^*, \tag{5}$$

with sample means $\bar{x}^* = \frac{1}{T}\sum_{s=0}^{T-1} x^*_{t-k-s}$ and $\bar{y}^* = \frac{1}{T}\sum_{s=0}^{T-1} y^*_{t-s}$. One important feature of the training sample is naturally the time shift between x and y due to the forecasting objective. The variable x_t is exploited to forecast a variable y_{t+k} that will only be measured k periods in the future.

We use the following notations in the paper. Vectors $\boldsymbol{v}$ and matrices $\boldsymbol{M}$ are written with bold fonts. For square matrices, we simplify the notation and use only one subscript: $\boldsymbol{I}_T$, $\mathbf{0}_T$ and $\mathbf{1}_T$ denote the T dimensional identity, zero (filled with 0) and unit (filled with 1) matrices, respectively, while $\mathbf{0}_{N,M}$ and $\mathbf{1}_{N,M}$ are $(N \times M)$ matrices filled with zeros and ones. $\boldsymbol{v}'$ and $\boldsymbol{M}'$ are the transpose of the corresponding vector and matrix. Finally, $\mathrm{tr}(\cdot)$ is the trace that operates on the set of square matrices and yields the sum of diagonal elements.

[7] For a detailed account on multistep forecasting, we refer to the survey Chevillon (2007).

Electronic copy available at: https://ssrn.com/abstract=3709412

One focal quantity in the present article is the mean squared error of the prediction, also called quadratic loss in ML parlance and defined by $\mathbb{E}[\hat{e}_{t+k}^2]$. First, we derive the conditional squared error based on all the information available at time t, namely $\mathcal{D}_t^*$ and x_t^*. It can be derived as follows

$$\begin{aligned}
L_t &:= L(k, \alpha_y, \rho_y, \sigma_y^2, \mathcal{D}_t^*, x_t^*) \qquad (6)\\
&= \mathbb{E}_t\left[\left(y_{t+k}^* - \hat{a}(\mathcal{D}_t^*) - \hat{b}(\mathcal{D}_t^*)x_t^*\right)^2\right]\\
&= \underbrace{\mathbb{E}_t\left[(y_{t+k}^* - \mathbb{E}_t[y_{t+k}^*])^2\right]}_{\mathrm{var}_t(y_{t+k}^*)} + \underbrace{\left(\mathbb{E}_t[y_{t+k}^*] - \hat{a}(\mathcal{D}_t^*) - \hat{b}(\mathcal{D}_t^*)x_t^*\right)^2}_{\mathrm{bias}_t^2}\\
&= \mathbb{E}_t\left[\left(\sum_{s=0}^{k-1}\rho_y^s e_{y^*,t+k-s}\right)^2\right] + \left(\alpha_y\sum_{s=0}^{k-1}\rho_y^s + \rho_y^k y_t^* - \hat{a}(\mathcal{D}_t^*) - \hat{b}(\mathcal{D}_t^*)x_t^*\right)^2\\
&= \sigma_y^2\sum_{s=0}^{k-1}\rho_y^{2s} + \left(\alpha_y\sum_{s=0}^{k-1}\rho_y^s + \rho_y^k y_t^* - \bar{y}^* + \hat{b}(\mathcal{D}_t^*)(\bar{x}^* - x_t^*)\right)^2. \qquad (7)
\end{aligned}$$

All items in the last equation are either model parameters, or elements of the training sample. More interestingly, this paper is focused on the unconditional prediction error, which depends only on model parameters. It is equal to

$$L := L(T, k, \alpha_x, \alpha_y, \rho_x, \rho_y, \rho, \sigma_x^2, \sigma_y^2) = \mathbb{E}\left[L_t\right] = \mathbb{E}\left[\left(y_{t+k}^* - \hat{a}(\mathcal{D}_t^*) - \hat{b}(\mathcal{D}_t^*)x_t^*\right)^2\right]. \qquad (8)$$

Our first result is the computation of the loss L when the slope b and intercept a parameters of the linear prediction model (3) are known. We also provide the expression of the minimal value achieved by the loss in that specific framework.

Lemma 1. *Consider two auto-regressive processes with correlated innovation processes given by (1) and (2), and the predictive regression model (3). The quadratic loss L of the predictive model satisfies the following properties:*

1. *If the intercept a and slope b are known, then*

$$L_{ab} = \frac{\sigma_y^2}{1-\rho_y^2} - 2b\rho_y^k\frac{\rho\sigma_x\sigma_y}{1-\rho_x\rho_y} + b^2\frac{\sigma_x^2}{1-\rho_x^2} \qquad (9)$$

2. *The minimal value of L_{ab} is achieved for*

$$b_o = \frac{cov(x_t, y_{t+k})}{var(x_t)} = \rho_y^k\frac{\rho\sigma_x\sigma_y}{1-\rho_x\rho_y}\frac{1-\rho_x^2}{\sigma_x^2}, \qquad (10)$$

and is equal to

$$L_o = \frac{\sigma_y^2}{1-\rho_y^2}\left(1 - \rho_y^{2k}\frac{\rho^2}{(1-\rho_x\rho_y)^2}(1-\rho_x^2)(1-\rho_y^2)\right). \qquad (11)$$

In practice, the slope b is not known but estimated using the training sample $\mathcal{D}_t^*$ which makes the computation of the loss much more cumbersome. The aim of the paper is to provide tractable and analytical formulae for L and to characterize its sensitivity to key parameters. Indeed, as will be proven later on, some characteristics of the data generating processes (1) and (2) have little interest because their impact on L is straightforward. In order to focus on the key elements, we will work with an alternative formulation which we specify below.

Electronic copy available at: https://ssrn.com/abstract=3709412

2.2. Reformulation of the problem

Replacing $\hat{a}(\mathcal{D}_t^*)$ with its expression in Equation (8) leads to

$$L = \mathbb{E}\left[\left(y_{t+k}^* - \bar{y}^* - \hat{b}(\mathcal{D}_t^*)(x_t^* - \bar{x}^*)\right)^2\right]. \tag{12}$$

The differences $y_{t+k}^* - \bar{y}^*$ and $x_t^* - \bar{x}^*$ in the two processes seem to indicate that the drift terms α_x and α_y will cancel out and that L will not be affected by these terms. Also, the slope term $\hat{b}(\mathcal{D}_t^*)$ is linear in the y process and inversely proportional to the x process, loosely speaking. Hence we expect that L will be linearly dependent on σ_y^2, but insensitive to σ_x^2. These claims are formally proven in our first theorem below.

Taking into account these observations, Equation (8) can be simplified to the equivalent definition

$$L := L(T, k, \rho_x, \rho_y, \rho, \sigma_y^2) = \sigma_y^2\, \mathbb{E}\left[\left(y_{t+k} - \hat{a}(\mathcal{D}_t) - \hat{b}(\mathcal{D}_t)x_t\right)^2\right], \tag{13}$$

where (x_t, y_{t+k}) and the training sample $\mathcal{D}_t = \{(x_{t-k-T+1}, y_{t-T+1}), ..., (x_{t-k}, y_t)\}$ are obtained from the two simplified auto-regressive processes x and y defined by

$$x_{t+1} = \rho_x x_t + e_{x,t+1}, \tag{14}$$

$$y_{t+1} = \rho_y y_t + e_{y,t+1}, \tag{15}$$

with autocorrelations ρ_x and ρ_y and correlated Gaussian white noise processes e_x and e_y with variances equal to 1 and correlation ρ.

The definitions of the estimators $\hat{a}$ and $\hat{b}$ remain the same as in (4) and (5) but applied to the new training sample $\mathcal{D}_t$. This means that, even though the constants of the two processes x and y are now equal to 0, the sample means $\bar{x}$ and $\bar{y}$ of processes x and y must not be forgotten in the computation of $\hat{a}(\mathcal{D}_t)$ and $\hat{b}(\mathcal{D}_t)$.

The new formulation (13) leads to

$$\begin{aligned} L &= \sigma_y^2\, \mathbb{E}\left[\left(y_{t+k} - \bar{y} - \hat{b}(\mathcal{D}_t)(x_t - \bar{x})\right)^2\right] && (16)\\ &= \sigma_y^2\, \mathbb{E}\left[(y_{t+k} - \bar{y})^2\right] - 2\mathbb{E}\left[(y_{t+k} - \bar{y})(x_t - \bar{x})\hat{b}(\mathcal{D}_t)\right] + \mathbb{E}\left[(x_t - \bar{x})^2 \hat{b}(\mathcal{D}_t)^2\right] \\ &= \sigma_y^2 \left(\underbrace{\mathbb{E}\left[\bar{\boldsymbol{z}}'\boldsymbol{E}\bar{\boldsymbol{z}}\right]}_{\text{variance of } y_{t+k}-\bar{y}} - \underbrace{2\mathbb{E}\left[(\bar{\boldsymbol{z}}'\boldsymbol{C}\bar{\boldsymbol{z}})\left(\frac{\bar{\boldsymbol{z}}'\boldsymbol{A}\bar{\boldsymbol{z}}}{\bar{\boldsymbol{z}}'\boldsymbol{B}\bar{\boldsymbol{z}}}\right)\right]}_{\text{cross term}} + \underbrace{\mathbb{E}\left[(\bar{\boldsymbol{z}}'\boldsymbol{D}\bar{\boldsymbol{z}})\left(\frac{\bar{\boldsymbol{z}}'\boldsymbol{A}\bar{\boldsymbol{z}}}{\bar{\boldsymbol{z}}'\boldsymbol{B}\bar{\boldsymbol{z}}}\right)^2\right]}_{\text{quadratic term}} \right), && (17) \end{aligned}$$

where $\bar{\boldsymbol{z}} = \boldsymbol{N}\boldsymbol{z} = [x_{t-k-T+1} - \bar{x}, \ldots, x_{t-k} - \bar{x}, x_t - \bar{x}, y_{t-T+1} - \bar{y}, \ldots, y_t - \bar{y}, y_{t+k} - \bar{y}]'$ is a Gaussian vector built from linear combinations of all the random variables of the system stacked in vector $\boldsymbol{z} = [x_{t-k-T+1}, \ldots, x_{t-k}, x_t, y_{t-T+1}, \ldots, y_t, y_{t+k}]'$ and where $\boldsymbol{N}$ is defined by

$$\boldsymbol{N} = \begin{bmatrix} \boldsymbol{N}_{T+1} & \boldsymbol{0}_{T+1} \\ \boldsymbol{0}_{T+1} & \boldsymbol{N}_{T+1} \end{bmatrix} \quad \text{with} \quad \boldsymbol{N}_{T+1} = \begin{bmatrix} \boldsymbol{M}_T & \boldsymbol{0}_{T,1} \\ -\frac{1}{T}\boldsymbol{1}_{1,T} & 1 \end{bmatrix} \quad \text{and} \quad \boldsymbol{M}_T = \boldsymbol{I}_T - \frac{1}{T}\boldsymbol{1}_T.$$

We note that $\boldsymbol{M}_T$ is a demeaning (i.e. centering) operator. Matrices $\boldsymbol{A}$, $\boldsymbol{B}$, $\boldsymbol{C}$, $\boldsymbol{D}$ and $\boldsymbol{E}$ in Equation (17) are given below:

Electronic copy available at: https://ssrn.com/abstract=3709412

$$\boldsymbol{A} = \frac{1}{2}\begin{bmatrix} \mathbf{0}_{T+1} & \boldsymbol{J}_{T+1} \\ \boldsymbol{J}_{T+1} & \mathbf{0}_{T+1} \end{bmatrix}, \quad \boldsymbol{B} = \begin{bmatrix} \boldsymbol{J}_{T+1} & \mathbf{0}_{T+1} \\ \mathbf{0}_{T+1} & \mathbf{0}_{T+1} \end{bmatrix} \quad \boldsymbol{C} = \frac{1}{2}\begin{bmatrix} \mathbf{0}_{T+1} & \boldsymbol{K}_{T+1} \\ \boldsymbol{K}_{T+1} & \mathbf{0}_{T+1} \end{bmatrix}$$

$$\boldsymbol{D} = \begin{bmatrix} \boldsymbol{K}_{T+1} & \mathbf{0}_{T+1} \\ \mathbf{0}_{T+1} & \mathbf{0}_{T+1} \end{bmatrix}, \quad \boldsymbol{E} = \begin{bmatrix} \mathbf{0}_{T+1} & \mathbf{0}_{T+1} \\ \mathbf{0}_{T+1} & \boldsymbol{K}_{T+1} \end{bmatrix}$$

where

$$\boldsymbol{J}_{T+1} = \begin{bmatrix} \boldsymbol{I}_T & \mathbf{0}_{T,1} \\ \mathbf{0}_{1,T} & 0 \end{bmatrix} \quad \text{and} \quad \boldsymbol{K}_{T+1} = \boldsymbol{I}_{T+1} - \boldsymbol{J}_{T+1} = \begin{bmatrix} \mathbf{0}_T & \mathbf{0}_{T,1} \\ \mathbf{0}_{1,T} & 1 \end{bmatrix}.$$

$\boldsymbol{z}$ is a stationary Gaussian vector with mean $\mathbf{0}_{2(T+1)}$ and covariance matrix $\boldsymbol{\Sigma}$ given by the following block form

$$\boldsymbol{\Sigma} = \begin{bmatrix} \boldsymbol{\Sigma}^{\boldsymbol{x}}_{T+1} & \left(\boldsymbol{\Sigma}^{\boldsymbol{xy}}_{T+1}\right)' \\ \boldsymbol{\Sigma}^{\boldsymbol{xy}}_{T+1} & \boldsymbol{\Sigma}^{\boldsymbol{y}}_{T+1} \end{bmatrix}, \tag{18}$$

where $\boldsymbol{\Sigma}^{\boldsymbol{x}}_{T+1}$ and $\boldsymbol{\Sigma}^{\boldsymbol{y}}_{T+1}$ are the two autocorrelation matrices of $\boldsymbol{x} = [x_{t-k-T+1}, \ldots, x_{t-k}, x_t]'$ and $\boldsymbol{y} = [y_{t-T+1}, \ldots, y_t, y_{t+k}]'$ and $\boldsymbol{\Sigma}^{\boldsymbol{xy}}_{T+1}$ the covariance matrix between $\boldsymbol{x}$ and $\boldsymbol{y}$. In the following, we will use subscript T and denote with $\boldsymbol{\Sigma}^{\boldsymbol{x}}_{T}$ (resp. $\boldsymbol{\Sigma}^{\boldsymbol{y}}_{T}$) the covariance matrix of the first T elements $[x_{t-k-T+1}, \ldots, x_{t-k}]'$ of vectors $\boldsymbol{x}$ (resp. the first T elements $[y_{t-T+1}, \ldots, y_t]'$ of vector $\boldsymbol{y}$). Moreover, we will denote with $\boldsymbol{\epsilon_x}$ (resp. $\boldsymbol{\epsilon_y}$) the covariance vector of x_t with the first T elements $[x_{t-k-T+1}, \ldots, x_{t-k}]'$ (resp. the covariance vector of y_{t+k} with the first T elements $[y_{t-T+1}, \ldots, y_t]'$). Therefore we can write[8]

$$\boldsymbol{\Sigma}^{\boldsymbol{x}}_{T+1} = \begin{bmatrix} \boldsymbol{\Sigma}^{\boldsymbol{x}}_{T} & \boldsymbol{\epsilon_x} \\ \boldsymbol{\epsilon}'_{\boldsymbol{x}} & (1-\rho_x^2)^{-1} \end{bmatrix} \quad \text{with} \quad \boldsymbol{\epsilon_x} = (1-\rho_x^2)^{-1}\left[\rho_x^{k+T-1}, \ldots, \rho_x^k\right]',$$

and where $\boldsymbol{\Sigma}^{\boldsymbol{x}}_{T}$ and $\boldsymbol{\Sigma}^{\boldsymbol{y}}_{T}$ are Toeplitz symmetric definite positive matrices whose elements on row i and column j are equal to $(1-\rho_x^2)^{-1} \times \rho_x^{|i-j|}$ and $(1-\rho_y^2)^{-1} \times \rho_y^{|i-j|}$ respectively.

Matrix $\boldsymbol{\Sigma}^{\boldsymbol{xy}}_{T+1}$ is a bit more complicated but can be written as

$$\boldsymbol{\Sigma}^{\boldsymbol{xy}}_{T+1} = \frac{\rho}{1-\rho_x\rho_y}\begin{bmatrix} \boldsymbol{\Xi}^{\boldsymbol{xy}}_{T} & \boldsymbol{\xi_1} \\ \boldsymbol{\xi}'_{\boldsymbol{2}} & \rho_y^k \end{bmatrix} \quad \text{with} \quad \boldsymbol{\xi_1} = \left[\rho_x^{T-1}, \ldots, \rho_x, 1\right]' \quad \text{and} \quad \boldsymbol{\xi_2} = \left[\rho_y^{2k+T-1}, \ldots, \rho_y^{2k}\right]', \tag{19}$$

and the element on row i and column j of matrix $\boldsymbol{\Xi}^{\boldsymbol{xy}}_{T}$ corresponds to the covariance between $x_{t-k-T+j}$ and y_{t-T+i} and is given by

$$\left[\boldsymbol{\Xi}^{\boldsymbol{xy}}_{T}\right]_{i,j} = \begin{cases} \rho_y^{i+k-j} & \text{for } i+k \geq j \\ \rho_x^{j-i-k} & \text{for } j \geq i+k \end{cases}.$$

Finally, we denote with $\bar{\boldsymbol{\Sigma}}$ the covariance matrix $\boldsymbol{N}\boldsymbol{\Sigma}\boldsymbol{N}'$ of the Gaussian vector $\bar{\boldsymbol{z}} = \boldsymbol{N}\boldsymbol{z}$ built from the linear combination $\boldsymbol{N}$ applied to all the random variables of the system stacked in vector $\boldsymbol{z} = [x_{t-k-T+1}, \ldots, x_{t-k}, x_t, y_{t-T+1}, \ldots, y_t, y_{t+k}]'$. We write Λ_i the eigenvalues of $\boldsymbol{B}\bar{\boldsymbol{\Sigma}}\boldsymbol{B}$ and $\boldsymbol{\Lambda}$ is the diagonal matrix that contains these values.

With all of these notations, we are equipped to proceed to our theoretical results.

[8] The same notations will be used for y.

Electronic copy available at: https://ssrn.com/abstract=3709412

2.3. Main results

Our first result holds in all generality. The first three points are simple yet useful properties of the quadratic loss, while the last item is its core decomposition, which we derive from the main theorem of Magnus (1990). The latter is recalled in Appendix H.

Theorem 2. *Consider two auto-regressive processes with correlated innovation processes given by (1) and (2), and the predictive regression model (3). The bias (average error) of the predictive model is zero. Moreover, its quadratic loss L of the predictive model, defined in (8), satisfies the following properties:*

1. *L is independent of the two constants α_x and α_y.*
2. *L is independent of σ_x^2, the variance of innovations in the predictor process x^*.*
3. *L is proportional to σ_y^2, the variance of innovations in the predicted process y^*.*
4. *L is equal to*

$$L(T, k, \rho_x, \rho_y, \rho, \sigma_y^2) = \sigma_y^2 \left(\mathrm{tr}(\boldsymbol{E}\bar{\boldsymbol{\Sigma}}) + \int_0^\infty |\boldsymbol{\Delta}| \, (t\, f_2(t) - 2 f_1(t))\, dt \right), \tag{20}$$

where $|\boldsymbol{\Delta}|$ denotes the determinant of matrix $\boldsymbol{\Delta} = (\boldsymbol{I}_{2(T+1)} + 2t\boldsymbol{\Lambda})^{-1/2}$, and the two functions f_1 and f_2 are given by

$$f_1(t) = \mathrm{tr}(\boldsymbol{A}^*)\mathrm{tr}(\boldsymbol{C}^*) + 2\mathrm{tr}(\boldsymbol{A}^*\boldsymbol{C}^*), \tag{21}$$

$$f_2(t) = 2\mathrm{tr}(\boldsymbol{A}^*\boldsymbol{A}^*)\mathrm{tr}(\boldsymbol{D}^*) + 8\mathrm{tr}(\boldsymbol{A}^*\boldsymbol{A}^*\boldsymbol{D}^*) + \mathrm{tr}(\boldsymbol{A}^*)^2\mathrm{tr}(\boldsymbol{D}^*) + 4\mathrm{tr}(\boldsymbol{A}^*)\mathrm{tr}(\boldsymbol{A}^*\boldsymbol{D}^*), \tag{22}$$

with $\boldsymbol{A}^* = \boldsymbol{A}\bar{\boldsymbol{\Sigma}}\boldsymbol{W}$, $\boldsymbol{C}^* = \boldsymbol{C}\bar{\boldsymbol{\Sigma}}\boldsymbol{W}$ and $\boldsymbol{D}^* = \boldsymbol{D}\bar{\boldsymbol{\Sigma}}\boldsymbol{W}$ where $\boldsymbol{W} = (\boldsymbol{I}_{2(T+1)} + 2t\boldsymbol{B}\bar{\boldsymbol{\Sigma}})^{-1}$.

The proof of the Theorem is located in Appendix B. The three terms we obtain in Equation (20) pertain to the decomposition obtained in (17): $\mathrm{tr}(\boldsymbol{E}\bar{\boldsymbol{\Sigma}})$ is the matrix representation of the variance of $y_{t+k} - \bar{y}$, while f_1 relates to the cross term, and f_2 stems from the quadratic term.[9]

In the literature on predictive models, a common yardstick for accuracy assessment is the out-of-sample R^2, which, in our setting is equal to $R^2 = 1 - L/\mathrm{var}(y^*_{t+k})$. Replacing the variance of y^*_{t+k} with its expression, Theorem 2 leads to

$$R^2(T, k, \rho_x, \rho_y, \rho) = 1 - (1 - \rho_y^2) \left(\mathrm{tr}(\boldsymbol{E}\bar{\boldsymbol{\Sigma}}) + \int_0^\infty |\boldsymbol{\Delta}| \, (t\, f_2(t) - 2 f_1(t))\, dt \right), \tag{23}$$

so that the R^2 does not depend on σ_y^2. In the theoretical section of the paper, we restrict our focus to the loss L, but the translation to the R^2 is immediate. Typically, if both the intercept a and the slope b of the linear model are known, the optimal R^2 is equal to

$$R_o^2 = \rho_y^{2k} \frac{\rho^2}{(1 - \rho_x\rho_y)^2}(1 - \rho_x^2)(1 - \rho_y^2), \tag{24}$$

[9] When the means of processes x^* and y^* are known, we can replace the sample means $\bar{x}^*$ and $\bar{y}^*$ in Equation (12) by the true means or equivalently replace the sample means $\bar{x}$ and $\bar{y}$ in Equation (16) by 0. Therefore, Theorem 2 remains true but with the simplification $\bar{\boldsymbol{\Sigma}} = \boldsymbol{\Sigma}$, which is equivalent to taking $\boldsymbol{N} = \boldsymbol{I}_{2(T+1)}$.

Electronic copy available at: https://ssrn.com/abstract=3709412

which is simply an affine transform of Equation (11). The only obvious conclusions from this form is that R_o^2 is never negative and that it increases with $|\rho|$. Obviously, if the two processes are unrelated, there is not much to learn from one another. Surprisingly, this may not be true with small samples.

The theorem below extends our results in several directions. First, it sheds light on an intuitive property of the loss: it is symmetric with respect to the correlation between the two innovations. Second, it underlines the simplifications that occur when this correlation, ρ, is equal to zero, that is, when the two innovations are uncorrelated. In particular, we are able to derive simpler explicit expressions for $|\boldsymbol{\Delta}_T|$ and for the functions f_1 and f_2.

Before we can formulate the result, we need to specify a matrix decomposition at the heart of some simplifications. If we denote with $\bar{\boldsymbol{\Sigma}}^{\boldsymbol{x}}_{\boldsymbol{T+1}}$ the covariance matrix $\boldsymbol{N_{T+1}}\boldsymbol{\Sigma}^{\boldsymbol{x}}_{\boldsymbol{T+1}}\boldsymbol{N}'_{\boldsymbol{T+1}}$ of the first $T+1$ elements of the Gaussian vector $\bar{\boldsymbol{z}}$, then a direct computation shows that

$$\bar{\boldsymbol{\Sigma}}^{\boldsymbol{x}}_{\boldsymbol{T+1}} = \begin{bmatrix} \bar{\boldsymbol{\Sigma}}^{\boldsymbol{x}}_{T} & \bar{\boldsymbol{\epsilon}}_{\boldsymbol{x}} \\ \bar{\boldsymbol{\epsilon}}'_{\boldsymbol{x}} & \bar{s}^2_x \end{bmatrix},$$

where the components are given by

$$\begin{aligned} \bar{\boldsymbol{\Sigma}}^{\boldsymbol{x}}_{T} &= \boldsymbol{M}_T \boldsymbol{\Sigma}^{\boldsymbol{x}}_{T} \boldsymbol{M}_T, \\ \bar{\boldsymbol{\epsilon}}_{\boldsymbol{x}} &= \boldsymbol{M}_T \left(\boldsymbol{\epsilon}_{\boldsymbol{x}} - \frac{1}{T} \boldsymbol{\Sigma}^{\boldsymbol{x}}_{T} \mathbf{1}_{T,1} \right), \\ \bar{s}^2_x &= \frac{1}{T^2} \mathbf{1}_{1,T} \boldsymbol{\Sigma}^{\boldsymbol{x}}_{T} \mathbf{1}_{T,1} - \frac{2}{T} \mathbf{1}_{1,T} \boldsymbol{\epsilon}_{\boldsymbol{x}} + (1-\rho_x^2)^{-1}. \end{aligned}$$

With these notations at hand, we can proceed to our second major result.

Theorem 3. *Under the same assumptions as those of Theorem 2, the quadratic loss L of the predictive model, defined in (8), satisfies the following properties:*

1. *L is a symmetric function of the correlation ρ between the innovations of the predictor x^* and the predicted y^* processes.*

2. *If the two processes x^* and y^* have uncorrelated innovations, i.e., $\rho = 0$, then L is equal to*

$$L(T, k, \rho_x, \rho_y, \rho, \sigma_y^2) = \sigma_y^2 \left(\bar{s}^2_y + \int_0^\infty |\boldsymbol{\Delta}_T| \, (t\, f_2(t) - 2 f_1(t))\, dt \right), \tag{25}$$

where $|\boldsymbol{\Delta}_T|$ denotes the determinant of matrix $\boldsymbol{\Delta}_T = (\boldsymbol{I}_T + 2t\boldsymbol{\Lambda}_T)^{-1/2}$ built from the diagonal matrix $\boldsymbol{\Lambda}_T$ containing the eigenvalues of $\bar{\boldsymbol{\Sigma}}^{\boldsymbol{x}}_T$, and the two functions f_1 and f_2 are given by

$$f_1(t) = \bar{\boldsymbol{\epsilon}}'_{\boldsymbol{x}} \boldsymbol{\Omega} \bar{\boldsymbol{\epsilon}}_{\boldsymbol{y}}, \tag{26}$$

$$f_2(t) = \mathrm{tr}(\bar{\boldsymbol{\Sigma}}^{\boldsymbol{x}}_T \boldsymbol{\Omega} \bar{\boldsymbol{\Sigma}}^{\boldsymbol{y}}_T) \left(\bar{s}^2_x - 2t\, \bar{\boldsymbol{\epsilon}}'_{\boldsymbol{x}} \boldsymbol{\Omega} \bar{\boldsymbol{\epsilon}}_{\boldsymbol{x}} \right) + 2 \bar{\boldsymbol{\epsilon}}'_{\boldsymbol{x}} \boldsymbol{\Omega} \bar{\boldsymbol{\Sigma}}^{\boldsymbol{y}}_T \boldsymbol{\Omega} \bar{\boldsymbol{\epsilon}}_{\boldsymbol{x}}, \tag{27}$$

where $\boldsymbol{\Omega} = (\boldsymbol{I}_T + 2t\bar{\boldsymbol{\Sigma}}^{\boldsymbol{x}}_T)^{-1}$.

3. *If we denote with $\boldsymbol{P}_T$ the orthogonal matrix such that $\bar{\boldsymbol{\Sigma}}^{\boldsymbol{x}}_T = \boldsymbol{P}_T \boldsymbol{\Lambda}_T \boldsymbol{P}'_T$ and with $\lambda_1, ..., \lambda_T$ the eigenvalues of $\bar{\boldsymbol{\Sigma}}^{\boldsymbol{x}}_T$ stacked on the diagonal of $\boldsymbol{\Lambda}_T$, then the determinant of $\boldsymbol{\Delta}_T$ and the*

Electronic copy available at: https://ssrn.com/abstract=3709412

functions f_1 and f_2 defined in Equations (26) and (27) are equal to

$$|\boldsymbol{\Delta}_T| = \prod_{j=1}^{T} (1+2t\lambda_j)^{-\frac{1}{2}} \tag{28}$$

$$f_1(t) = \sum_{j=1}^{T} \frac{p_j q_j}{1+2t\lambda_j}, \tag{29}$$

$$f_2(t) = \sum_{j=1}^{T} \frac{Q_{jj}\lambda_j}{1+2t\lambda_j} \left(\bar{s}_x^2 - 2t \sum_{j=1}^{T} \frac{p_j^2}{1+2t\lambda_j} \right) + 2 \sum_{i=1}^{T} \sum_{j=1}^{T} \frac{Q_{ij} p_i p_j}{(1+2t\lambda_i)(1+2t\lambda_j)}, \tag{30}$$

where the two vectors $\boldsymbol{p}$ and $\boldsymbol{q}$ denote the quantities $\boldsymbol{P}_T' \bar{\boldsymbol{\epsilon}}_{\boldsymbol{x}}$ and $\boldsymbol{P}_T' \bar{\boldsymbol{\epsilon}}_{\boldsymbol{y}}$ respectively and matrix $\boldsymbol{Q}$ denotes the product $\boldsymbol{P}_T' \bar{\boldsymbol{\Sigma}}_T^{\boldsymbol{y}} \boldsymbol{P}_T$.

The results[10] of Theorem 3 and especially Equations (29) and (30) are of particular interest. Indeed, we see that Equation (25) involves quantities of the form $I = \int_0^\infty h(t)dt$ and, splitting the integral into two terms $I = \int_0^u h(t)dt + \int_u^\infty h(t)dt$ allows us to compute the first part via Riemann trapezoidal sums and, if u is large enough, to approximate the second term as the integral of a polynomial function of t where we infer from both Equations (29) and (30) the behavior of the two integrands when t converges to infinity.[11]

Our final theoretical result pertains to the impact of the forecasting horizon on the loss. Intuitively, we would expect that the loss increases with k because predictions further away in time are subject to more uncertainty. Under strong assumptions, we are able to prove this fact in the following lemma.

Lemma 4. *Under the assumptions of Theorem 2, if we impose that the processes x^* and y^* have known means, that they have uncorrelated innovations ($\rho = 0$) and that they have identical autocorrelation satisfying $0 \leq \rho_x = \rho_y \leq 1$, then the MSE increments for two consecutive values of k is positive, i.e.*

$$\Delta_k L = L(T, k+1, \rho_x, \rho_y, \rho, \sigma_y^2) - L(T, k, \rho_x, \rho_y, \rho, \sigma_y^2) > 0 \quad \textit{for } k \geq 1. \tag{31}$$

Our theoretical results need to be confronted to numerical exercises, which is the purpose of the next section.

3. Numerical experiments

This section has multiple purposes. First, it warrants that the analytical formulae from Section 2.3 are valid. Second, it shows the computational efficiency of closed forms like the ones derived in the previous section. Third, it helps capture the impact of parameters on the R^2 and infer the configurations when predictions are reliable.

3.1. Methods and computational efficiency

We test three numerical routes to calculate the MSE defined in Equation (8). For simplicity we will refer to them with one word:

[10] When the means of processes x^* and y^* are known, Theorem 3 still holds but with the simplifications $\bar{\boldsymbol{\Sigma}}_{T+1}^{\boldsymbol{x}} = \boldsymbol{\Sigma}_{T+1}^{\boldsymbol{x}}$ and $\bar{\boldsymbol{\Sigma}}_{T+1}^{\boldsymbol{y}} = \boldsymbol{\Sigma}_{T+1}^{\boldsymbol{y}}$ or equivalently $\bar{\boldsymbol{\Sigma}}_T^{\boldsymbol{x}} = \boldsymbol{\Sigma}_T^{\boldsymbol{x}}$, $\bar{\boldsymbol{\Sigma}}_T^{\boldsymbol{y}} = \boldsymbol{\Sigma}_T^{\boldsymbol{y}}$, $\bar{\boldsymbol{\epsilon}}_x = \boldsymbol{\epsilon}_x$, $\bar{\boldsymbol{\epsilon}}_y = \boldsymbol{\epsilon}_y$, $\bar{s}_x^2 = (1-\rho_x^2)^{-1}$ and $\bar{s}_y^2 = (1-\rho_y^2)^{-1}$.

[11] More details about the behavior of the two integrands when t converges to infinity are given in Appendix F.

Electronic copy available at: https://ssrn.com/abstract=3709412

- **process**: we run simulations of the processes x^* and y^*, compute the OLS estimates and the MSE à la Monte Carlo using Equation (8);

- **multivariate**: we run simulations of the Gaussian variate $\boldsymbol{z}$ according to the covariance matrix (18) followed again by an aggregation via a Monte Carlo averaging of Equation (17);

- **integral**: we use the pseudo closed-form solution (20) with numerical evaluation of the integral. In the case where the two processes have uncorrelated innovations (i.e. $\rho = 0$), we can resort to the simpler form of Equation (25).

The first two methods are straightforward to implement and both heavily rely on Monte Carlo simulations. The last one requires further clarification which we provide in Appendix F.

In Figure 1, we plot the MSE values obtained for different levels of discretizations (for the **integral** method) and Monte Carlo (MC) scenarios (for both the **process** and **multivariate** methods). The sample size is equal to $T = 36$ and the lag equal to $k = 12$ in the left panel.[12] Given the large number of discretization points, the integral method yields a constant result. In contrast, the other methods generate oscillating patterns, which is a clear indication of the superiority of the integral formulae. In the right panel, we plot the computation times (on CPU) obtained for different training sample size T. The **multivariate** approach is not shown because it is too slow and not competitive.

Because the speed of convergence of MC averaging is $n^{-1/2}$, the magnitude of the error is expected to have an order of 10^{-3} for a number of points equal to 10^6 - which is what we observe to the right of the left panel. Computation times (CPU) are exceedingly large for the **multivariate** method which is the reason why we discard this method from the right panel of Figure 1 and from the remaining of the paper. One reason for this is that it involves lengthy matrix operations, which is not the case for the **process** technique. In term of computation times, the **integral** and **process** are hardly comparable. The fastest one is the integral evaluation, by far, which, as expected, highlights the gains stemming from closed-form expressions.

While the mean squared error is a common performance indicator, we will henceforth report R^2 values, for two reasons. First, by Equation (23), it reduces the degrees of freedom by one unit because we do not need to worry about σ_y^2. Second, it has no particular scale, which makes it easy to compare across different configurations.

3.2. Comparison to univariate learning

A natural question that arises in predictive modelling is whether resorting to an exogenous variable, x, is useful compared to relying solely on y. What we call univariate learning consists in simply predicting y^*_{t+k} based on the current level of y^*_t without using process x^*. This topic is studied in Fuller and Hasza (1980), Hoque et al. (1988), Magnus and Pesaran (1989, 1991) and more recently in Ing et al. (2009), where analytical results are derived.

Given a trial sample $\mathcal{D}^*_t = \{y^*_{t-T}, ..., y^*_t\}$ of size $T+1$, the least-squares estimators of α_y and ρ_y are given by:

$$\hat{\rho}_y(\mathcal{D}^*_t) = \frac{\sum_{s=0}^{T-1}(y^*_{t-s-1} - \bar{y}^*_1)(y^*_{t-s} - \bar{y}^*_2)}{\sum_{s=0}^{T-1}(y^*_{t-s-1} - \bar{y}^*_1)^2} \tag{32}$$

$$\hat{\alpha}_y(\mathcal{D}^*_t) = \bar{y}^*_2 - \hat{\rho}_y(\mathcal{D}^*_t)\bar{y}^*_1, \tag{33}$$

[12] Results for other sample sizes are not particularly enlightening. We invite the interested reader to check the online *comparison* notebook on the paper's website: http://www.gcoqueret.com/PRs.html for further details.

Electronic copy available at: https://ssrn.com/abstract=3709412

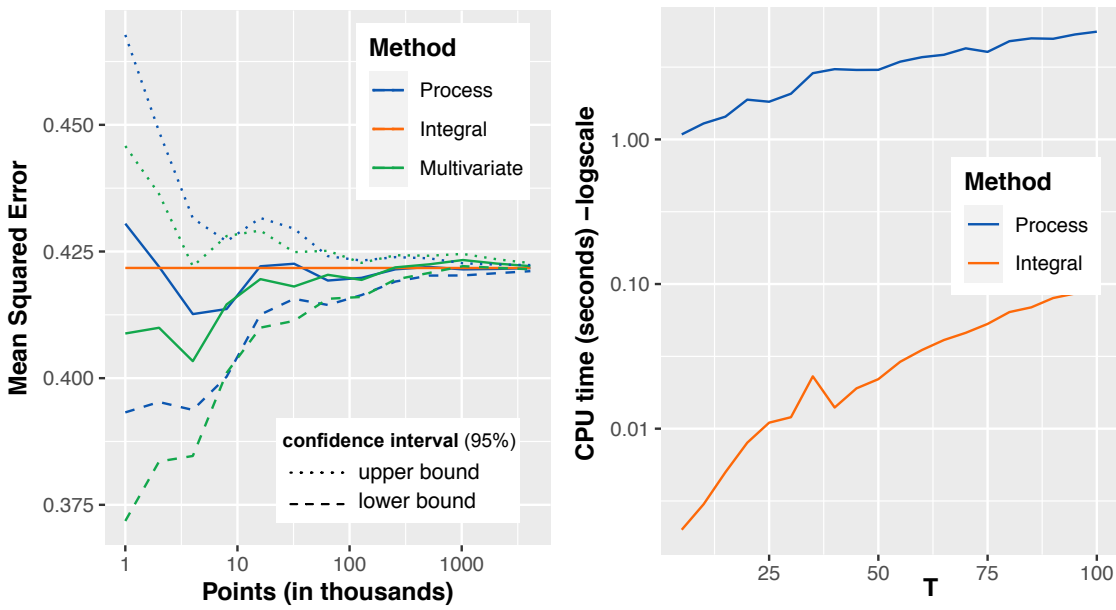

Fig. 1. **Comparison of methods - speed and accuracy**. In the left panel, we plot the values obtained for the MSE as a function of the computational cost n expressed by the number of Monte Carlo simulations (for both the **process** and **multivariate** methods) or the number of discretization points in the Riemann integral (for the **integral** method). The cost ranges from $n = 10^3$ up to $n \approx 4 \times 10^6$. We use a training sample size $T = 36$ - equivalent to 3 years of monthly data. In the right graph, we show the CPU time as a function of the training sample size T using a number of points equal to 10^3 for the discretization of the **integral** method and a number of simulations equal to 10^6 for the **process** method. The other parameters are fixed and equal to $\rho_x = 0.9$, $\rho_y = 0.7$, $\rho = 0$ and $k = 12$. For the Integral method, the upper bound for the trapezoidal approximation is taken equal to $u = 100\lfloor \log(n) \rfloor$ - see Appendix F for more details. The confidence intervals around the means are defined by ± 1.96 times the standard deviations of simulated squared errors, divided by $\sqrt{n}$.

with sample means $\bar{y}_1^* = \frac{1}{T}\sum_{s=0}^{T-1} y_{t-s-1}^*$ and $\bar{y}_2^* = \frac{1}{T}\sum_{s=0}^{T-1} y_{t-s}^*$. A simple computation shows that the numerator of the estimator in Equation (32) is also equal to

$$\begin{aligned}
\sum_{s=0}^{T-1}(y_{t-s-1}^* - \bar{y}_1^*)(y_{t-s}^* - \bar{y}_2^*) &= \sum_{s=0}^{T-1}(y_{t-s-1}^* - \bar{y}_1^*)(y_{t-s}^* - \bar{y}_1^* + \bar{y}_1^* - \bar{y}_2^*) \\
&= \sum_{s=0}^{T-1}(y_{t-s-1}^* - \bar{y}_1^*)(y_{t-s}^* - \bar{y}_1^*) + (\bar{y}_1^* - \bar{y}_2^*)\sum_{s=0}^{T-1}(y_{t-s-1}^* - \bar{y}_1^*) \\
&= \sum_{s=0}^{T-1}(y_{t-s-1}^* - \bar{y}_1^*)(y_{t-s}^* - \bar{y}_1^*),
\end{aligned}$$

which leads to the simple formulation

$$\hat{\rho}_y(\mathcal{D}_t^*) = \frac{\bar{\boldsymbol{y}}'\boldsymbol{F}_{T+1}\bar{\boldsymbol{y}}}{\bar{\boldsymbol{y}}'\boldsymbol{J}_{T+1}\bar{\boldsymbol{y}}}, \tag{34}$$

where we have used the notation $\boldsymbol{y} = [y_{t-T}^*, ..., y_t^*]'$, the identity $\bar{\boldsymbol{y}} = \boldsymbol{N}_{T+1}\boldsymbol{y}$ where $\boldsymbol{N}_{T+1}$ and $\boldsymbol{J}_{T+1}$

Electronic copy available at: https://ssrn.com/abstract=3709412

are the same matrices as in Section 2 and where matrix $\boldsymbol{F}_{T+1}$ is given by

$$\boldsymbol{F}_{T+1} = \frac{1}{2}\begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 1 & \ddots & \ddots & & \vdots \\ 0 & \ddots & & 1 & 0 \\ \vdots & & 1 & 0 & 1 \\ 0 & \dots & 0 & 1 & 0 \end{bmatrix}. \tag{35}$$

We see that Equation (34) is the ratio of two quadratic forms so its expectation is given by Magnus (1986) and with our notations reads[13]

$$\mathbb{E}\left[\hat{\rho}_y(\mathcal{D}_t^*)\right] = \int_0^\infty |\boldsymbol{\Delta}_{T+1}| \operatorname{tr}(\boldsymbol{F}^*) dt, \tag{36}$$

where $|\boldsymbol{\Delta}_{T+1}|$ denotes the determinant of matrix $\boldsymbol{\Delta}_{T+1} = (\boldsymbol{I}_{T+1} + 2t\boldsymbol{\Lambda}_{T+1})^{-1/2}$ built from the diagonal matrix $\boldsymbol{\Lambda}_{T+1}$ containing the eigenvalues of $\boldsymbol{J}_{T+1}\bar{\boldsymbol{\Sigma}}^{\boldsymbol{y}}_{T+1}\boldsymbol{J}_{T+1}$, and $\boldsymbol{F}^* = \boldsymbol{F}_{T+1}\bar{\boldsymbol{\Sigma}}^{\boldsymbol{y}}_{T+1}(\boldsymbol{I}_{T+1} + 2t\boldsymbol{J}_{T+1}\bar{\boldsymbol{\Sigma}}^{\boldsymbol{y}}_{T+1})^{-1}$. More general moment identities for $\hat{\rho}_y$ are developed in Kan and Wang (2010).

Given the number of degrees of freedom in the models (ρ_x, ρ_y, ρ, T and k), we chose to focus on a small group of values. Henceforth, we set $\rho = 0.5$ because working with alternative levels does not alter the conclusions (further results are available on the paper's website). We consider three forecasting horizons ($k = 1$, 12 and 24), as well as two autocorrelation values for the dependent variable ($\rho_y = 0.3$ for mild persistence and $\rho_y = 0.9$ for high persistence).

In Figure 2, we plot the R^2 as a function of the training sample size. We show its value for five levels of ρ_x (bivariate learning), represented with color dots, while the black line symbolizes the results for univariate learning. Plainly, the latter is hard to improve when the forecasting horizon is short ($k = 1$) as the black line is always above the color points. However, when $k = 12$ or 24, it is possible to beat univariate learning, especially for low autocorrelations ρ_x of the predictor x and high autocorrelation ρ_y of the predicted variable y.

3.3. Sensitivity analysis

As emphasized in Equation (20), the loss function depends on 6 parameters and the R^2 in Equation (23), on 5. Because it is hard to grasp sensitivity in 4 dimensions, we agglomerate the results in Figures 3 for $k = 1$, i.e. short-term forecasting, and in Figure 4 for long-term predictions with $k = 12$.

The major takeaway from Figure 3 is that ρ_y matters much more than ρ_x because all rows are fairly similar, while the columns exhibit significantly different patterns. The most striking effect is when ρ_y increases to large levels: in this case, the impact of the training sample size reverses. When y is not too autocorrelated, it is preferable to resort to long training samples, as they increase the R^2. However, when the dependent variable is very persistent, short training samples yield better results. When the prediction horizon increases, see Figure 4 with $k = 12$, these considerations do not hold anymore, as all curves behave alike, i.e. longer training samples imply larger R-squared.

It is worth noting that almost all the curves in Figures 3 and 4 seem to converge toward an asymptotic value of R-squared that is close to zero when T increases. In order to understand this feature, we refer to Equation (24) which gives the expression of the R^2 in the hypothetical case where the slope b of the predictive linear model is equal to the value $b_o = \operatorname{cov}(x_t, y_{t+k})/\operatorname{var}(x_t)$ that

[13] A proof of this expression is provided in Appendix E.

Electronic copy available at: https://ssrn.com/abstract=3709412

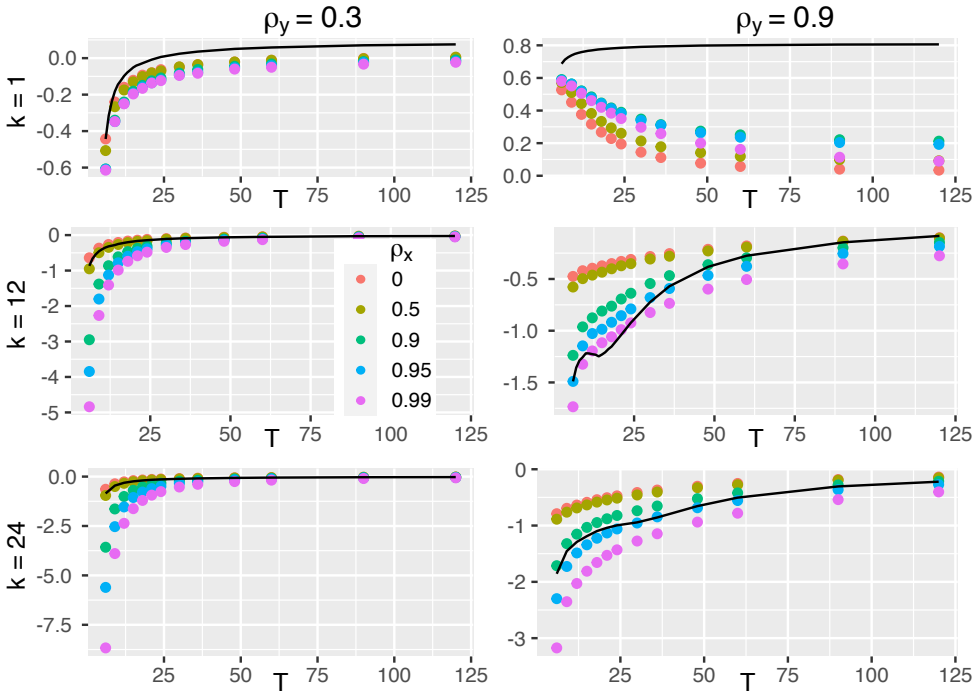

Fig. 2. **Comparison with univariate learning**. We plot the R^2 for different choices of T (x-axis) and ρ_x (color). In the graph, we show the values computed via one million Monte Carlo simulations. The parameters are equal to $\sigma_x^2 = \sigma_y^2 = 1$, $\rho = 0.5$, $\rho_y \in \{0.3, 0.9\}$ (columns) and $k \in \{1, 12, 24\}$ (rows). All color points pertain to bivariate learning while the black lines relate to univariate learning.

minimizes the MSE (and maximizes the R^2). Of course, this is not the case in practice since the slope has to be estimated over training samples, but when the training sample size T increases, we show in Appendix G that the estimated slope converges in probability to the constant slope b_o.[14] Hence, from Equation (24), we see that when k increases to $+\infty$, and when ρ or ρ_y decrease to 0, the R^2 converges to zero.

It is only for the sets of parameters $k = 1, \rho_y \geq 0.5, \rho \geq 0.5$ that we observe slightly positive asymptotic values for the R^2. In short, learning is most effective when the forecasting horizon is short and when the dependent is persistent. Finally, comparing the ranges of R^2 in Figures 3 and 4, we have a strong confirmation that long horizons penalize forecasting accuracy, as was signalled in Lemma 4.

3.4. The memory trade-off

From the previous subsection, the best cocktail of parameters seems to be a combination of low k (short term forecasting), with high autocorrelation in the dependent variable. Sadly, in practice, both variable are often linked. The time lag k pertains to the horizon at which returns are computed. When $k = 1$, we seek to predict monthly returns while for $k = 12$, we consider annual returns. The problem is that short term returns are not very autocorrelated, in contrast to long terms ones, because the latter are evaluated on overlapping windows. Thus, low (*resp.* high) k are associated

[14]This pattern is investigated further in Section 3.5.

Electronic copy available at: https://ssrn.com/abstract=3709412

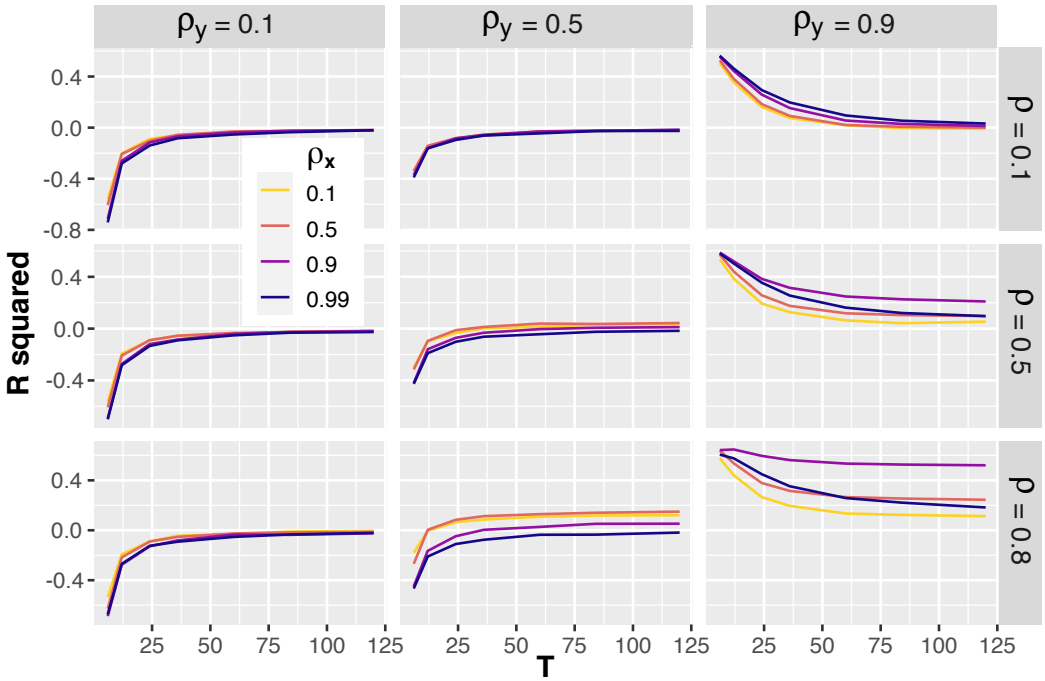

Fig. 3. **Sensitivity of R^2 with regard to autocorrelations and sample size (short term horizon)**. We plot the R^2 values for different choices of ρ, ρ_x, ρ_y and T with a fixed short-term lag $k = 1$. These values are computed via one million Monte Carlo simulations with $\sigma_y^2 = \sigma_x^2 = 1$.

with low (*resp.* high) ρ_y.[15] This relationship is illustrated for the US equity premium (S&P500 minus TBill returns) in the left graph of Figure 6.

The reason why large k deteriorate the prediction model is because they push the training samples further in the past to make sure they do not incorporate forward-looking data. This is suboptimal because the model coefficients are fitted on old data and become obsolete as k increases. This effect is depicted in the diagram of Figure 5. Thus, two opposing effects are at work: the gains from increasing the autocorrelation via higher k may be lost due to model imprecision if coefficients are outdated. When k is low, the model is reactive, but the loss does not benefit from high autocorrelation. When k is large, series may be slightly easier to forecast, but the accuracy of the estimation fades. This is what we refer to as the *memory tradeoff*.

To demonstrate this tradeoff, we consider the equity premium (defined as returns on the S&P500 index minus the Treasury bill rate - see Section 4) and choose five horizons for this premium: 1 month, 3 months, 6 months, 12 months, and 24 months. Table 3 shows the autoregression parameters estimated for each variable over the period of data availability (between January 1871 and December 2019). We fit the empirical data observed for k and ρ_y to a simple inverse function $f(x) = 1 - 0.9/x$ and obtain a very good match, as is shown in the left panel of Figure 6.

In the right panel fo Figure 6, we plot the R^2 as a function of the lag k for various training sample size T and we impose the relationship $\rho_y = 1 - 0.9/k$. We observe that only the curves plotted for $T > 30$ are monotonous with respect to k. This is a clear illustration of the memory tradeoff because we see that for small training sample size ($T \leq 30$), predictive models using longer

[15]This link is obviously different for other financial metrics. Volatility is one notable counterexample, as it is already very autocorrelated at short horizons.

Electronic copy available at: https://ssrn.com/abstract=3709412

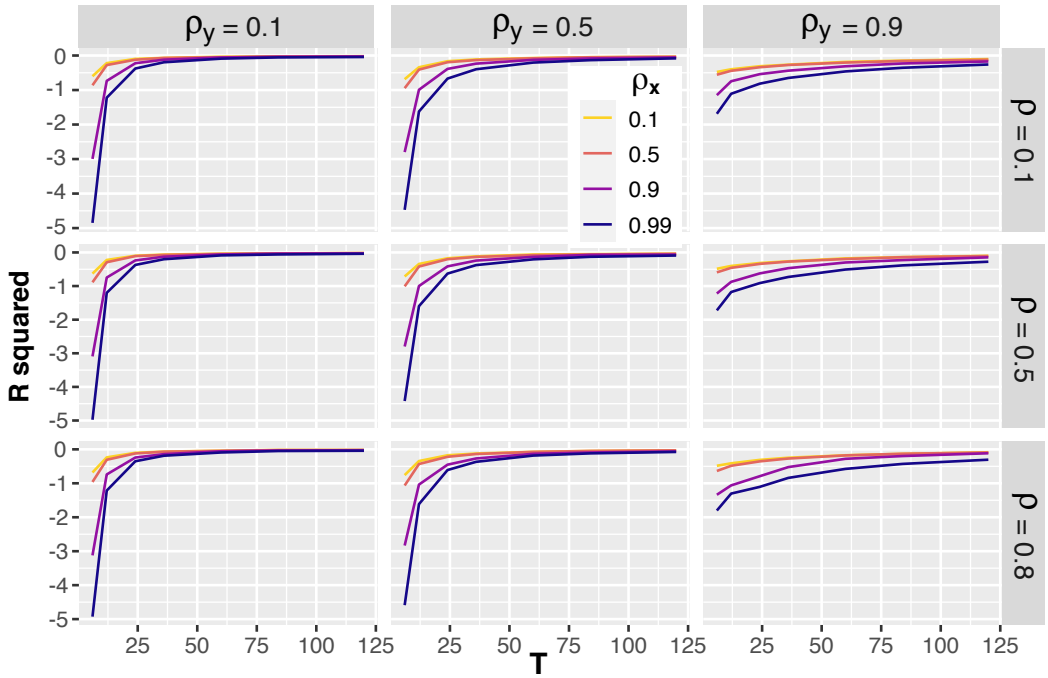

Fig. 4. **Sensitivity of R^2 with regard to autocorrelations and sample size (long term horizon)**. We plot the R^2 values for different choices of ρ, ρ_x, ρ_y and T with a fixed short-term lag $k = 12$. These values are computed via one million Monte Carlo simulations with $\sigma_y^2 = \sigma_x^2 = 1$.

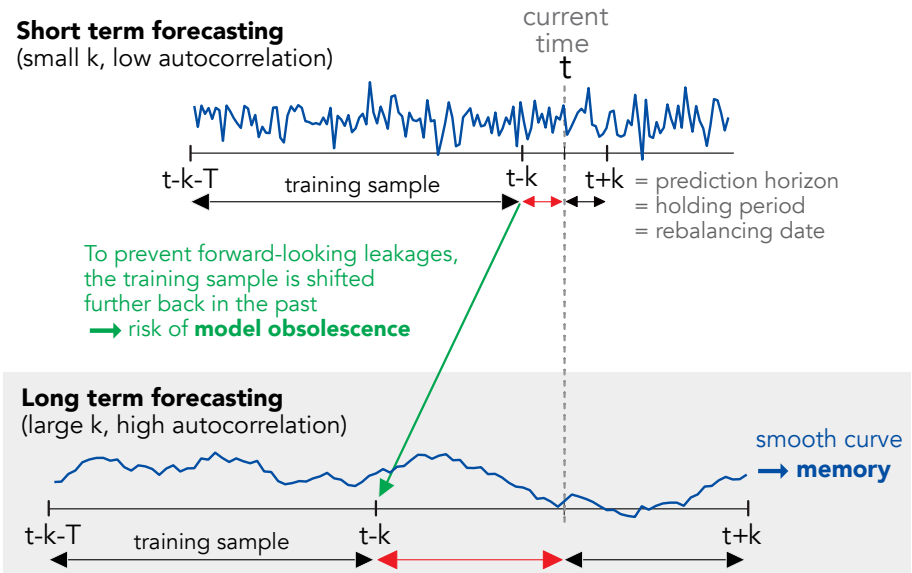

Fig. 5. **Short term versus long term forecasting**.

horizons for the returns can lead to a better estimation accuracy than models with shorter horizons, thus illustrating the domination of the gains coming from an increase in the autocorrelation over the loss of precision due to outdated coefficients. However, for training samples ($T > 30$), the

Electronic copy available at: https://ssrn.com/abstract=3709412

obsolescence of the predictive models due to larger horizons clearly dominates and leads to poorer estimation accuracy.[16]

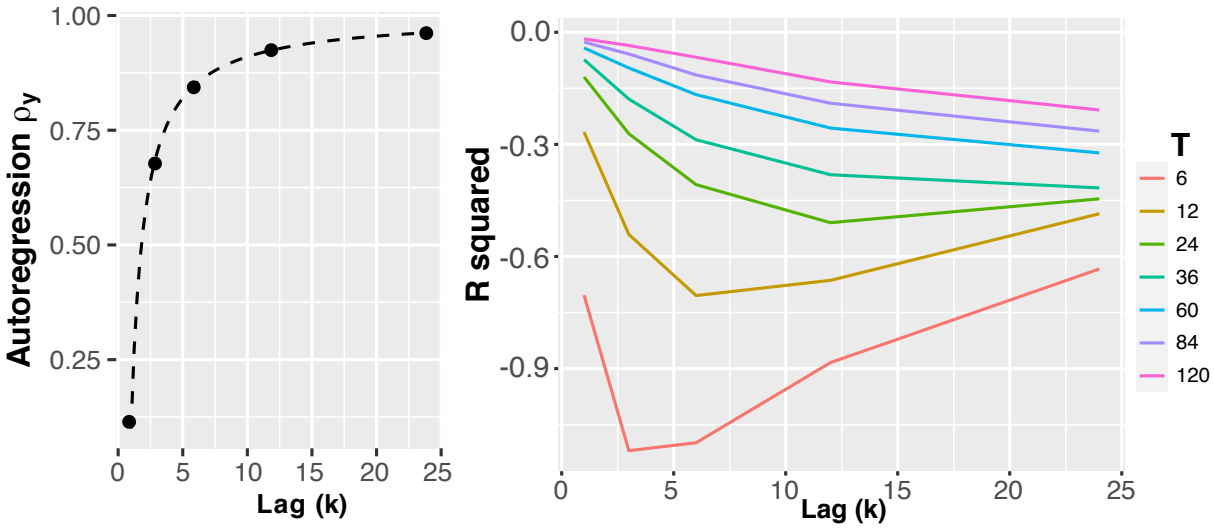

Fig. 6. **Memory tradeoff illustration**. The left panel shows the relationship $\hat{\rho}_y = 1 - 0.9/k$ to explain how the auto-regressive parameters of the equity premium (S&P500 minus TBill returns) depend on the forecasting horizon k. The right panel shows the R^2 as a function of the forecasting horizon k (x-axis) and training sample size T (color). We computed the R^2 using the **process** method with 10^6 simulations. The other parameters are fixed and equal to $\sigma_x^2 = \sigma_y^2 = 1$, $\rho_x = 0.9$, $\rho = 0.5$, and ρ_y is obtained from the variable k using the function $\rho_y = 1 - 0.9/k$ calibrated to the equity premium.

Unlike in Figures 3 and 4 where ρ_y and k were independent of each other, Figure 7 proposes a sensitivity analysis by imposing the constraint $\rho_y = 1 - 0.9/k$ that binds autocorrelation and forecasting horizon k. First, we notice that when ρ_y, or, equivalently, k, is low, the level of autocorrelation ρ_x in the predictor has no impact on the R-squared. However, this observation no longer holds as ρ_y or k increases, in which case the R^2 obtained with a predictor with low autocorrelation is systematically higher than that obtained with a predictor with high autocorrelation. Therefore, with finite samples, choosing predictors with high autocorrelation can lead to predictive models with lower performance, especially when the dependent variable is persistent. Finally, the impact of ρ remains quite unnoticed as the shape and level of the R^2 curves remain quite similar for the three values of ρ under study.

One last conclusion is that in contrast to the right plots of Figure 3, there are no more cases where the R-squared decreases with T. Thus, it seems preferable to favor long training samples in this case.

3.5. Convergence towards optimal loss

Our final analysis pertains to the speed of convergence towards asymptotic values defined in Equation (24). When the training sample increases, we would expect that, despite the absence of independence in the training data set, the estimated coefficient of the models is consistent with

[16] In all generality, finding optimal sample sizes is hard and contingent on many externalities. We refer for instance to the theoretical work of Inoue et al. (2017) on this subject.

Electronic copy available at: https://ssrn.com/abstract=3709412

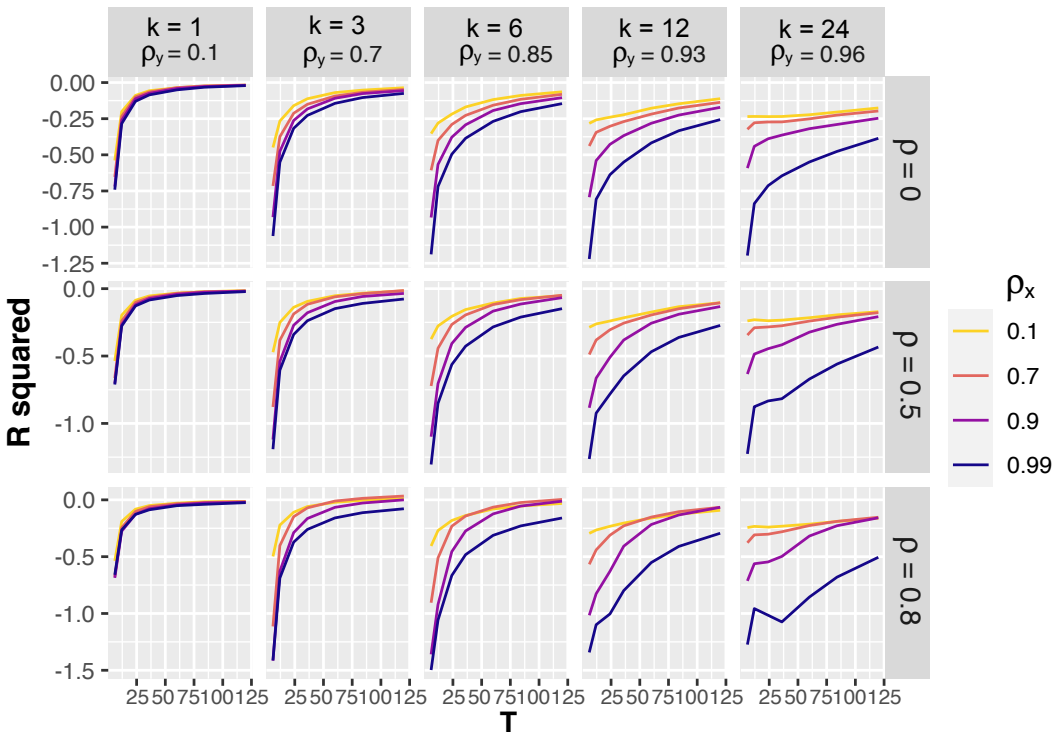

Fig. 7. **Sensitivity under memory tradeoff constraints**. We plot the R^2 for different choices of ρ, ρ_x, k and T. The autocorrelation ρ_y is obtained from the relationship $\rho_y = 1 - 0.9/k$ calibrated on the equity premium of the S&P500. The values are computed via Monte Carlo simulations with $\sigma_y^2 = \sigma_x^2 = 1$.

the classical OLS form (defined by Equation (10) for $\hat{b}$). The lemma below states a result in this direction.

Lemma 5. *It holds that*

$$\hat{b}(\mathcal{D}_t^*) \xrightarrow[T \to +\infty]{P} b_o,$$

where $\xrightarrow{P}$ *stands for convergence in probability and* b_o *is defined in* (10).

The proof is located in Appendix G. Under an additional integrability condition, the R-squared is thus expected to converge to its corresponding value R_o^2 (see Equation 24). This is an important subject, because most R^2 we report in the previous subsections are negative, whereas the R_o^2 are not. Hence, with sufficient data, the estimates are expected to push the R^2 above zero even in the least favorable configurations, as long as $R_o^2 > 0$.

In Figure 8, we illustrate this convergence for two memory-tradeoff constraints. The first is the one (left panel) we observe in financial returns. The second one ($\rho_y = 1 - 0.1/k$ in the right panel) starts directly with high autocorrelation even for small forecasting horizons ($\rho_y = 0.9$ for $k = 1$). One first conclusion that holds for the two cases is that it takes *very* large samples to reach the limiting values, especially when k is large.

The second finding is that the two configurations yield contrasting shapes. In the left panel, the R-squared progressively increases to the optimal performance level. Nevertheless, in the second plot, predictive ability deteriorates when the training samples are longer. Thus, unexpectedly, small

Electronic copy available at: https://ssrn.com/abstract=3709412

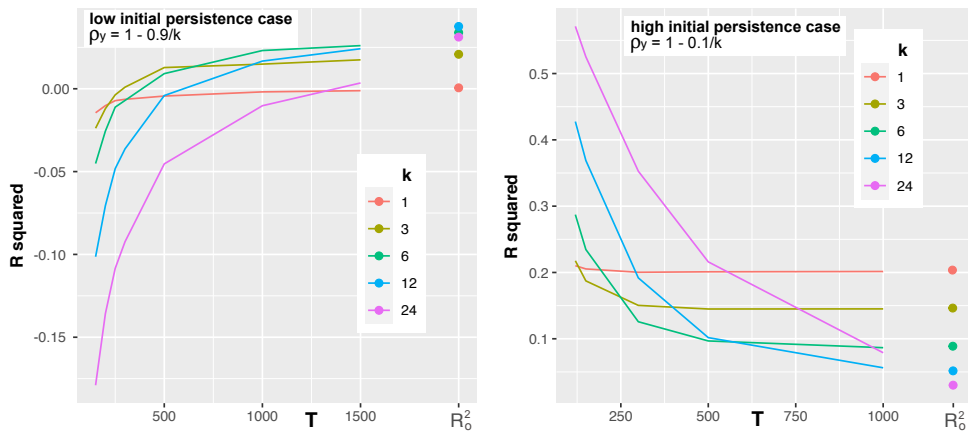

Fig. 8. **Convergence towards R_o^2.** We show the limiting values of the R-squared when sample sizes increase. The asymptotic R_o^2 from Equation (24) are shown as points on the right of the graph. The special case is for $\rho_x = 0.9$ and $\rho = 0.5$. The autocovariance of the dependent variable is given by $\rho_y = 1 - 0.9/k$ (left) and $\rho_y = 1 - 0.1/k$ (right). The R^2 values are computed via Monte Carlo simulations with $\sigma_y^2 = \sigma_x^2 = 1$.

samples are found to increase forecasting accuracy to levels that are even higher than those deemed optimal (R_o^2). However, this requires very high levels of persistence in the dependent variable.

4. Empirical evidence

4.1. Data

One of the academic standards in return predictability is the study of Welch and Goyal (2008). Consequently, we resort to the updated version of their dataset for our empirical analysis.[17] Since the main purpose of the paper is to study the impact of predictor persistence, we remove a few variables because many are highly autocorrelated. This would generate unnecessary redundancies in the results. We also complement our dataset with one of the mildly persistent predictors tested in Novy-Marx (2014): **temp**, the index of global temperature anomalies.[18] The list of all variables is located in Table 2.

To produce more diversity in autocorrelation, we introduce the difference in the first three of these variables (dp, ep and de). The new variables are simply defined by $\Delta x_t = x_t - x_{t-1}$. The dependent variables will be returns on the S&P500 index minus returns on US Treasury bills. We choose five horizons (k) for these returns: 1 month, 3 months, 6 months, 12 months, and 24 months. The descriptive statistics of all variables used in the study can be found in Table 3.

4.2. Baseline results

We briefly describe the protocol we follow. For each predictor x, label y, sample size T and point in time t where the training sample $\mathcal{D}_t$ is well defined (i.e., with no missing point), we:

[17] Available on Amit Goyal's website: http://www.hec.unil.ch/agoyal/docs/PredictorData2019.xlsx

[18] The other climate related variables in Novy-Marx (2014) are highly persistent and do not add value, we thus omit them. The Combined Land-Surface Air and Sea-Surface Water Temperature Anomalies can be accessed at https://data.giss.nasa.gov/gistemp/tabledata_v4/GLB.Ts+dSST.csv.

Electronic copy available at: https://ssrn.com/abstract=3709412

Short name	Brief description	Academic source
dp	log of dividend-price ratio	Welch and Goyal (2008)
ep	log of earnings-price ratio	Welch and Goyal (2008)
de	log of dividend-earnings ratio	Welch and Goyal (2008)
svar	stock variance	Welch and Goyal (2008)
tbl	treasury bill rate	Welch and Goyal (2008)
lty	long term government bond yield	Welch and Goyal (2008)
ltr	long term rate of bond returns	Welch and Goyal (2008)
tms	bond term spread	Welch and Goyal (2008)
dfr	default return spread	Welch and Goyal (2008)
dfy	default yield spread (BAA minus AAA)	Welch and Goyal (2008)
bm	book-to-market ratio (aggregate)	Welch and Goyal (2008)
ntis	net equity expansion	Welch and Goyal (2008)
temp	Global temperature	Novy-Marx (2014)

Table 2: **List of predictors**. We list the predictors used in the empirical study, along with their sources. We keep the same compact notation as in Welch and Goyal (2008), from which most predictors originate. **temp** is documented in Novy-Marx (2014).

varable	begins	ends	min	max	mean	median	std. dev.	$\hat{\rho}_v$	$\hat{\sigma}_v^2$
PANEL A: Predictors (features)									
Panel A1: High persistence ($\hat{\rho}_x > 0.8$)									
bm	1921-03	2019-12	0.121	2.028	0.559	0.531	0.260	0.987	0.002
de	1871-01	2019-12	-1.244	1.380	-0.549	-0.557	0.315	0.993	0.001
dfy	1919-01	2019-12	0.003	0.056	0.012	0.009	0.007	0.977	0.000
dp	1871-01	2019-12	-4.524	-1.873	-3.228	-3.155	0.441	0.995	0.002
ep	1871-01	2019-12	-4.836	-1.670	-2.679	-2.691	0.377	0.989	0.003
lty	1919-01	2018-12	0.018	0.148	0.050	0.042	0.027	0.996	0.000
tbl	1920-02	2019-12	0.000	0.163	0.034	0.031	0.030	0.993	0.000
tms	1920-02	2018-12	-0.044	0.046	0.016	0.017	0.013	0.947	0.000
Panel A2: Moderate persistence									
Δde	1871-02	2019-12	-0.725	0.523	-0.000	0.000	0.037	0.772	0.001
svar	1885-02	2019-12	0.000	0.071	0.002	0.001	0.005	0.620	0.000
temp	1880-01	2019-12	-0.410	5.190	0.070	-0.020	0.547	0.733	0.147
Panel A3: Low persistence ($\hat{\rho}_x < 0.3$)									
Δdp	1871-02	2019-12	-0.363	0.341	-0.001	-0.003	0.048	0.145	0.002
Δep	1871-02	2019-12	-0.536	0.724	-0.000	-0.001	0.057	0.290	0.003
dfr	1926-01	2019-12	-0.098	0.074	0.000	0.001	0.014	-0.119	0.000
ltr	1926-01	2019-12	-0.112	0.152	0.005	0.003	0.024	0.039	0.001
PANEL B: Dependent variables (labels)									
r01m	1871-01	2019-11	-0.299	0.422	0.005	0.006	0.047	0.117	0.002
r03m	1871-01	2019-09	-0.461	0.877	0.015	0.017	0.089	0.680	0.004
r06m	1871-01	2019-06	-0.529	0.959	0.030	0.034	0.124	0.846	0.004
r12m	1871-01	2018-12	-0.701	1.463	0.062	0.066	0.188	0.927	0.005
r24m	1871-01	2017-12	-0.817	1.214	0.126	0.123	0.273	0.964	0.005

Table 3: **Descriptive statistics of the dataset**. In the last columns, $\hat{\rho}_v$ stands for the sample autoregressive parameter of variable v and $\hat{\sigma}_v^2$ refers to the variance of the residuals of the corresponding estimation. The increment variables are defined as $\Delta v_t = v_t - v_{t-1}$ for each variable v.

Electronic copy available at: https://ssrn.com/abstract=3709412

1. estimate Equations (1) for x and (2) for y over a past sample of 120 months (10 years), this yields $\hat{\rho}_{x,t}$, $\hat{\rho}_{y,t}$, $\hat{\rho}_t$, $\hat{\sigma}^2_{x,t}$ and $\hat{\sigma}^2_{y,t}$, which are local estimates for the underlying processes;
2. estimate Equation (3) based on the sample $\mathcal{D}_t$ (and store the residuals);
3. make a prediction for y_{t+k} based on $\hat{a}$ and $\hat{b}$;
4. store the error with the realized y_{t+k}: $e_t(x, y, T)$.

From the above quantities, we are able to compute the out-of-sample R^2, defined as

$$R^2_{\text{oos}}(x, y, t) = 1 - \frac{\sum_{s=1}^{S} e_s(x, y, T)^2}{\sum_{s=1}^{S} (y_{s+k} - \bar{y})^2}, \tag{37}$$

where $s = 1, \ldots S$ are the indices of the available out-of-sample dates.

We report results for 7 choices of T: 6, 12, 24, 36, 60, 84 and 120 months. Over all combinations of predictors, labels, sample sizes and dates, we obtain 856,800 predictions. In Figure 9, we show the impact of the chronological average of $\hat{\rho}_{x,t}$ and $\hat{\rho}_{y,t}$ on the out-of-sample R-squared. The results echo those of Figure 7. Forecasting precision improves with sample size, and predictors with low persistence fare better than those with high autocorrelation. The impact of ρ is not significant, which explains why we present aggregate results.

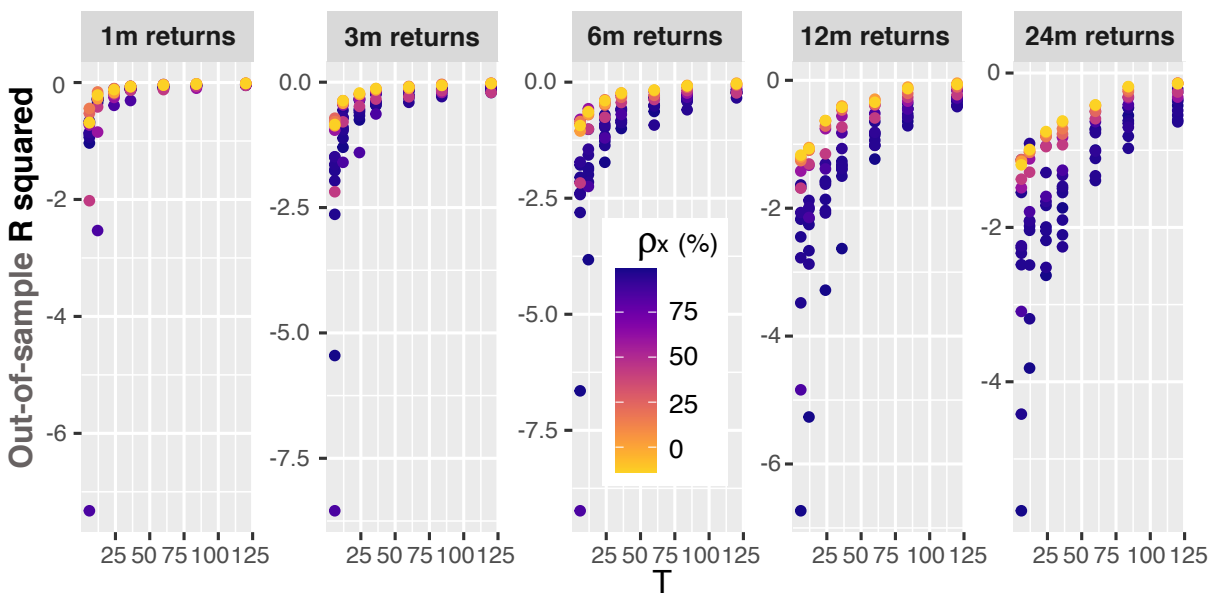

Fig. 9. **Relationship between R^2_{oos} and persistence**. We plot the out-of-sample R-squared as a function of averages of $\hat{\rho}^2_{x,t}$ (across time). Colors relate to autocorrelation in predictors. Before averaging, we remove all outliers in predictions, i.e, those values that lie outside the $[-100\%, +300\%]$ interval. Each panel pertains to one forecasting horizon, which, in turn, corresponds to one persistence level for the dependent variable.

4.3. Economic constraints

Because the equity premium is positive on the long run, researchers (see, e.g., Campbell and Thompson (2008) and Pettenuzzo et al. (2014)) have proposed to impose economy-based restrictions on predictions. The simplest idea, which we follow here, is to truncate the negative part of forecasts,

Electronic copy available at: https://ssrn.com/abstract=3709412

akin to Rectified Linear Unit activations in neural networks. Whenever, a forecast is negative, it is set to zero, because it violates the assumption that the equity premium should be positive. More sophisticated transformations are developed in the aforementioned papers but we do not follow their route.

In Figure 10, we compare the baseline out-of-sample R-squared (x-axis) to those obtained when setting the negative predictions to zero (y-axis). The conclusion is unequivocal: the constraint adds value in all situations, across all predictors, sample sizes, and dependent variables. The gains in R^2 seem more pronounced for small sample sizes as illustrated in the left panel. Indeed, the constraint helps mitigate the risk of unreliably negative predictions when T is small. The right panel indicates that gains are also larger when predictors are persistent. One reason for this is simply that losses are more substantial when ρ_x is large in the first place, which, en route, confirms one of the theoretical conjecture from Figure 7.

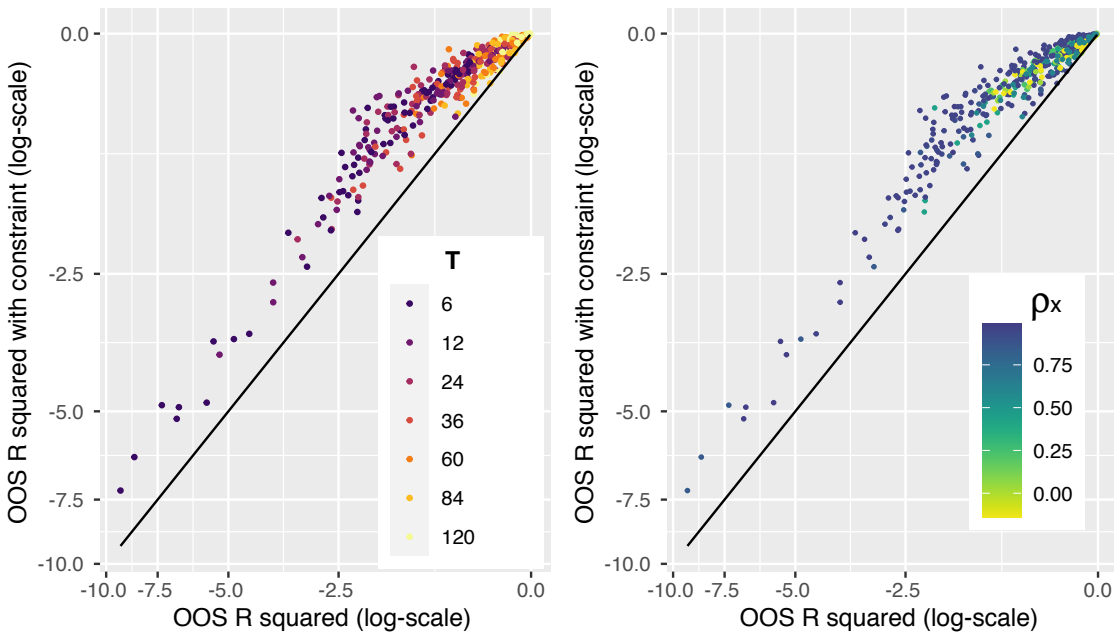

Fig. 10. **Relationship between the R-squared computed with and without constraints**. We plot the R^2_{oos} values against their counterparts where predicted premium is set to 0 if its value is negative. In the left panel, each color corresponds to one sample size T, whereas in the right panel colors relate to the persistence of predictors.

4.4. Portfolio timing implications

The most straightforward application of equity premium prediction is portfolio timing. We proceed to two simple allocation exercises. In the first one, we test heuristic binary strategies which choose to take position (fully invested) either in the aggregate index (S&P500), or the risk-free asset. When the prediction of the equity premium is positive, the strategy is long the equity index, while if it is negative, the strategy is defensive and invested in government TBills (which are expected to fare better than the equity index). We report two indicators: the hit-ratio (proportion of times where the prediction makes a correct bet) and the average return of the strategy. More precisely, if we consider a set of Q predictors indexed by q and the corresponding sample dates $\mathcal{S}_q = 1, \ldots, S_q$,

Electronic copy available at: https://ssrn.com/abstract=3709412

the mean hit ratio is equal to

$$\mathrm{HR} = \frac{1}{Q}\sum_{q=1}^{Q}\frac{1}{S_q}\sum_{t=1}^{S_q} 1_{\{\hat{y}_{t+k}^{(q)} y_{t+k} > 0\}}, \tag{38}$$

where $\hat{y}_{t+k}^{(q)}$ is the time-t predicted value of the time-$(t+k)$ equity premium based on predictor q and a sample size of $T = 120$ months, while y_{t+k} is its realized value. In short, HR evaluates the proportion of times the forecast was in the right direction. The notation $1_{\{\cdot\}}$ stands for the indicator function. For each group of predictors, we also compute the average portfolio return

$$\mathrm{R} = \frac{1}{Q}\sum_{q=1}^{Q}\frac{1}{S_q}\sum_{t=1}^{S_q}\left(\underbrace{r_{t+k}^{\mathrm{SP500}} 1_{\{\hat{y}_{t+k}^{(q)} > 0\}}}_{\text{return if positive pred.}} + \underbrace{r_{t+k}^{\mathrm{TBill}} 1_{\{\hat{y}_{t+k}^{(q)} < 0\}}}_{\text{return if negative pred.}}\right), \tag{39}$$

where the returns are self-explanatory. We compute both HR and R when the sets of predictors are those defined in Table 3 (low, medium and high persistence). In addition, we test the wisdom of crowds property of the predictions by averaging all forecasts across predictors all into an ensemble, in which case

$$\tilde{y}_{t+k} = \frac{1}{Q}\sum_{q=1}^{Q}\hat{y}_{t+k}^{(q)} \tag{40}$$

is the prediction used in the protocols.

Table 4 gathers the results, which confirm the higher accuracy that is derived from predictors with low persistence. They exhibit the largest sample hit ratio across four of the five dependent variables. Nevertheless, the ensemble prediction is superior for return horizons equal to 1 month, 3 months and 24 months. For long horizons (12 and 24 months), no predictive engine is able to beat the heuristic benchmark. This conclusion holds for average returns, which makes sense considering the equity premium: it is always more profitable to be long on the equity market over durable investment horizons.

The aggregation of individual predictions is not the only way to process a cross-section of trading signals. Another, smoother, option is to tune the intensity with which the strategy invests in the risky asset based on whether the signals agree across all the predictors. If all predictions forecast a positive equity premium, then it makes sense to be fully invested in the equity index, while if all point towards a negative premium, then it seems wiser to conservatively allocate to the risk free asset. The intermediate situation is when half of the predictions disagree with the other half, in which case a 50/50 allocation is a balanced solution. The second exercise thus defines the equity pocket as

$$w_{t,k,T} = \frac{1}{Q}\sum_{q=1}^{Q} 1_{\left\{\hat{y}_{t+k}^{(q,T)} > 0\right\}}, \tag{41}$$

which is equal to one if all predictions are positive and to zero if they are all negative. We have added a T index in the notation of predictions because in this last batch of results, we test the impact of the sample size on portfolio performance. Realized portfolio returns are thus equal to

$$r_{t+k}^{(T)} = w_{t,k,T} \times r_{t+k}^{\mathrm{SP500}} + (1 - w_{t,k,T}) \times r_{t+k}^{\mathrm{TBill}}. \tag{42}$$

Electronic copy available at: https://ssrn.com/abstract=3709412

PANEL A: Hit ratio					
Persistence	Return horizon				
	1 month	3 months	6 months	12 months	24 months
low	0.558	0.603	0.629	0.611	0.611
medium	0.563	0.592	0.622	0.596	0.569
high	0.546	0.570	0.577	0.580	0.608
baseline	0.570	0.594	0.628	0.643	0.643
aggregate	0.589	0.605	0.610	0.602	0.641
PANEL B: Average return of long-only portfolio (%))					
Persistence	Return horizon				
	1 month	3 months	6 months	12 months	24 months
low	0.603	1.797	3.710	7.527	15.502
medium	0.623	1.855	3.832	7.186	14.109
high	0.567	1.703	3.364	7.062	14.827
baseline	0.626	1.876	3.858	8.002	16.850
aggregate	0.665	1.848	3.519	7.210	16.108

Table 4: **Prediction performance**. We report the hit ratio (upper panel, defined in (38)) and mean portfolio return (lower panel, defined in (39)) of binary trading strategies based on predictions with 10 years of sample sizes ($T = 120$). The baseline rows in the table show the sample averages. In the upper panel, they compute the proportion of times the equity premium was positive. In the lower panel, they amount to the average value of the dependent variable. The *aggregate* row pertains to the predictions which are averages over all forecasts, across all predictors (see (40)). The three groups of persistence are defined in Table 3. To ensure that all predictors are evaluated on the same samples, the exercise runs from January 1937 onwards (which allows for a 120 month buffer period to compute the first estimates).

Following Brandt et al. (2009), we proxy transaction costs by the simple formula

$$TC_{k,T} = \frac{1}{S-k} \sum_{t=k+1}^{S} |w_{t-k,k,T} - w_{t,k,T}| \times \frac{0.0035}{k},$$

where the scaling constant is linked to high liquidity (numerator) and the rebalancing frequency (k, in the denominator). The transaction cost adjusted Sharpe ratio is then computed as

$$\mathrm{SR}_{k,T} = \frac{\mu_{k,T} - TC_{k,T}}{\sigma_{k,T}}, \tag{43}$$

where $\mu_{k,T}$ is the sample mean of $r_{t+k}^{(T)}$ defined above and $\sigma_{k,T}$ its standard deviation over all dates.

In Table 5, we gather the average returns and the Sharpe ratios of the strategies, for each dependent variable, and each sample size. Plainly, the raw profitabilities of the strategies are comparable to that of the S&P500 benchmark. The return obtained with the 84 and 120 month sample size surpass that of the S&P500 for all forecasting horizons. In terms of risk-adjusted performance, all forecast-based strategies beat the benchmark, which means that, even after transaction costs, they are able to provide the same level of returns, but with only a fraction of the uncertainty. This indicates that there is value in resorting to a pool of predictive regressions when attempting to time the market.

Electronic copy available at: https://ssrn.com/abstract=3709412

Forecasting horizon (rebalancing freq.)	Sample size 6	12	24	36	60	84	120	SP500
PANEL A: Average returns (%)								
1 month	0.567	0.579	0.539	0.564	0.582	0.590	0.629	0.561
3 months	1.670	1.726	1.661	1.749	1.743	1.819	1.913	1.723
6 months	3.424	3.237	3.285	3.328	3.330	3.600	3.824	3.469
12 months	5.887	5.738	6.255	6.503	6.528	7.576	7.548	7.219
24 months	13.691	14.085	14.560	14.166	14.802	15.855	15.216	14.851
PANEL B: Transaction cost adjusted Sharpe ratios								
1 month	0.086	0.088	0.072	0.078	0.078	0.081	0.098	0.054
3 months	0.163	0.170	0.154	0.163	0.147	0.165	0.176	0.091
6 months	0.238	0.207	0.205	0.200	0.184	0.220	0.243	0.132
12 months	0.203	0.177	0.206	0.220	0.222	0.317	0.286	0.189
24 months	0.323	0.336	0.364	0.341	0.372	0.426	0.363	0.270

Table 5: **Portfolio performance.** We report the mean of portfolio returns and the transaction cost adjusted Sharpe ratio (43) related to the investment policy defined by (42).

4.5. Variance prediction

Realized variance (or volatility), unlike future returns, is easier to predict because it is autocorrelated by construction. Thus, it fits the "high ρ_y and low k" combination that is expected to yield the best results (see Figures 3 and 8). To test this statement, we run PRs in which the one month *ahead* svar is the dependent variable and it is regressed against all other predictors, one-by-one:

$$\text{svar}_{t+1} = a + bx_t + e_{t+1}. \tag{44}$$

Predictions are then made accordingly for the next period value and the out-of-sample R^2 is derived. In Figure 11, we plot this OOS R^2 as a function of sample size T (x-axis) and predictor persistence (color). This graph confirms two stylized fact unveiled in the numerical section. First, when $k = 1$ and ρ_y is moderately high, the R^2 can be positive, which essentially means that the predictions fare better than the constant forecast equal to the out-of-sample mean of y. Second, the R^2 decreases with T. In contrast to the equity premium prediction, it is here preferable to stick to short training samples. However, we do not find any marked impact of predictors persistence, which, according to Figure 3, may indicate that the innovations of both sides of the PRs are weakly correlated.

5. Conclusion

In this article, we analyze the accuracy of simple predictive regressions when the two sides of the equation are AR(1) processes with correlated innovations. We provide closed-form expressions for the mean squared error (and R^2) when coefficients are given by the sample OLS estimates. In addition, we document the sensitivity of the R^2 to key quantities, such as autocorrelation (both in the dependent variable and the predictors), forecasting horizon, and sample size.

Our results reveal that some configurations are preferable from the econometrician's point of view. First, autocorrelation in the dependent variable is often associated to reduction in loss. However, in returns, higher persistence is obtained at the cost of longer horizons, which generates a tradeoff between autocorrelation (memory) and model obsolescence (required to avoid forward-looking leakages). At long horizons, positive R^2 can then only be reached with very large samples.

Electronic copy available at: https://ssrn.com/abstract=3709412

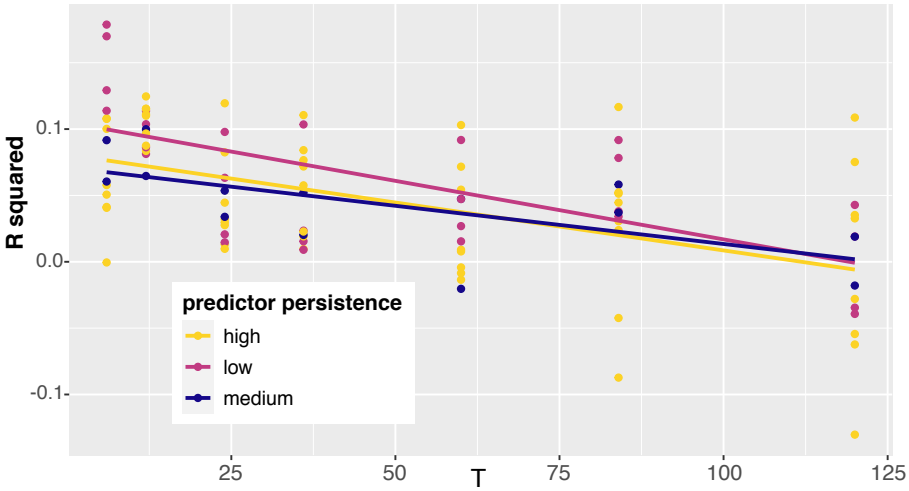

Fig. 11. **Variance prediction**. We plot the out-of-sample R^2 stemming from PRs in which the S&P500 one month ahead variance is the dependent variable. All other predictors are used, one at a time. For each value of T, they correspond to the 14 vertical dots. The persistence categorization (coded with colors) is taken from Table 3. For each color (persistence category), a linear model is fit that maps training sample to R^2.

In contrast, volatility is by construction persistent even at short horizons, and, in this case, the best results are obtained for short training samples, which appear to be more reactive.

Our theoretical findings are confirmed in an empirically study of return predictability where the dependent variable is the S&P500 index and the predictive variables are taken from the study from Welch and Goyal (2008) and Novy-Marx (2014). We show that the predictive performance improves when economic constraints require a post-hoc treatment of the predictions, which corroborates similar analyses on the benefits of such constraints (Campbell and Thompson (2008) and Pettenuzzo et al. (2014)). We also propose applications of return predictability to the design of portfolio strategies delivering higher risk-adjusted performances compared to a plain investment in the S&P500. When we move to the prediction of the S&P500 volatility, we observe that short training samples yield larger R-squared. Although it may appears as an unexpected empirical conclusion, it actually confirms the results we obtained in the sensitivity analysis.

The generalization of our results to higher dimensions is left for future work. In particular, three extensions are of interest. Predictive regressions with many predictors are harder to handle analytically because of the inverse matrix in the sample coefficients, but they seem to be a promising direction for research. Panel approaches are another suggestion, as they would allow to explain dependent variables for a cross-section of assets. Finally and more generally, the generalization to nonlinear models (e.g., tree methods and neural networks) is of interest, though likely out of reach analytically.[19] The joint impact of variable persistence and sample size on the performance of these technical tools remains an open question.

[19] A few studies, e.g., Turmon and Fine (1995) and Alwosheel et al. (2018), assess the effect of training sample length on the predictive quality of neural networks.

Electronic copy available at: https://ssrn.com/abstract=3709412

Appendix A. Proof of Lemma 1

We go back to the definition of the mean squared error of the prediction as $L = \mathbb{E}[e_{t+k}^2]$. As the error e_{t+k} has a zero mean by construction, if we assume that both a and b are known, we have

$$\begin{aligned} L_{ab} &= \text{var}\left(e_{t+k}^2\right) = \text{var}\left(y_{t+k}^* - a - bx_t^*\right) = \text{var}\left(y_{t+k}^* - bx_t^*\right) = \text{var}\left(\boldsymbol{b}'\boldsymbol{z}\right) = \boldsymbol{b}'\text{var}\left(\boldsymbol{z}\right)\boldsymbol{b} \\ &= \text{var}(y_{t+k}^*) - 2b\,\text{cov}(x_t, y_{t+k}) + b^2\text{var}(x_t^*), \end{aligned}$$

where we have defined $\boldsymbol{b} = (1, -b)'$ and $\boldsymbol{z} = (y_{t+k}^*, x_t^*)'$ is a Gaussian vector whose covariance between its two elements is given by

$$\text{cov}(x_t^*, y_{t+k}^*) = \rho_y^k \text{cov}(x_t^*, y_t^*) = \rho_y^k \frac{\rho\sigma_x\sigma_y}{1 - \rho_x\rho_y}.$$

Plugging the covariance expression together with both variances for x_t^* and y_{t+k}^* into the above expression of L_{ab} leads to

$$L_{ab} = \frac{\sigma_y^2}{1 - \rho_y^2} - 2b\,\rho_y^k \frac{\rho\sigma_x\sigma_y}{1 - \rho_x\rho_y} + b^2 \frac{\sigma_x^2}{1 - \rho_x^2}.$$

We easily see that the derivative of L_{ab} with respect to b achieves 0 for $b_o = \text{cov}(x_t, y_{t+k})/\text{var}(x_t)$. Replacing b with its optimal value b_o in Equation (9) and rearranging the terms leads to Equation (11) and concludes the proof.

Appendix B. Proof of Theorem 2

The proof of the zero bias is postponed to Section B.4.

Consider the two auto-regressive processes with correlated white noise processes defined by (1) and (2). For each time t, we center and normalize the two processes as follows

$$x_t = \left(x_t^* - \frac{\alpha_x}{1 - \rho_x}\right)/\sigma_x \quad \text{and} \quad y_t = \left(y_t^* - \frac{\alpha_y}{1 - \rho_y}\right)/\sigma_y,$$

and note that the resulting transformations coincide with the definition of the two auto-regressive processes given by (14) and (15). Now we consider the definition of the estimate for b given in Equation (4) and replace, for each time step s, the quantities x_s^* and y_s^* by $\sigma_x x_s + \frac{\alpha_x}{1-\rho_x}$ and $\sigma_y y_s + \frac{\alpha_y}{1-\rho_y}$ respectively. This leads to

$$\hat{b}(\mathcal{D}_t^*) = \frac{\sum_{s=0}^{T-1}(x_{t-k-s}^* - \bar{x}^*)(y_{t-s}^* - \bar{y}^*)}{\sum_{s=0}^{T-1}(x_{t-k-s}^* - \bar{x}^*)^2} = \frac{\sigma_y\sigma_x \sum_{s=0}^{T-1}(x_{t-k-s} - \bar{x})(y_{t-s} - \bar{y})}{\sigma_x^2 \sum_{s=0}^{T-1}(x_{t-k-s} - \bar{x})^2} = \frac{\sigma_y}{\sigma_x}\hat{b}(\mathcal{D}_t).$$

Finally, we consider the quadratic loss defined in Equation (12) and perform the same substitution to obtain

$$L = \mathbb{E}\left[\left(\sigma_y(y_{t+k} - \bar{y}) - \frac{\sigma_y}{\sigma_x}\hat{b}(\mathcal{D}_t)\sigma_x(x_t - \bar{x})\right)^2\right] = \sigma_y^2\mathbb{E}\left[\left(y_{t+k} - \bar{y} - \hat{b}(\mathcal{D}_t)(x_t - \bar{x})\right)^2\right],$$

which shows that Problem (8) is indeed equivalent to Problem (13) and concludes the proof of items 1 to 3 of Theorem 2. Then, we propose to split the proof of item 4 into three parts for clarity.

Electronic copy available at: https://ssrn.com/abstract=3709412

B.1. Preliminary identities

Since $\boldsymbol{\Sigma}$ is definite positive, it admits the following decomposition $\boldsymbol{L}\boldsymbol{L}'$ (where $\boldsymbol{L}$ is for instance obtained from the Cholesky decomposition or the diagonalization of $\boldsymbol{\Sigma}$). Therefore $\boldsymbol{z}$, which has the following distribution $\mathcal{N}\left(\boldsymbol{0}_{2(T+1)}, \boldsymbol{\Sigma}\right)$, can be written as $\boldsymbol{z} = \boldsymbol{L}\boldsymbol{u}$, where $\boldsymbol{u} \sim \mathcal{N}\left(\boldsymbol{0}_{2(T+1)}, \boldsymbol{I}_{2(T+1)}\right)$. This leads to $\bar{\boldsymbol{z}} = \boldsymbol{N}\boldsymbol{z} = \boldsymbol{N}\boldsymbol{L}\boldsymbol{u}$, which then leads to the following reformulation of Equation (17)

$$L = \sigma_y^2 \left(\mathbb{E}\left[\boldsymbol{u}'\bar{\boldsymbol{E}}\boldsymbol{u}\right] - 2\mathbb{E}\left[(\boldsymbol{u}'\bar{\boldsymbol{C}}\boldsymbol{u})\left(\frac{\boldsymbol{u}'\bar{\boldsymbol{A}}\boldsymbol{u}}{\boldsymbol{u}'\bar{\boldsymbol{B}}\boldsymbol{u}}\right)\right] + \mathbb{E}\left[(\boldsymbol{u}'\bar{\boldsymbol{D}}\boldsymbol{u})\left(\frac{\boldsymbol{u}'\bar{\boldsymbol{A}}\boldsymbol{u}}{\boldsymbol{u}'\bar{\boldsymbol{B}}\boldsymbol{u}}\right)^2\right]\right), \tag{45}$$

where we denote with $\bar{\boldsymbol{X}}$ the matrix product $\boldsymbol{L}'\boldsymbol{N}'\boldsymbol{X}\boldsymbol{N}\boldsymbol{L}$ for each matrix $\boldsymbol{X} = \boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}, \boldsymbol{D}$ or $\boldsymbol{E}$.

The first term of Equation (45) can easily be computed as

$$\mathbb{E}\left[\boldsymbol{u}'\bar{\boldsymbol{E}}\boldsymbol{u}\right] = \mathrm{tr}(\bar{\boldsymbol{E}}) = \mathrm{tr}(\boldsymbol{L}'\boldsymbol{N}'\boldsymbol{E}\boldsymbol{N}\boldsymbol{L}) = \mathrm{tr}(\boldsymbol{E}\boldsymbol{N}\boldsymbol{\Sigma}\boldsymbol{N}') = \mathrm{tr}(\boldsymbol{E}\bar{\boldsymbol{\Sigma}}),$$

where we use the fact that the trace is invariant under cyclic permutations.

B.2. On ratios of quadratic forms

From Appendix B.1, we know that the following properties are true

1. $\boldsymbol{u}$ is a Gaussian vector with mean $\boldsymbol{0}_{2(T+1)}$ and covariance matrix $\boldsymbol{I}_{2(T+1)}$;
2. $\bar{\boldsymbol{A}}, \bar{\boldsymbol{C}}$ and $\bar{\boldsymbol{D}}$ are symmetric matrices since $\boldsymbol{A}, \boldsymbol{C}$ and $\boldsymbol{D}$ are themselves symmetric;
3. $\bar{\boldsymbol{B}}$, which can also be written as $(\boldsymbol{B}\boldsymbol{N}\boldsymbol{L})'\boldsymbol{B}\boldsymbol{N}\boldsymbol{L}$, is a nontrivial positive semi-definite matrix.

The last property above shows that $\bar{\boldsymbol{B}}$ is diagonalizable with non-negative eigenvalues. Hence, there exist an orthogonal matrix $\boldsymbol{P}$ and a diagonal matrix $\boldsymbol{\Lambda}$ such that $\boldsymbol{P}'\bar{\boldsymbol{B}}\boldsymbol{P} = \boldsymbol{\Lambda}$. Therefore, we can apply Theorem 5 from Magnus (1990), which we recall in Appendix H, to compute the last two terms of Equation (45)

$$\mathbb{E}\left[(\boldsymbol{u}'\bar{\boldsymbol{C}}\boldsymbol{u})\left(\frac{\boldsymbol{u}'\bar{\boldsymbol{A}}\boldsymbol{u}}{\boldsymbol{u}'\bar{\boldsymbol{B}}\boldsymbol{u}}\right)\right] = \gamma_1(\boldsymbol{\nu})\int_0^\infty |\boldsymbol{\Delta}|\, g_1(t,\boldsymbol{\nu})dt \tag{46}$$

$$\mathbb{E}\left[(\boldsymbol{u}'\bar{\boldsymbol{D}}\boldsymbol{u})\left(\frac{\boldsymbol{u}'\bar{\boldsymbol{A}}\boldsymbol{u}}{\boldsymbol{u}'\bar{\boldsymbol{B}}\boldsymbol{u}}\right)^2\right] = \sum_{\boldsymbol{\nu}} \gamma_2(\boldsymbol{\nu})\int_0^\infty t\,|\boldsymbol{\Delta}|\, g_2(t,\boldsymbol{\nu})dt. \tag{47}$$

The elements of vector $\boldsymbol{\nu} = (n_1, \ldots, n_s)$ are subject to the condition $\sum_{j=1}^s jn_j = s$, where s is equal to 1 or 2 and represents the exponent of the ratio of quadratic forms. Below, we list their possible values and corresponding γ and g functions.

$$\boldsymbol{\nu} = (1) \Rightarrow \gamma_1(\boldsymbol{\nu}) = 1 \quad \text{and} \quad g_1(t,\boldsymbol{\nu}) = \mathrm{tr}(\boldsymbol{R})\mathrm{tr}(\boldsymbol{\Gamma_C}) + 2\mathrm{tr}(\boldsymbol{R}\boldsymbol{\Gamma_C}),$$

and

$$\begin{aligned}
\boldsymbol{\nu}_1 = (0,1) \Rightarrow \gamma_2(\boldsymbol{\nu}_1) = 2 \quad &\text{and} \quad g_2(t,\boldsymbol{\nu}_1) = \mathrm{tr}(\boldsymbol{R}^2)\mathrm{tr}(\boldsymbol{\Gamma_D}) + 4\mathrm{tr}(\boldsymbol{R}^2\boldsymbol{\Gamma_D}),\\
\boldsymbol{\nu}_2 = (2,0) \Rightarrow \gamma_2(\boldsymbol{\nu}_2) = 1 \quad &\text{and} \quad g_2(t,\boldsymbol{\nu}_2) = \mathrm{tr}(\boldsymbol{R})^2\mathrm{tr}(\boldsymbol{\Gamma_D}) + 4\mathrm{tr}(\boldsymbol{R})\mathrm{tr}(\boldsymbol{R}\boldsymbol{\Gamma_D}).
\end{aligned}$$

Electronic copy available at: https://ssrn.com/abstract=3709412

The matrices $\boldsymbol{R}$, $\boldsymbol{\Gamma_C}$ and $\boldsymbol{\Gamma_D}$ are given by

$$\boldsymbol{R} = \boldsymbol{\Delta P' \bar{A} P \Delta}, \quad \boldsymbol{\Gamma_C} = \boldsymbol{\Delta P' \bar{C} P \Delta} \quad \boldsymbol{\Gamma_D} = \boldsymbol{\Delta P' \bar{D} P \Delta},$$

where

$$\boldsymbol{\Delta} = (\boldsymbol{I}_{2(T+1)} + 2t\boldsymbol{\Lambda})^{-1/2}.$$

The expressions in Equations (46) and (47) can therefore be expressed as

$$\mathbb{E}\left[(\boldsymbol{u'\bar{C}u})\left(\frac{\boldsymbol{u'\bar{A}u}}{\boldsymbol{u'\bar{B}u}}\right)\right] = \int_0^\infty |\boldsymbol{\Delta}| \left[\mathrm{tr}(\boldsymbol{R})\mathrm{tr}(\boldsymbol{\Gamma_C}) + 2\mathrm{tr}(\boldsymbol{R\Gamma_C})\right] dt \tag{48}$$

$$\mathbb{E}\left[(\boldsymbol{u'\bar{D}u})\left(\frac{\boldsymbol{u'\bar{A}u}}{\boldsymbol{u'\bar{B}u}}\right)^2\right] = \int_0^\infty t\,|\boldsymbol{\Delta}| \left[2\mathrm{tr}(\boldsymbol{R}^2)\mathrm{tr}(\boldsymbol{\Gamma_D}) + 8\mathrm{tr}(\boldsymbol{R}^2\boldsymbol{\Gamma_D}) + \mathrm{tr}(\boldsymbol{R})^2\mathrm{tr}(\boldsymbol{\Gamma_D}) + 4\mathrm{tr}(\boldsymbol{R})\mathrm{tr}(\boldsymbol{R\Gamma_D})\right] dt \tag{49}$$

B.3. Independence with respect to $\boldsymbol{L}$ and $\boldsymbol{P}$

From Equations (48) and (49), we see that the result depends on the traces of matrices $\boldsymbol{R}$, $\boldsymbol{\Gamma_C}$, $\boldsymbol{R\Gamma_C}$, $\boldsymbol{R}^2$, $\boldsymbol{\Gamma_D}$ and $\boldsymbol{R\Gamma_D}$ and $\boldsymbol{R}^2\boldsymbol{\Gamma_D}$, which all seem to depend on the choice of $\boldsymbol{L}$ that is used in the decomposition of the covariance matrix $\boldsymbol{\Sigma}$ and therefore on the two matrices $\boldsymbol{P}$ and $\boldsymbol{\Lambda}$ obtained from the diagonalization of $\boldsymbol{\bar{B}}$. In this section, we will show through matrix manipulations that the result is in fact independent from $\boldsymbol{L}$ and $\boldsymbol{P}$.

As an example, we first consider the computation of that trace of matrix $\boldsymbol{R}$

$$\mathrm{tr}(\boldsymbol{R}) = \mathrm{tr}\left(\boldsymbol{\Delta P' \bar{A} P \Delta}\right) = \mathrm{tr}\left(\boldsymbol{\Delta P' L' N' A N L P \Delta}\right) = \mathrm{tr}\left(\boldsymbol{A N L P \Delta^2 P' L' N'}\right),$$

and focus on the computation of

$$\begin{aligned}
\boldsymbol{N L P \Delta^2 P' L' N'} &= \boldsymbol{N L P}\left(\boldsymbol{I}_{2(T+1)} + 2t\boldsymbol{\Lambda}\right)^{-1}\boldsymbol{P' L' N'} \\
&= \boldsymbol{N L P}\left(\boldsymbol{I}_{2(T+1)} + 2t\boldsymbol{P' L' N' B N L P}\right)^{-1}\boldsymbol{P' L' N'} \\
&= \boldsymbol{N L P P' L' N'}\left(\boldsymbol{I}_{2(T+1)} + 2t\boldsymbol{B N L P P' L' N'}\right)^{-1} \\
&= \boldsymbol{\bar{\Sigma}}\left(\boldsymbol{I}_{2(T+1)} + 2t\boldsymbol{B\bar{\Sigma}}\right)^{-1},
\end{aligned}$$

where we use the push-through matrix identity. Therefore, if we denote with $\boldsymbol{W}$ the following matrix inverse $(\boldsymbol{I}_{2(T+1)} + 2t\boldsymbol{B\bar{\Sigma}})^{-1}$, then we obtain that $\mathrm{tr}(\boldsymbol{R}) = \mathrm{tr}\left(\boldsymbol{A\bar{\Sigma}W}\right)$. Applying the above matrix manipulation to each trace involved in Equations (48) and (49) leads to

$$-2\mathbb{E}\left[(\boldsymbol{u'\bar{C}u})\left(\frac{\boldsymbol{u'\bar{A}u}}{\boldsymbol{u'\bar{B}u}}\right)\right] + \mathbb{E}\left[(\boldsymbol{u'\bar{D}u})\left(\frac{\boldsymbol{u'\bar{A}u}}{\boldsymbol{u'\bar{B}u}}\right)^2\right] = \int_0^\infty |\boldsymbol{\Delta}| \left(t\,f_2(t) - 2f_1(t)\right) dt,$$

with

$$\begin{aligned}
f_1(t) &= \mathrm{tr}(\boldsymbol{A\bar{\Sigma}W})\mathrm{tr}(\boldsymbol{C\bar{\Sigma}W}) + 2\mathrm{tr}(\boldsymbol{A\bar{\Sigma}WC\bar{\Sigma}W}), \\
f_2(t) &= 2\mathrm{tr}(\boldsymbol{A\bar{\Sigma}WA\bar{\Sigma}W})\mathrm{tr}(\boldsymbol{D\bar{\Sigma}W}) + 8\mathrm{tr}(\boldsymbol{A\bar{\Sigma}WA\bar{\Sigma}WD\bar{\Sigma}W}) \\
&\quad + \mathrm{tr}(\boldsymbol{A\bar{\Sigma}W})^2\mathrm{tr}(\boldsymbol{D\bar{\Sigma}W}) + 4\mathrm{tr}(\boldsymbol{A\bar{\Sigma}W})\mathrm{tr}(\boldsymbol{A\bar{\Sigma}WD\bar{\Sigma}W}),
\end{aligned}$$

Electronic copy available at: https://ssrn.com/abstract=3709412

which shows that f_1 and f_2 are independent of the choice of $\boldsymbol{L}$ and $\boldsymbol{P}$. However, the quadratic loss is a function of the determinant of matrix $\boldsymbol{\Delta}$ which depends on the eigenvalues of matrix $\bar{\boldsymbol{B}} = (\boldsymbol{BNL})'\boldsymbol{BNL}$ through the diagonal matrix $\boldsymbol{\Lambda}$. Since, for any square matrix $\boldsymbol{M}$, $\boldsymbol{MM}'$ and $\boldsymbol{M}'\boldsymbol{M}$ have the same eigenvalues, we know that $\bar{\boldsymbol{B}}$ and $\boldsymbol{BNL}(\boldsymbol{BNL})' = \boldsymbol{B}\bar{\boldsymbol{\Sigma}}\boldsymbol{B}$ have the same eigenvalues which proves that the quadratic loss depends neither on the choice of $\boldsymbol{L}$ nor on the choice of $\boldsymbol{P}$ obtained from the diagonalization of $\bar{\boldsymbol{B}}$ but only on the eigenvalues $\boldsymbol{\Lambda}$.

B.4. Unbiased predictions: zero average error

Following the notations introduced previously, the average error of the predictive model, denoted B for bias, is given by

$$B = \sigma_y \mathbb{E}\left[y_{t+k} - \bar{y} - \hat{b}(\mathcal{D}_t)(x_t - \bar{x})\right] = \sigma_y \mathbb{E}\left[\hat{b}(\mathcal{D}_t)(x_t - \bar{x})\right] = \mathbb{E}\left[(\bar{\boldsymbol{a}}'\boldsymbol{u})\left(\frac{\boldsymbol{u}'\bar{\boldsymbol{A}}\boldsymbol{u}}{\boldsymbol{u}'\bar{\boldsymbol{B}}\boldsymbol{u}}\right)\right],$$

where $\bar{\boldsymbol{a}}$ is a $2(T+1) \times 1$ vector given by the linear transformation $\bar{\boldsymbol{a}} = \boldsymbol{L}'\boldsymbol{N}'\boldsymbol{a}$ of the vector $\boldsymbol{a}$ that is equal to zero except for the entry $T+1$ corresponding to element $x_t - \bar{x}$ in vector $\bar{\boldsymbol{z}}$ which is equal to 1.

From Appendix B.2, we know that the assumptions for applying Theorem 5 from Magnus (1990), which we recall in Appendix H, are verified, and therefore we can compute the bias using Equation (53) applied to the simple case $s = 1$ and $\boldsymbol{\mu} = \boldsymbol{0}_{2(T+1)}$. This leads to

$$B = \sigma_y \mathbb{E}\left[\hat{b}(\mathcal{D}_t)(x_t - \bar{x})\right] = \mathbb{E}\left[(\bar{\boldsymbol{a}}'\boldsymbol{u})\left(\frac{\boldsymbol{u}'\bar{\boldsymbol{A}}\boldsymbol{u}}{\boldsymbol{u}'\bar{\boldsymbol{B}}\boldsymbol{u}}\right)\right] = 0, \tag{50}$$

as it is clear from the definition $f_1(t, \boldsymbol{\nu})$ in Theorem 5 from Magnus (1990) that $f_1(t, \boldsymbol{\nu}) = 0$. Indeed, vector $\boldsymbol{\xi}$ is a linear transformation of $\boldsymbol{\mu} = \boldsymbol{0}_{2(T+1)}$ and therefore equal to $\boldsymbol{0}_{2(T+1)}$ as well.

Appendix C. Proof of Theorem 3

The first term on the right-hand side of Equation (20), which corresponds to the variance of $(y_{t+k} - \bar{y})^2$, can be easily obtained as

$$\mathrm{tr}(\boldsymbol{E}\bar{\boldsymbol{\Sigma}}) = \mathrm{tr}\left(\boldsymbol{K_{T+1}}\bar{\boldsymbol{\Sigma}}^{\boldsymbol{y}}_{T+1}\right) = \bar{\epsilon}^2_y.$$

Then $|\boldsymbol{\Delta}|$ can be simplified by noticing that

$$\boldsymbol{B}\bar{\boldsymbol{\Sigma}}\boldsymbol{B} = \begin{bmatrix} \bar{\boldsymbol{\Sigma}}^{\boldsymbol{x}}_T & \boldsymbol{0}_{T,T+2} \\ \boldsymbol{0}_{T+2,T} & \boldsymbol{0}_{T+2} \end{bmatrix},$$

which shows that the $2(T+1)$ eigenvalues of matrix $\boldsymbol{B}\bar{\boldsymbol{\Sigma}}\boldsymbol{B}$ are composed of the eigenvalues of matrix $\bar{\boldsymbol{\Sigma}}^{\boldsymbol{x}}_T$ and $T+2$ eigenvalues equal to 0. Therefore, the determinant of $\boldsymbol{\Delta}$ can be rewritten as the determinant of the smaller matrix $\boldsymbol{\Delta}_T = (\boldsymbol{I}_T + 2t\boldsymbol{\Lambda}_T)^{-1/2}$ where the diagonal matrix $\boldsymbol{\Lambda}_T$ contains the eigenvalues of $\bar{\boldsymbol{\Sigma}}^{\boldsymbol{x}}_T$.

Electronic copy available at: https://ssrn.com/abstract=3709412

C.1. Proof of item 1.

So far we have shown that both $\mathrm{tr}(\boldsymbol{E}\bar{\boldsymbol{\Sigma}})$ and $|\boldsymbol{\Delta}|$ are indepdendent from the correlation parameter ρ. It remains to study the dependence of the two functions f_1 and f_2 given in Equations (21) and (22) with respect to ρ. To do so, we propose to compute the traces of matrices $\boldsymbol{A}^*$, $\boldsymbol{C}^*$ and $\boldsymbol{D}^*$. First, we compute matrix $\boldsymbol{W}$ whose expression if given in Theorem 2 by

$$\boldsymbol{W} = (\boldsymbol{I}_{2(T+1)} + 2t\boldsymbol{B}\bar{\boldsymbol{\Sigma}})^{-1} = \begin{bmatrix} \boldsymbol{I}_{T+1} + 2t\boldsymbol{J}_{T+1}\boldsymbol{N}_{T+1}\boldsymbol{\Sigma}^{\boldsymbol{x}}_{T+1}\boldsymbol{N}'_{T+1} & 2t\boldsymbol{J}_{T+1}\boldsymbol{N}_{T+1}\left(\boldsymbol{\Sigma}^{\boldsymbol{xy}}_{T+1}\right)'\boldsymbol{N}'_{T+1} \\ \boldsymbol{0}_{T+1} & \boldsymbol{I}_{T+1} \end{bmatrix}^{-1}.$$

We introduce the following notations

$$\begin{aligned} \boldsymbol{W}^{\boldsymbol{x}}_{T+1} &= \left(\boldsymbol{I}_{T+1} + 2t\boldsymbol{J}_{T+1}\boldsymbol{N}_{T+1}\boldsymbol{\Sigma}^{\boldsymbol{x}}_{T+1}\boldsymbol{N}'_{T+1}\right)^{-1} \\ \bar{\boldsymbol{\Sigma}}^{\boldsymbol{x}}_{T+1} &= \boldsymbol{N}_{T+1}\boldsymbol{\Sigma}^{\boldsymbol{x}}_{T+1}\boldsymbol{N}'_{T+1} \\ \bar{\boldsymbol{\Sigma}}^{\boldsymbol{y}}_{T+1} &= \boldsymbol{N}_{T+1}\boldsymbol{\Sigma}^{\boldsymbol{y}}_{T+1}\boldsymbol{N}'_{T+1} \\ \bar{\boldsymbol{\Sigma}}^{\boldsymbol{xy}}_{T+1} &= \boldsymbol{N}_{T+1}\boldsymbol{\Sigma}^{\boldsymbol{xy}}_{T+1}\boldsymbol{N}'_{T+1}, \end{aligned}$$

which lead to the following result for $\boldsymbol{W}$

$$\boldsymbol{W} = \begin{bmatrix} \boldsymbol{W}^{\boldsymbol{x}}_{T+1} & -2t\boldsymbol{W}^{\boldsymbol{x}}_{T+1}\boldsymbol{J}_{T+1}\left(\bar{\boldsymbol{\Sigma}}^{\boldsymbol{xy}}_{T+1}\right)' \\ \boldsymbol{0}_{T+1} & \boldsymbol{I}_{T+1} \end{bmatrix},$$

and the expressions for matrices $\boldsymbol{A}^*$, $\boldsymbol{C}^*$ and $\boldsymbol{D}^*$ are given by

$$\boldsymbol{A}^* = \frac{1}{2}\begin{bmatrix} \boldsymbol{J}_{T+1}\bar{\boldsymbol{\Sigma}}^{\boldsymbol{xy}}_{T+1}\boldsymbol{W}^{\boldsymbol{x}}_{T+1} & -2t\boldsymbol{J}_{T+1}\bar{\boldsymbol{\Sigma}}^{\boldsymbol{xy}}_{T+1}\boldsymbol{W}^{\boldsymbol{x}}_{T+1}\boldsymbol{J}_{T+1}\left(\bar{\boldsymbol{\Sigma}}^{\boldsymbol{xy}}_{T+1}\right)' + \boldsymbol{J}_{T+1}\bar{\boldsymbol{\Sigma}}^{\boldsymbol{y}}_{T+1} \\ \boldsymbol{J}_{T+1}\bar{\boldsymbol{\Sigma}}^{\boldsymbol{x}}_{T+1}\boldsymbol{W}^{\boldsymbol{x}}_{T+1} & -2t\boldsymbol{J}_{T+1}\bar{\boldsymbol{\Sigma}}^{\boldsymbol{x}}_{T+1}\boldsymbol{W}^{\boldsymbol{x}}_{T+1}\boldsymbol{J}_{T+1}\left(\bar{\boldsymbol{\Sigma}}^{\boldsymbol{xy}}_{T+1}\right)' + \boldsymbol{J}_{T+1}\left(\bar{\boldsymbol{\Sigma}}^{\boldsymbol{xy}}_{T+1}\right)' \end{bmatrix},$$

$$\boldsymbol{C}^* = \frac{1}{2}\begin{bmatrix} \boldsymbol{K}_{T+1}\bar{\boldsymbol{\Sigma}}^{\boldsymbol{xy}}_{T+1}\boldsymbol{W}^{x}_{T+1} & -2t\boldsymbol{K}_{T+1}\bar{\boldsymbol{\Sigma}}^{\boldsymbol{xy}}_{T+1}\boldsymbol{W}^{\boldsymbol{x}}_{T+1}\boldsymbol{J}_{T+1}\left(\bar{\boldsymbol{\Sigma}}^{\boldsymbol{xy}}_{T+1}\right)' + \boldsymbol{K}_{T+1}\bar{\boldsymbol{\Sigma}}^{\boldsymbol{y}}_{T+1} \\ \boldsymbol{K}_{T+1}\bar{\boldsymbol{\Sigma}}^{\boldsymbol{x}}_{T+1}\boldsymbol{W}^{x}_{T+1} & -2t\boldsymbol{K}_{T+1}\bar{\boldsymbol{\Sigma}}^{\boldsymbol{x}}_{T+1}\boldsymbol{W}^{\boldsymbol{x}}_{T+1}\boldsymbol{J}_{T+1}\left(\bar{\boldsymbol{\Sigma}}^{\boldsymbol{xy}}_{T+1}\right)' + \boldsymbol{K}_{T+1}\left(\bar{\boldsymbol{\Sigma}}^{\boldsymbol{xy}}_{T+1}\right)' \end{bmatrix},$$

$$\boldsymbol{D}^* = \begin{bmatrix} \boldsymbol{K}_{T+1}\bar{\boldsymbol{\Sigma}}^{\boldsymbol{x}}_{T+1}\boldsymbol{W}^{\boldsymbol{x}}_{T+1} & -2t\boldsymbol{K}_{T+1}\bar{\boldsymbol{\Sigma}}^{\boldsymbol{x}}_{T+1}\boldsymbol{W}^{\boldsymbol{x}}_{T+1}\boldsymbol{J}_{T+1}\left(\bar{\boldsymbol{\Sigma}}^{\boldsymbol{xy}}_{T+1}\right)' + \boldsymbol{K}_{T+1}\left(\bar{\boldsymbol{\Sigma}}^{\boldsymbol{xy}}_{T+1}\right)' \\ \boldsymbol{0}_{T+1} & \boldsymbol{0}_{T+1} \end{bmatrix}.$$

Since Equation (19) shows that $\bar{\boldsymbol{\Sigma}}^{\boldsymbol{xy}}_{T+1}$ is linear in ρ, we can make explicit the dependence of matrices $\boldsymbol{A}^*, \boldsymbol{B}^*$ and $\boldsymbol{C}^*$ with respect to parameter ρ as

$$\boldsymbol{A}^* = \begin{bmatrix} \rho\boldsymbol{A}_{11} & \rho^2\boldsymbol{A}_{12,1} + \boldsymbol{A}_{12,2} \\ \boldsymbol{A}_{21} & \rho\boldsymbol{A}_{22} \end{bmatrix}, \quad \boldsymbol{C}^* = \begin{bmatrix} \rho\boldsymbol{C}_{11} & \rho^2\boldsymbol{C}_{12,1} + \boldsymbol{C}_{12,2} \\ \boldsymbol{C}_{21} & \rho\boldsymbol{C}_{22} \end{bmatrix}, \quad \boldsymbol{D}^* = \begin{bmatrix} \boldsymbol{D}_{11} & \rho\boldsymbol{D}_{12} \\ \boldsymbol{0}_{T+1} & \boldsymbol{0}_{T+1} \end{bmatrix}.$$

Therefore, we can derive the following identities

$$\begin{aligned} \mathrm{tr}\left(\boldsymbol{A}^*\right) &= \rho(\boldsymbol{A}_{11} + \boldsymbol{A}_{22}), \\ \mathrm{tr}\left(\boldsymbol{C}^*\right) &= \rho(\boldsymbol{C}_{11} + \boldsymbol{C}_{22}), \\ \mathrm{tr}\left(\boldsymbol{D}^*\right) &= \boldsymbol{D}_{11}, \\ \mathrm{tr}\left(\boldsymbol{A}^*\boldsymbol{C}^*\right) &= \rho^2(\boldsymbol{A}_{11}\boldsymbol{C}_{11} + \boldsymbol{A}_{22}\boldsymbol{C}_{22}) + \boldsymbol{C}_{21}(\rho^2\boldsymbol{A}_{12,1} + \boldsymbol{A}_{12,2}) + \boldsymbol{A}_{21}(\rho^2\boldsymbol{C}_{12,1} + \boldsymbol{C}_{12,2}), \\ \mathrm{tr}\left(\boldsymbol{A}^*\boldsymbol{D}^*\right) &= \rho(\boldsymbol{A}_{11}\boldsymbol{D}_{11} + \boldsymbol{A}_{21}\boldsymbol{D}_{12}), \\ \mathrm{tr}\left(\boldsymbol{A}^*\boldsymbol{A}^*\right) &= \rho^2\boldsymbol{A}_{11}^2 + \rho^4\boldsymbol{A}_{12,1}^2 + 2\rho^2\boldsymbol{A}_{12,1}\boldsymbol{A}_{12,2} + \boldsymbol{A}_{12,2}^2 + \boldsymbol{A}_{21}^2 + \rho^2\boldsymbol{A}_{22}^2, \\ \mathrm{tr}\left(\boldsymbol{A}^*\boldsymbol{A}^*\boldsymbol{D}^*\right) &= \boldsymbol{D}_{11}\left[\rho^2\boldsymbol{A}_{11}^2 + \boldsymbol{A}_{21}\left(\rho^2\boldsymbol{A}_{12,1} + \boldsymbol{A}_{12,2}\right)\right] + \rho^2\boldsymbol{D}_{12}\boldsymbol{A}_{21}\left(\boldsymbol{A}_{11} + \boldsymbol{A}_{22}\right), \end{aligned}$$

Electronic copy available at: https://ssrn.com/abstract=3709412

which leads to

$$\begin{aligned} f_1(t) &= \mathrm{tr}(\boldsymbol{A}^*)\mathrm{tr}(\boldsymbol{C}^*) + 2\mathrm{tr}(\boldsymbol{A}^*\boldsymbol{C}^*) \\ &= \rho^2\left[(\boldsymbol{A}_{11}+\boldsymbol{A}_{22})(\boldsymbol{C}_{11}+\boldsymbol{C}_{22}) + 2\left(\boldsymbol{A}_{11}\boldsymbol{C}_{11}+\boldsymbol{A}_{22}\boldsymbol{C}_{22}+\boldsymbol{C}_{21}\boldsymbol{A}_{12,1}+\boldsymbol{A}_{21}\boldsymbol{C}_{12,1}\right)\right] \\ &\quad + 2\left(\boldsymbol{C}_{21}\boldsymbol{A}_{12,2}+\boldsymbol{A}_{21}\boldsymbol{C}_{12,2}\right), \end{aligned}$$

and

$$\begin{aligned} f_2(t) &= 2\mathrm{tr}(\boldsymbol{A}^*\boldsymbol{A}^*)\mathrm{tr}(\boldsymbol{D}^*) + 8\mathrm{tr}(\boldsymbol{A}^*\boldsymbol{A}^*\boldsymbol{D}^*) + \mathrm{tr}(\boldsymbol{A}^*)^2\mathrm{tr}(\boldsymbol{D}^*) + 4\mathrm{tr}(\boldsymbol{A}^*)\mathrm{tr}(\boldsymbol{A}^*\boldsymbol{D}^*) \\ &= 2\rho^4\boldsymbol{A}_{12,1}^2\boldsymbol{D}_{11} + \rho^2\left[\boldsymbol{D}_{11}\left(15\boldsymbol{A}_{11}^2+3\boldsymbol{A}_{22}^2+6\boldsymbol{A}_{11}\boldsymbol{A}_{22}+4\boldsymbol{A}_{12,1}(\boldsymbol{A}_{12,2}+2\boldsymbol{A}_{21})\right)+12\boldsymbol{D}_{12}\boldsymbol{A}_{21}\left(\boldsymbol{A}_{11}+\boldsymbol{A}_{22}\right)\right] \\ &\quad + 2\boldsymbol{D}_{11}\left(\boldsymbol{A}_{12,2}^2+\boldsymbol{A}_{21}^2+4\boldsymbol{A}_{21}\boldsymbol{A}_{12,2}\right). \end{aligned}$$

From there, we conclude that the integrand in Equation (20) is symmetric in ρ, which concludes the proof of item 1.

C.2. Proof of item 2.

In case of uncorrelated innovations between two auto-regressive processes, i.e. $\rho = 0$, the anti-diagonal blocks $\bar{\boldsymbol{\Sigma}}_{T+1}^{\boldsymbol{xy}}$ and $\left(\bar{\boldsymbol{\Sigma}}_{T+1}^{\boldsymbol{xy}}\right)'$ in the covariance matrix $\boldsymbol{\Sigma}$ of vector $\boldsymbol{z}$ are equal to 0. Therefore, we easily compute matrix $\boldsymbol{W}$ as

$$\boldsymbol{W} = \begin{bmatrix} \boldsymbol{W}_{\boldsymbol{T+1}}^{\boldsymbol{x}} & \boldsymbol{0}_{T+1} \\ \boldsymbol{0}_{T+1} & \boldsymbol{I}_{T+1} \end{bmatrix},$$

and the expressions for matrices $\boldsymbol{A}^*$, $\boldsymbol{C}^*$ and $\boldsymbol{D}^*$ simplify into

$$\boldsymbol{A}^* = \frac{1}{2}\begin{bmatrix} \boldsymbol{0}_{T+1} & \boldsymbol{J}_{\boldsymbol{T+1}}\bar{\boldsymbol{\Sigma}}_{\boldsymbol{T+1}}^{\boldsymbol{y}} \\ \boldsymbol{J}_{\boldsymbol{T+1}}\bar{\boldsymbol{\Sigma}}_{\boldsymbol{T+1}}^{\boldsymbol{x}}\boldsymbol{W}_{\boldsymbol{T+1}}^{\boldsymbol{x}} & \boldsymbol{0}_{T+1} \end{bmatrix}, \quad \boldsymbol{C}^* = \frac{1}{2}\begin{bmatrix} \boldsymbol{0}_{T+1} & \boldsymbol{K}_{\boldsymbol{T+1}}\bar{\boldsymbol{\Sigma}}_{\boldsymbol{T+1}}^{\boldsymbol{y}} \\ \boldsymbol{K}_{\boldsymbol{T+1}}\bar{\boldsymbol{\Sigma}}_{\boldsymbol{T+1}}^{\boldsymbol{x}}\boldsymbol{W}_{\boldsymbol{T+1}}^{\boldsymbol{x}} & \boldsymbol{0}_{T+1} \end{bmatrix},$$

$$\boldsymbol{D}^* = \begin{bmatrix} \boldsymbol{K}_{\boldsymbol{T+1}}\bar{\boldsymbol{\Sigma}}_{\boldsymbol{T+1}}^{\boldsymbol{x}}\boldsymbol{W}_{\boldsymbol{T+1}}^{\boldsymbol{x}} & \boldsymbol{0}_{T+1} \\ \boldsymbol{0}_{T+1} & \boldsymbol{0}_{T+1} \end{bmatrix},$$

which shows immediately that $\mathrm{tr}\left(\boldsymbol{A}^*\right) = \mathrm{tr}\left(\boldsymbol{C}^*\right) = 0$, and leads to the following simplifications for functions f_1 and f_2

$$\begin{aligned} f_1(t) &= 2\mathrm{tr}(\boldsymbol{A}^*\boldsymbol{C}^*), \\ f_2(t) &= 2\mathrm{tr}(\boldsymbol{A}^*\boldsymbol{A}^*)\mathrm{tr}(\boldsymbol{D}^*) + 8\mathrm{tr}(\boldsymbol{A}^*\boldsymbol{A}^*\boldsymbol{D}^*). \end{aligned}$$

Finally, we focus on the computation of the traces of $\boldsymbol{A}^*\boldsymbol{C}^*$, $\boldsymbol{A}^*\boldsymbol{A}^*$, $\boldsymbol{D}^*$ and $\boldsymbol{A}^*\boldsymbol{A}^*\boldsymbol{D}^*$ involved in the two functions f_1 and f_2. First, we easily observe from the expression of $\boldsymbol{D}^*$ above that

$$\mathrm{tr}\left(\boldsymbol{D}^*\right) = \mathrm{tr}\left(\boldsymbol{K}_{\boldsymbol{T+1}}\bar{\boldsymbol{\Sigma}}_{\boldsymbol{T+1}}^{\boldsymbol{x}}\boldsymbol{W}_{\boldsymbol{T+1}}^{\boldsymbol{x}}\right).$$

Simple matrix computations also lead to

$$\begin{aligned} \mathrm{tr}\left(\boldsymbol{A}^*\boldsymbol{C}^*\right) &= \frac{1}{4}\left[\mathrm{tr}\left(\boldsymbol{J}_{\boldsymbol{T+1}}\bar{\boldsymbol{\Sigma}}_{\boldsymbol{T+1}}^{\boldsymbol{y}}\boldsymbol{K}_{\boldsymbol{T+1}}\bar{\boldsymbol{\Sigma}}_{\boldsymbol{T+1}}^{\boldsymbol{x}}\boldsymbol{W}_{\boldsymbol{T+1}}^{\boldsymbol{x}}\right) + \mathrm{tr}\left(\boldsymbol{J}_{\boldsymbol{T+1}}\bar{\boldsymbol{\Sigma}}_{\boldsymbol{T+1}}^{\boldsymbol{x}}\boldsymbol{W}_{\boldsymbol{T+1}}^{\boldsymbol{x}}\boldsymbol{K}_{\boldsymbol{T+1}}\bar{\boldsymbol{\Sigma}}_{\boldsymbol{T+1}}^{\boldsymbol{y}}\right)\right], \\ \mathrm{tr}\left(\boldsymbol{A}^*\boldsymbol{A}^*\right) &= \frac{1}{2}\mathrm{tr}\left(\boldsymbol{J}_{\boldsymbol{T+1}}\bar{\boldsymbol{\Sigma}}_{\boldsymbol{T+1}}^{\boldsymbol{y}}\boldsymbol{J}_{\boldsymbol{T+1}}\bar{\boldsymbol{\Sigma}}_{\boldsymbol{T+1}}^{\boldsymbol{x}}\boldsymbol{W}_{\boldsymbol{T+1}}^{\boldsymbol{x}}\right), \\ \mathrm{tr}\left(\boldsymbol{A}^*\boldsymbol{A}^*\boldsymbol{D}^*\right) &= \frac{1}{4}\mathrm{tr}\left(\boldsymbol{J}_{\boldsymbol{T+1}}\bar{\boldsymbol{\Sigma}}_{\boldsymbol{T+1}}^{\boldsymbol{y}}\boldsymbol{J}_{\boldsymbol{T+1}}\bar{\boldsymbol{\Sigma}}_{\boldsymbol{T+1}}^{\boldsymbol{x}}\boldsymbol{W}_{\boldsymbol{T+1}}^{\boldsymbol{x}}\boldsymbol{K}_{\boldsymbol{T+1}}\bar{\boldsymbol{\Sigma}}_{\boldsymbol{T+1}}^{\boldsymbol{x}}\boldsymbol{W}_{\boldsymbol{T+1}}^{\boldsymbol{x}}\right). \end{aligned}$$

Electronic copy available at: https://ssrn.com/abstract=3709412

The computation of $\mathrm{tr}\left(\boldsymbol{A^*C^*}\right)$ can be simplified by noticing that

$$\begin{aligned}
\mathrm{tr}\left(\boldsymbol{J_{T+1}}\bar{\boldsymbol{\Sigma}}^{\boldsymbol{y}}_{\boldsymbol{T+1}}\boldsymbol{K_{T+1}}\bar{\boldsymbol{\Sigma}}^{\boldsymbol{x}}_{\boldsymbol{T+1}}\boldsymbol{W}^{\boldsymbol{x}}_{\boldsymbol{T+1}}\right) &= \mathrm{tr}\left(\left(\boldsymbol{J_{T+1}}\bar{\boldsymbol{\Sigma}}^{\boldsymbol{y}}_{\boldsymbol{T+1}}\boldsymbol{K_{T+1}}\bar{\boldsymbol{\Sigma}}^{\boldsymbol{x}}_{\boldsymbol{T+1}}\boldsymbol{W}^{\boldsymbol{x}}_{\boldsymbol{T+1}}\right)'\right) \\
&= \mathrm{tr}\left(\left(\boldsymbol{W}^{\boldsymbol{x}}_{\boldsymbol{T+1}}\right)'\bar{\boldsymbol{\Sigma}}^{\boldsymbol{x}}_{\boldsymbol{T+1}}\boldsymbol{K_{T+1}}\bar{\boldsymbol{\Sigma}}^{\boldsymbol{y}}_{\boldsymbol{T+1}}\boldsymbol{J_{T+1}}\right) \\
&= \mathrm{tr}\left(\boldsymbol{J_{T+1}}\left(\boldsymbol{W}^{\boldsymbol{x}}_{\boldsymbol{T+1}}\right)'\bar{\boldsymbol{\Sigma}}^{\boldsymbol{x}}_{\boldsymbol{T+1}}\boldsymbol{K_{T+1}}\bar{\boldsymbol{\Sigma}}^{\boldsymbol{y}}_{\boldsymbol{T+1}}\right) \\
&= \mathrm{tr}\left(\boldsymbol{J_{T+1}}\bar{\boldsymbol{\Sigma}}^{\boldsymbol{x}}_{\boldsymbol{T+1}}\boldsymbol{W}^{\boldsymbol{x}}_{\boldsymbol{T+1}}\boldsymbol{K_{T+1}}\bar{\boldsymbol{\Sigma}}^{\boldsymbol{y}}_{\boldsymbol{T+1}}\right),
\end{aligned}$$

where we used sequentially the following properties: $\mathrm{tr}(\boldsymbol{M}) = \mathrm{tr}(\boldsymbol{M'})$, the symmetry of matrices $\bar{\boldsymbol{\Sigma}}^{\boldsymbol{x}}_{\boldsymbol{T+1}}$, $\boldsymbol{K_{T+1}}$, $\bar{\boldsymbol{\Sigma}}^{\boldsymbol{y}}_{\boldsymbol{T+1}}$, $\boldsymbol{J_{T+1}}$, the invariance of the trace under cyclic permutations, and the push-through identity applied to the matrix product $\left(\boldsymbol{W}^{\boldsymbol{x}}_{\boldsymbol{T+1}}\right)'\bar{\boldsymbol{\Sigma}}^{\boldsymbol{x}}_{\boldsymbol{T+1}} = \bar{\boldsymbol{\Sigma}}^{\boldsymbol{x}}_{\boldsymbol{T+1}}\boldsymbol{W}^{\boldsymbol{x}}_{\boldsymbol{T+1}}$. This leads to

$$\mathrm{tr}\left(\boldsymbol{A^*C^*}\right) = \frac{1}{2}\mathrm{tr}\left(\boldsymbol{J_{T+1}}\bar{\boldsymbol{\Sigma}}^{\boldsymbol{y}}_{\boldsymbol{T+1}}\boldsymbol{K_{T+1}}\bar{\boldsymbol{\Sigma}}^{\boldsymbol{x}}_{\boldsymbol{T+1}}\boldsymbol{W}^{\boldsymbol{x}}_{\boldsymbol{T+1}}\right).$$

In order to develop further the matrix products involved in each trace, we notice that the inverse matrix $\boldsymbol{W}^{\boldsymbol{x}}_{T+1}$ can be expressed as

$$\boldsymbol{W}^{\boldsymbol{x}}_{T+1} = \left(\boldsymbol{I}_{T+1} + 2t\boldsymbol{J_{T+1}}\bar{\boldsymbol{\Sigma}}^{\boldsymbol{x}}_{T+1}\right)^{-1} = \begin{bmatrix} \boldsymbol{I}_T + 2t\bar{\boldsymbol{\Sigma}}^{\boldsymbol{x}}_T & 2t\bar{\boldsymbol{\epsilon}}_{\boldsymbol{x}} \\ \boldsymbol{0}_{1,T} & 1 \end{bmatrix}^{-1} = \begin{bmatrix} \boldsymbol{\Omega} & -2t\boldsymbol{\Omega}\bar{\boldsymbol{\epsilon}}_{\boldsymbol{x}} \\ \boldsymbol{0}_{1,T} & 1 \end{bmatrix},$$

which leads to the following results

$$\begin{aligned}
\mathrm{tr}\left(\boldsymbol{D^*}\right) &= \mathrm{tr}\left(\boldsymbol{K_{T+1}}\bar{\boldsymbol{\Sigma}}^{\boldsymbol{x}}_{\boldsymbol{T+1}}\boldsymbol{W}^{\boldsymbol{x}}_{\boldsymbol{T+1}}\right) = \bar{s}_x^2 - 2t\,\bar{\boldsymbol{\epsilon}}'_{\boldsymbol{x}}\boldsymbol{\Omega}\bar{\boldsymbol{\epsilon}}_{\boldsymbol{x}}, \\
\mathrm{tr}\left(\boldsymbol{A^*C^*}\right) &= \frac{1}{2}\mathrm{tr}\left(\boldsymbol{J_{T+1}}\bar{\boldsymbol{\Sigma}}^{\boldsymbol{y}}_{\boldsymbol{T+1}}\boldsymbol{K_{T+1}}\bar{\boldsymbol{\Sigma}}^{\boldsymbol{x}}_{\boldsymbol{T+1}}\boldsymbol{W}^{\boldsymbol{x}}_{\boldsymbol{T+1}}\right) = \frac{1}{2}\bar{\boldsymbol{\epsilon}}'_{\boldsymbol{x}}\boldsymbol{\Omega}\bar{\boldsymbol{\epsilon}}_{\boldsymbol{y}}, \\
\mathrm{tr}\left(\boldsymbol{A^*A^*}\right) &= \frac{1}{2}\mathrm{tr}\left(\boldsymbol{J_{T+1}}\bar{\boldsymbol{\Sigma}}^{\boldsymbol{y}}_{\boldsymbol{T+1}}\boldsymbol{J_{T+1}}\bar{\boldsymbol{\Sigma}}^{\boldsymbol{x}}_{\boldsymbol{T+1}}\boldsymbol{W}^{\boldsymbol{x}}_{\boldsymbol{T+1}}\right) = \frac{1}{2}\mathrm{tr}(\bar{\boldsymbol{\Sigma}}^{\boldsymbol{x}}_T\boldsymbol{\Omega}\bar{\boldsymbol{\Sigma}}^{\boldsymbol{y}}_T), \\
\mathrm{tr}\left(\boldsymbol{A^*A^*D^*}\right) &= \frac{1}{4}\mathrm{tr}\left(\boldsymbol{J_{T+1}}\bar{\boldsymbol{\Sigma}}^{\boldsymbol{y}}_{\boldsymbol{T+1}}\boldsymbol{J_{T+1}}\bar{\boldsymbol{\Sigma}}^{\boldsymbol{x}}_{\boldsymbol{T+1}}\boldsymbol{W}^{\boldsymbol{x}}_{\boldsymbol{T+1}}\boldsymbol{K_{T+1}}\bar{\boldsymbol{\Sigma}}^{\boldsymbol{x}}_{\boldsymbol{T+1}}\boldsymbol{W}^{\boldsymbol{x}}_{\boldsymbol{T+1}}\right) = \frac{1}{4}\bar{\boldsymbol{\epsilon}}'_{\boldsymbol{x}}\boldsymbol{\Omega}\bar{\boldsymbol{\Sigma}}^{\boldsymbol{y}}_T\boldsymbol{\Omega}\bar{\boldsymbol{\epsilon}}_{\boldsymbol{x}},
\end{aligned}$$

and concludes the proof of item 2.

C.3. Proof of item 3.

As $\bar{\boldsymbol{\Sigma}}^{\boldsymbol{x}}_T$ is a positive semi-definite matrix, then there exist an orthogonal matrix $\boldsymbol{P}_T$ and a diagonal matrix $\boldsymbol{\Lambda}_T$ such that $\bar{\boldsymbol{\Sigma}}^{\boldsymbol{x}}_T = \boldsymbol{P}_T\boldsymbol{\Lambda}_T\boldsymbol{P}_T$. If we denote with $\lambda_1, ..., \lambda_T$ the eigenvalues ot matrix $\bar{\boldsymbol{\Sigma}}^{\boldsymbol{x}}_T$, then it is straightforward to compute the determinant

$$|\boldsymbol{\Delta}_T| = \prod_{j=1}^{T}\left(1 + 2t\lambda_j\right)^{-\frac{1}{2}}.$$

Moreover, we have

$$\begin{aligned}
\boldsymbol{\Omega} &= (\boldsymbol{I}_T + 2t\boldsymbol{P}_T\boldsymbol{\Lambda}_T\boldsymbol{P}'_T)^{-1} = \boldsymbol{P}_T\boldsymbol{P}'_T(\boldsymbol{I}_T + 2t\boldsymbol{P}_T\boldsymbol{\Lambda}_T\boldsymbol{P}'_T)^{-1}\boldsymbol{P}_T\boldsymbol{P}'_T \\
&= \boldsymbol{P}_T(\boldsymbol{I}_T + 2t\boldsymbol{P}'_T\boldsymbol{P}_T\boldsymbol{\Lambda}_T)^{-1}\boldsymbol{P}'_T\boldsymbol{P}_T\boldsymbol{P}'_T = \boldsymbol{P}_T\boldsymbol{\Delta}_T^2\boldsymbol{P}'_T,
\end{aligned}$$

Electronic copy available at: https://ssrn.com/abstract=3709412

where we have used the property of orthogonal matrices and the push-through matrix identity. Then, we use the expression of the trace of $\boldsymbol{A}^*\boldsymbol{C}^*$ provided in Appendix (C.2) and show that

$$\operatorname{tr}(\boldsymbol{A}^*\boldsymbol{C}^*) = \frac{1}{2}\bar{\boldsymbol{\epsilon}}_{\boldsymbol{x}}'\boldsymbol{\Omega}\bar{\boldsymbol{\epsilon}}_{\boldsymbol{y}} = \frac{1}{2}\bar{\boldsymbol{\epsilon}}_{\boldsymbol{x}}'\boldsymbol{P}_T\boldsymbol{\Delta}_T^2\boldsymbol{P}_T'\bar{\boldsymbol{\epsilon}}_{\boldsymbol{y}} = \frac{1}{2}\boldsymbol{p}'\boldsymbol{\Delta}_T^2\boldsymbol{q} = \frac{1}{2}\sum_{j=1}^{T}\frac{p_j q_j}{1+2t\lambda_j},$$

where $\boldsymbol{p}$ and $\boldsymbol{q}$ are defined by $\boldsymbol{P}_T'\bar{\boldsymbol{\epsilon}}_{\boldsymbol{x}}$ and $\boldsymbol{P}_T'\bar{\boldsymbol{\epsilon}}_{\boldsymbol{y}}$ respectively, and conclude that

$$f_1(t) = \sum_{j=1}^{T}\frac{p_j q_j}{1+2t\lambda_j}.$$

Using the expressions of the traces of $\boldsymbol{D}^*$, $\boldsymbol{A}^*\boldsymbol{A}^*$ and $\boldsymbol{A}^*\boldsymbol{A}^*\boldsymbol{D}^*$ from Appendix (C.2) also leads to

$$\begin{aligned}
\operatorname{tr}(\boldsymbol{D}^*) &= \bar{s}_x^2 - 2t\,\bar{\boldsymbol{\epsilon}}_{\boldsymbol{x}}'\boldsymbol{\Omega}\bar{\boldsymbol{\epsilon}}_{\boldsymbol{x}} = \bar{s}_x^2 - 2t\,\bar{\boldsymbol{\epsilon}}_{\boldsymbol{x}}'\boldsymbol{P}_T\boldsymbol{\Delta}_T^2\boldsymbol{P}_T'\bar{\boldsymbol{\epsilon}}_{\boldsymbol{x}} = \bar{s}_x^2 - 2t\,\boldsymbol{p}'\boldsymbol{\Delta}_T^2\boldsymbol{p} \\
&= \bar{s}_x^2 - 2t\sum_{j=1}^{T}\frac{p_j^2}{1+2t\lambda_j}, \\
\operatorname{tr}(\boldsymbol{A}^*\boldsymbol{A}^*) &= \frac{1}{2}\operatorname{tr}(\bar{\boldsymbol{\Sigma}}_T^{\boldsymbol{x}}\boldsymbol{\Omega}\bar{\boldsymbol{\Sigma}}_T^{\boldsymbol{y}}) = \frac{1}{2}\operatorname{tr}(\boldsymbol{P}_T\boldsymbol{\Lambda}_T\boldsymbol{P}_T'\boldsymbol{P}_T\boldsymbol{\Delta}_T^2\boldsymbol{P}_T'\bar{\boldsymbol{\Sigma}}_T^{\boldsymbol{y}}) = \frac{1}{2}\operatorname{tr}(\boldsymbol{\Lambda}_T\boldsymbol{\Delta}_T^2\boldsymbol{P}_T'\bar{\boldsymbol{\Sigma}}_T^{\boldsymbol{y}}\boldsymbol{P}_T) \\
&= \frac{1}{2}\operatorname{tr}(\boldsymbol{\Lambda}_T\boldsymbol{\Delta}_T^2\boldsymbol{Q}) = \frac{1}{2}\sum_{j=1}^{T}\frac{Q_{jj}\lambda_j}{1+2t\lambda_j},
\end{aligned}$$

where matrix $\boldsymbol{Q}$ denotes $\boldsymbol{P}_T'\bar{\boldsymbol{\Sigma}}_T^{\boldsymbol{y}}\boldsymbol{P}_T$. Finally, we have

$$\begin{aligned}
\operatorname{tr}(\boldsymbol{A}^*\boldsymbol{A}^*\boldsymbol{D}^*) &= \frac{1}{4}\bar{\boldsymbol{\epsilon}}_{\boldsymbol{x}}'\boldsymbol{\Omega}\bar{\boldsymbol{\Sigma}}_T^{\boldsymbol{y}}\boldsymbol{\Omega}\bar{\boldsymbol{\epsilon}}_{\boldsymbol{x}} = \frac{1}{4}\bar{\boldsymbol{\epsilon}}_{\boldsymbol{x}}'\boldsymbol{P}_T\boldsymbol{\Delta}_T^2\boldsymbol{P}_T'\bar{\boldsymbol{\Sigma}}_T^{\boldsymbol{y}}\boldsymbol{P}_T\boldsymbol{\Delta}_T^2\boldsymbol{P}_T'\bar{\boldsymbol{\epsilon}}_{\boldsymbol{x}} = \frac{1}{4}\boldsymbol{p}'\boldsymbol{\Delta}_T^2\boldsymbol{Q}\boldsymbol{\Delta}_T^2\boldsymbol{p} \\
&= \frac{1}{4}\sum_{i=1}^{T}\sum_{j=1}^{T}\frac{Q_{ij}p_i p_j}{(1+2t\lambda_i)(1+2t\lambda_j)},
\end{aligned}$$

which leads to the following expression for f_2 and concludes the proof of item 3

$$f_2(t) = \sum_{j=1}^{T}\frac{Q_{jj}\lambda_j}{1+2t\lambda_j}\left(\bar{s}_x^2 - 2t\sum_{j=1}^{T}\frac{p_j^2}{1+2t\lambda_j}\right) + 2\sum_{i=1}^{T}\sum_{j=1}^{T}\frac{Q_{ij}p_i p_j}{(1+2t\lambda_i)(1+2t\lambda_j)}.$$

Appendix D. Proof of Lemma 4.

As matrix $\bar{\boldsymbol{\Sigma}}_T^{\boldsymbol{x}}$ is independent of k, we can conclude the same for matrix $|\boldsymbol{\Delta}_T|$. Therefore, we can express the difference between the MSE for $k+1$ and the MSE for k as

$$L(T,k+1,\rho_x,\rho_y,\rho,\sigma_y^2) - L(T,k,\rho_x,\rho_y,\rho,\sigma_y^2) = \sigma_y^2\left(\Delta_k\bar{s}_y^2 + \int_0^\infty |\boldsymbol{\Delta}_T|\,(t\,\Delta_k f_2(t) - 2\Delta_k f_1(t))\,dt\right),$$

As the dependence with respect to k only appears in $\boldsymbol{\epsilon}_{\boldsymbol{x}}$ and $\boldsymbol{\epsilon}_{\boldsymbol{y}}$, which by convention are assumed to be the quantities computed with a lag equal to k, then we can derive

$$\Delta_k\bar{s}_y^2 = \bar{s}_y^2(k+1) - \bar{s}_y^2(k) = -\frac{2}{T}(\rho_y - 1)\,\mathbf{1}_{1,T}\boldsymbol{\epsilon}_{\boldsymbol{y}}$$

Electronic copy available at: https://ssrn.com/abstract=3709412

$$\begin{aligned}\Delta_k f_1(t) &= f_1(t,k+1) - f_1(t,k)\\ &= (\rho_x\rho_y - 1)\,\boldsymbol{\epsilon}_{\boldsymbol{x}}'\boldsymbol{M}_T\boldsymbol{\Omega}\boldsymbol{M}_T\boldsymbol{\epsilon}_{\boldsymbol{y}} - \frac{1}{T}\mathbf{1}_{1,T}\left[(\rho_y-1)\boldsymbol{\Sigma}_T^{\boldsymbol{x}}\boldsymbol{M}_T\boldsymbol{\Omega}\boldsymbol{M}_T\boldsymbol{\epsilon}_{\boldsymbol{y}} + (\rho_x-1)\boldsymbol{\Sigma}_T^{\boldsymbol{y}}\boldsymbol{M}_T\boldsymbol{\Omega}\boldsymbol{M}_T\boldsymbol{\epsilon}_{\boldsymbol{x}}\right],\end{aligned}$$

$$\begin{aligned}\Delta_k f_2(t) &= f_2(t,k+1) - f_2(t,k)\\ &= \mathrm{tr}(\bar{\boldsymbol{\Sigma}}_T^{\boldsymbol{x}}\boldsymbol{\Omega}\bar{\boldsymbol{\Sigma}}_T^{\boldsymbol{y}})\,(\rho_x-1)\left[-\frac{2}{T}\mathbf{1}_{1,T}\boldsymbol{\epsilon}_{\boldsymbol{x}} - 2t\left((\rho_x+1)\boldsymbol{\epsilon}_{\boldsymbol{x}}'\boldsymbol{M}_T\boldsymbol{\Omega}\boldsymbol{M}_T\boldsymbol{\epsilon}_{\boldsymbol{x}} - \frac{2}{T}\mathbf{1}_{1,T}\boldsymbol{\Sigma}_T^{\boldsymbol{x}}\boldsymbol{M}_T\boldsymbol{\Omega}\boldsymbol{M}_T\boldsymbol{\epsilon}_{\boldsymbol{x}}\right)\right]\\ &\quad + 2\,(\rho_x-1)\left((\rho_x+1)\boldsymbol{\epsilon}_{\boldsymbol{x}}'\boldsymbol{M}_T\boldsymbol{\Omega}\bar{\boldsymbol{\Sigma}}_T^{\boldsymbol{y}}\boldsymbol{\Omega}\boldsymbol{M}_T\boldsymbol{\epsilon}_{\boldsymbol{x}} - \frac{2}{T}\mathbf{1}_{1,T}\boldsymbol{\Sigma}_T^{\boldsymbol{x}}\boldsymbol{M}_T\boldsymbol{\Omega}\bar{\boldsymbol{\Sigma}}_T^{\boldsymbol{y}}\boldsymbol{\Omega}\boldsymbol{M}_T\boldsymbol{\epsilon}_{\boldsymbol{x}}\right).\end{aligned}$$

When the means of processes x^* and y^* are known, we have $\bar{\boldsymbol{\Sigma}}_T^{\boldsymbol{x}} = \boldsymbol{\Sigma}_T^{\boldsymbol{x}}$, $\bar{\boldsymbol{\Sigma}}_T^{\boldsymbol{y}} = \boldsymbol{\Sigma}_T^{\boldsymbol{y}}$, $\bar{\boldsymbol{\epsilon}}_x = \boldsymbol{\epsilon}_x$, $\bar{\boldsymbol{\epsilon}}_y = \boldsymbol{\epsilon}_y$, $\bar{s}_x^2 = (1-\rho_x^2)^{-1}$ and $\bar{s}_y^2 = (1-\rho_y^2)^{-1}$, and the above MSE increment with respect to k simplifies into

$$\Delta_k L = L(T,k+1,\rho_x,\rho_y,\rho,\sigma_y^2) - L(T,k,\rho_x,\rho_y,\rho,\sigma_y^2) \quad = \sigma_y^2\int_0^\infty |\boldsymbol{\Delta}_T|\,\left(t\,\Delta_k f_2(t) - 2\Delta_k f_1(t)\right)dt,$$

where

$$\begin{aligned}\Delta_k f_1(t) &= f_1(t,k+1) - f_1(t,k) = (\rho_x\rho_y - 1)\,\boldsymbol{\epsilon}_{\boldsymbol{x}}'\boldsymbol{\Omega}\boldsymbol{\epsilon}_{\boldsymbol{y}}\\ \Delta_k f_2(t) &= f_2(t,k+1) - f_2(t,k) = -\mathrm{tr}(\boldsymbol{\Sigma}_T^{\boldsymbol{x}}\boldsymbol{\Omega}\boldsymbol{\Sigma}_T^{\boldsymbol{y}})\left(\rho_x^2-1\right)2t\,\boldsymbol{\epsilon}_{\boldsymbol{x}}'\boldsymbol{\Omega}\boldsymbol{\epsilon}_{\boldsymbol{x}} + 2\left(\rho_x^2-1\right)\boldsymbol{\epsilon}_{\boldsymbol{x}}'\boldsymbol{\Omega}\boldsymbol{\Sigma}_T^{\boldsymbol{y}}\boldsymbol{\Omega}\boldsymbol{\epsilon}_{\boldsymbol{x}}.\end{aligned}$$

Assuming $\rho_x = \rho_y$ leads to $\boldsymbol{\Sigma}_T^{\boldsymbol{x}} = \boldsymbol{\Sigma}_T^{\boldsymbol{y}}$ and $\boldsymbol{\epsilon}_x = \boldsymbol{\epsilon}_y$. Therefore the MSE increment can be written as

$$\Delta_k L = -2\left(\rho_x^2-1\right)\sigma_y^2\int_0^\infty |\boldsymbol{\Delta}_T|\,\boldsymbol{\epsilon}_{\boldsymbol{x}}'\bar{\boldsymbol{\Omega}}\boldsymbol{\epsilon}_{\boldsymbol{x}}dt,$$

where

$$\bar{\boldsymbol{\Omega}} = \left(1 + t^2\mathrm{tr}(\boldsymbol{\Sigma}_T^{\boldsymbol{x}}\boldsymbol{\Omega}\boldsymbol{\Sigma}_T^{\boldsymbol{x}})\right)\boldsymbol{\Omega} - t\boldsymbol{\Omega}\boldsymbol{\Sigma}_T^{\boldsymbol{x}}\boldsymbol{\Omega}.$$

As $\boldsymbol{\Sigma}_T^{\boldsymbol{x}}$ is a positive definite matrix, then there exist an orthogonal matrix $\boldsymbol{P}_T$ and a diagonal matrix $\boldsymbol{\Lambda}_T$ such that $\boldsymbol{\Sigma}_T^{\boldsymbol{x}} = \boldsymbol{P}_T\boldsymbol{\Lambda}_T\boldsymbol{P}_T'$. Therefore, we know that

$$\boldsymbol{I}_T + 2t\boldsymbol{\Sigma}_T^{\boldsymbol{x}} = \boldsymbol{P}_T\left(\boldsymbol{I}_T + 2t\boldsymbol{\Lambda}_T\right)\boldsymbol{P}_T' \quad \Leftrightarrow \quad \boldsymbol{\Omega} = \boldsymbol{P}_T\left(\boldsymbol{I}_T + 2t\boldsymbol{\Lambda}_T\right)^{-1}\boldsymbol{P}_T',$$

which leads to

$$\begin{aligned}\mathrm{tr}(\boldsymbol{\Sigma}_T^{\boldsymbol{x}}\boldsymbol{\Omega}\boldsymbol{\Sigma}_T^{\boldsymbol{x}}) &= \mathrm{tr}(\boldsymbol{P}_T\boldsymbol{\Lambda}_T\boldsymbol{P}_T'\boldsymbol{P}_T\left(\boldsymbol{I}_T + 2t\boldsymbol{\Lambda}_T\right)^{-1}\boldsymbol{P}_T'\boldsymbol{P}_T\boldsymbol{\Lambda}_T\boldsymbol{P}_T') = \mathrm{tr}(\boldsymbol{P}_T\boldsymbol{\Lambda}_T\left(\boldsymbol{I}_T + 2t\boldsymbol{\Lambda}_T\right)^{-1}\boldsymbol{\Lambda}_T\boldsymbol{P}_T')\\ &= \mathrm{tr}(\boldsymbol{\Lambda}_T\left(\boldsymbol{I}_T + 2t\boldsymbol{\Lambda}_T\right)^{-1}\boldsymbol{\Lambda}_T) = \sum_{j=1}^T\frac{\lambda_j^2}{(1+2t\lambda_j)} > 0.\end{aligned}$$

Moreover, we can also write

$$\boldsymbol{\Omega}\boldsymbol{\Sigma}_T^{\boldsymbol{x}}\boldsymbol{\Omega} = \boldsymbol{P}_T\left(\boldsymbol{I}_T + 2t\boldsymbol{\Lambda}_T\right)^{-1}\boldsymbol{\Lambda}_T\left(\boldsymbol{I}_T + 2t\boldsymbol{\Lambda}_T\right)^{-1}\boldsymbol{P}_T',$$

which shows that

$$\bar{\boldsymbol{\Omega}} = \boldsymbol{P}_T\left[\left(1 + t^2\sum_{j=1}^T\frac{\lambda_j^2}{(1+2t\lambda_j)}\right)\left(\boldsymbol{I}_T + 2t\boldsymbol{\Lambda}_T\right)^{-1} - t\left(\boldsymbol{I}_T + 2t\boldsymbol{\Lambda}_T\right)^{-1}\boldsymbol{\Lambda}_T\left(\boldsymbol{I}_T + 2t\boldsymbol{\Lambda}_T\right)^{-1}\right]\boldsymbol{P}_T'.$$

Electronic copy available at: https://ssrn.com/abstract=3709412

From the above expression, we obtain the following identity for the eigenvalues $\bar{\omega}_1, ..., \bar{\omega}_T$ of matrix $\bar{\boldsymbol{\Omega}}$

$$\bar{\omega}_j = \frac{1}{(1+2t\lambda_j)^2}\left(1 + t\lambda_j + (1+2t\lambda_j)t^2 \sum_{j=1}^{T} \frac{\lambda_j^2}{(1+2t\lambda_j)}\right) > 0,$$

which shows that they are all positive since $t \geq 0$ and $\lambda_j > 0$ for all j. Therefore, as the determinant $|\boldsymbol{\Delta}_T|$ is also positive, we can conclude that the integrand satisfies $|\boldsymbol{\Delta}_T|\, \boldsymbol{\epsilon}_{\boldsymbol{x}}' \bar{\boldsymbol{\Omega}} \boldsymbol{\epsilon}_{\boldsymbol{x}} > 0$ for all $t \geq 0$. Assuming that $0 \leq \rho_x \leq 1$ finally leads to $\Delta_k L > 0$.

Appendix E. Proof of Expression 36

We proceed as we did in Appendix B.1. Since $\boldsymbol{\Sigma}_{T+1}^{\boldsymbol{y}}$ is definite positive, it admits the following decomposition $\boldsymbol{L}_{T+1}\boldsymbol{L}_{T+1}'$ (where $\boldsymbol{L}_{T+1}$ is for instance obtained from the Cholesky decomposition or the diagonalization of $\boldsymbol{\Sigma}$). Therefore $\bar{\boldsymbol{y}}$, which has the following distribution $\mathcal{N}\left(\boldsymbol{0}_{T+1}, \bar{\boldsymbol{\Sigma}}_{T+1}^{\boldsymbol{y}}\right)$, can be written as $\boldsymbol{N}_{T+1}\boldsymbol{L}_{T+1}\boldsymbol{u}$, where $\boldsymbol{u} \sim \mathcal{N}\left(\boldsymbol{0}_{T+1}, \boldsymbol{I}_{T+1}\right)$. This leads to

$$\hat{\rho}_y(\mathcal{D}_t^*) = \frac{\boldsymbol{u}'\bar{\boldsymbol{F}}_{T+1}\boldsymbol{u}}{\boldsymbol{u}'\bar{\boldsymbol{J}}_{T+1}\boldsymbol{u}},$$

where we denote with $\bar{\boldsymbol{X}}$ the matrix product $\boldsymbol{L}_{T+1}'\boldsymbol{N}_{T+1}'\boldsymbol{X}\boldsymbol{N}_{T+1}\boldsymbol{L}_{T+1}$ for each matrix $\boldsymbol{X} = \boldsymbol{F}_{T+1}$ and $\boldsymbol{J}_{T+1}$.

The following properties are true

1. $\boldsymbol{u}$ is a Gaussian vector with mean $\boldsymbol{0}_{T+1}$ and covariance matrix $\boldsymbol{I}_{T+1}$;
2. $\bar{\boldsymbol{F}}_{T+1}$ is symmetric since $\boldsymbol{F}_{T+1}$ is itself symmetric;
3. $\bar{\boldsymbol{J}}_{T+1}$ is a nontrivial positive semi-definite matrix.

The last property above shows that $\bar{\boldsymbol{J}}_{T+1}$ is diagonalizable with non-negative eigenvalues. Hence, there exist an orthogonal matrix $\boldsymbol{P}_{T+1}$ and a diagonal matrix $\boldsymbol{\Lambda}_{T+1}$ such that $\boldsymbol{P}_{T+1}'\bar{\boldsymbol{J}}_{T+1}\boldsymbol{P}_{T+1} = \boldsymbol{\Lambda}_{T+1}$. Therefore, we can apply Theorem 6 from Magnus (1986) and obtain

$$\mathbb{E}\left[\frac{\boldsymbol{u}'\bar{\boldsymbol{F}}_{T+1}\boldsymbol{u}}{\boldsymbol{u}'\bar{\boldsymbol{J}}_{T+1}\boldsymbol{u}}\right] = \int_0^\infty |\boldsymbol{\Delta}_{T+1}|\, \mathrm{tr}(\boldsymbol{R})dt.$$

where $\boldsymbol{R} = \boldsymbol{\Delta}_{T+1}\boldsymbol{P}_{T+1}'\bar{\boldsymbol{F}}_{T+1}\boldsymbol{P}_{T+1}\boldsymbol{\Delta}_{T+1}$ and $\boldsymbol{\Delta}_{T+1} = (\boldsymbol{I}_{T+1} + 2t\boldsymbol{\Lambda}_{T+1})^{-1/2}$.

Finally, we use the same matrix manipulation as in Section B.3 to show that $\boldsymbol{R} = \mathrm{tr}(\boldsymbol{F}^*)$ where $\boldsymbol{F}^* = \boldsymbol{F}_{T+1}\bar{\boldsymbol{\Sigma}}_{T+1}^{\boldsymbol{y}}(\boldsymbol{I}_{T+1} + 2t\boldsymbol{J}_{T+1}\bar{\boldsymbol{\Sigma}}_{T+1}^{\boldsymbol{y}})^{-1}$ which concludes the proof.

Electronic copy available at: https://ssrn.com/abstract=3709412

Appendix F. Details on the discretized integrals

In both cases ($\rho = 0$ and $\rho \neq 0$), the main task is to evaluate the integral in Equation (20) which we generically write $I = \int_0^\infty h(t)dt$. We split it in two $I = \int_0^u h(t)dt + \int_u^\infty h(t)dt$. The first part is approximated via Riemann trapezoidal sums. If u is large enough, then the second part is negligible and can be discarded, which is what we do in the general case.

When $\rho = 0$, via Equations (28) to (30), we can infer the behavior of the two integrands when t converges to infinity. Thus, it is possible to precisely approximate $\int_u^\infty h(t)dt$. More precisely, we have

$$-2|\boldsymbol{\Delta}_T| f_1(t) \overset{t\to\infty}{\sim} -2\frac{\sum_{j=1}^{T-1} p_j q_j/\lambda_j}{\prod_{j=1}^{T-1}\sqrt{\lambda_j}}(2t)^{-\frac{T+1}{2}}$$

$$|\boldsymbol{\Delta}_T|\, t\, f_2(t) \overset{t\to\infty}{\sim} \frac{\left(\bar{s}_x^2 - \sum_{j=1}^{T-1} p_j^2/\lambda_j\right)\sum_{j=1}^{T-1} Q_{jj}}{2\prod_{j=1}^{T-1}\sqrt{\lambda_j}}(2t)^{-\frac{T-1}{2}},$$

where p_T is the value within the vector $\boldsymbol{p}$ that is related to the null eigenvalue (λ_T).[20] The above values are well defined because all eigenvalues except the last one are strictly positive. We have used the notation $f(x) \overset{x\to a}{\sim} g(x)$ which means $f(x)/g(x) \to 1$ when $x \to a$. The areas under the tails can be accurately approximated by

$$\int_u^\infty -2|\boldsymbol{\Delta}_T| f_1(t)dt \approx -2\frac{\sum_{j=1}^{T-1} p_j q_j/\lambda_j}{\prod_{j=1}^{T-1}\sqrt{\lambda_j}}\frac{(2u)^{-\frac{T-1}{2}}}{T-1} \qquad \text{if } T > 1, \tag{51}$$

$$\int_u^\infty |\boldsymbol{\Delta}_T|\, t\, f_2(t)dt \approx \frac{\left(\bar{s}_x^2 - \sum_{j=1}^{T-1} p_j^2/\lambda_j\right)\sum_{j=1}^{T-1} Q_{jj}}{2\prod_{j=1}^{T-1}\sqrt{\lambda_j}}\frac{(2u)^{-\frac{T-3}{2}}}{T-3} \qquad \text{if } T > 3. \tag{52}$$

The only subtlety is that the grid is not uniform because we want to split the error over all sections and not concentrate it in the leftmost part of the function. Thus, the integration grid is $(\alpha i)^{(T-1)/2}$, for $i = 0, \ldots, n$, where n is the number of points used for the discretization, and $\alpha = u^{2/(T-1)}/n$. The upper bound u is chosen to be large enough that the tails are small.

[20] It can be shown that $p_T = q_T = 0$ because the eigenvector $\boldsymbol{v}_T$ of $\bar{\boldsymbol{\Sigma}}_T^{\boldsymbol{x}}$ corresponding to the null eigenvalue λ_T is equal to $\mathbf{1}_{T,1}/\sqrt{T}$ leading to $p_T = \boldsymbol{v}_T'\bar{\boldsymbol{\epsilon}}_{\boldsymbol{x}} = 0$ and $q_T = \boldsymbol{v}_T'\bar{\boldsymbol{\epsilon}}_{\boldsymbol{y}} = 0$.

Electronic copy available at: https://ssrn.com/abstract=3709412

Appendix G. Proof of the slope consistency

First, we recall a result attributed to S.N. Bernstein about some sufficient condition for the Weak Law of Large Numbers to apply:

Lemma 6. *Let $X_1, X_2, ...$ be a sequence of random variables with finite expectation $E(X_i) = \mu < \infty$ and uniformly bounded variance $var(X_i) < K < \infty$ for $j \geq 1$ and covariances satisfying $cov(X_i, X_j) \underset{|i-j|\to+\infty}{\longrightarrow} 0$, then the weak law of large number applies, i.e.*

$$\frac{1}{n}\sum_{i=1}^{n} X_i \underset{n\to+\infty}{\overset{P}{\longrightarrow}} \mu.$$

We know that the error is equal to

$$\hat{e}_{t+k} = y^*_{t+k} - \hat{a}(\mathcal{D}^*_t) - \hat{b}(\mathcal{D}^*_t)x^*_t = y^*_{t+k} - \bar{y}^* - \hat{b}(\mathcal{D}^*_t)(x^*_t - \bar{x}^*)$$

where

$$\hat{b}(\mathcal{D}^*_t) = \frac{\sum_{s=0}^{T-1}(x^*_{t-k-s} - \bar{x}^*)(y^*_{t-s} - \bar{y}^*)}{\sum_{s=0}^{T-1}(x^*_{t-k-s} - \bar{x}^*)^2} \qquad \text{with} \qquad \bar{x}^* = \frac{1}{T}\sum_{s=0}^{T-1} x^*_{t-k-s} \quad \text{and} \quad \bar{y}^* = \frac{1}{T}\sum_{s=0}^{T-1} y^*_{t-s}.$$

For simplicity, we denote with u and v the following centered random processes $u_t = x^*_{t-k-s}$ and $v_t = y^*_{t-s}$, so that

$$\hat{b}(\mathcal{D}^*_t) = \frac{\frac{1}{T}\sum_{s=0}^{T-1} u_s v_s - \bar{x}^*\bar{y}^*}{\frac{1}{T}\sum_{s=0}^{T-1} u_s^2 - (\bar{x}^*)^2} \qquad \text{with} \qquad \bar{x}^* = \frac{1}{T}\sum_{s=0}^{T-1} u_s \quad \text{and} \quad \bar{y}^* = \frac{1}{T}\sum_{s=0}^{T-1} v_s.$$

Moreover, from the properties of processes x^* and y^*, we know that

$$\mu_x = E[u_s] = \frac{\alpha_x}{1-\rho_x}, \text{var}(u_s) = \frac{\sigma_x^2}{1-\rho_x^2} \quad \text{for all} \quad s \geq 0, \quad \text{and} \quad \gamma_x(h) = \text{cov}(u_s, u_{s+h}) = \rho_x^{|h|}\frac{\sigma_x^2}{1-\rho_x^2} \underset{|h|\to+\infty}{\longrightarrow} 0,$$

$$\mu_y = E[v_s] = \frac{\alpha_y}{1-\rho_y}, \text{var}(v_s) = \frac{\sigma_y^2}{1-\rho_y^2} \quad \text{for all} \quad s \geq 0, \quad \text{and} \quad \gamma_y(h) = \text{cov}(v_s, v_{s+h}) = \rho_y^{|h|}\frac{\sigma_y^2}{1-\rho_y^2}, \underset{|h|\to+\infty}{\longrightarrow} 0.$$

Therefore, we can apply Lemma 6 and obtain that

$$\bar{x}^* \underset{T\to+\infty}{\overset{P}{\longrightarrow}} \mu_x \quad \text{and} \quad \bar{y}^* \underset{T\to+\infty}{\overset{P}{\longrightarrow}} \mu_y.$$

So we are left with the convergence of $\frac{1}{T}\sum_{s=0}^{T-1} u_s v_s$ and $\frac{1}{T}\sum_{s=0}^{T-1} u_s^2$. To derive the WLLN for the these quantities, we first introduce the following notation for the covariance between u and v:

$$\text{cov}(u_s, v_{s+h}) = \xi(h) = \left(\rho_y^{|h+k|}\mathbf{1}_{h+k\geq 0} + \rho_x^{|h+k|}\mathbf{1}_{h+k<0}\right)\frac{\rho\sigma_x\sigma_y}{1-\rho_x\rho_y} \underset{|h|\to+\infty}{\longrightarrow} 0.$$

and then use Wecker (1978) which shows that

$$\mu_{xy} = E[u_s v_u] = \xi(0) + \mu_x\mu_y, \qquad \text{var}(u_s v_s) = \mu_x^2\gamma_y(0) + \mu_y^2\gamma_x(0) + 2\mu_x\mu_y\xi(0) + \gamma_x(0)\gamma_y(0) + \xi(0)^2,$$

$$\gamma_{xy}(h) = \text{cov}(u_s v_s, u_{s+h}v_{s+h}) = \mu_x^2\gamma_y(h) + \mu_y^2\gamma_x(h) + \mu_x\mu_y\left(\xi(h) + \xi(-h)\right) + \gamma_x(h)\gamma_y(h) + \xi(h)\xi(-h)$$
$$\underset{|h|\to+\infty}{\longrightarrow} 0.$$

Electronic copy available at: https://ssrn.com/abstract=3709412

Therefore we can apply Lemma 6 to obtain the WLLN for $\frac{1}{T}\sum_{s=0}^{T-1} u_s v_s$ and also for $\frac{1}{T}\sum_{s=0}^{T-1} u_s^2$ by simply replacing v by u, leading to

$$\frac{1}{T}\sum_{s=0}^{T-1} u_s v_s \xrightarrow[T\to+\infty]{P} \xi(0) + \mu_x\mu_y \quad \text{and} \quad \frac{1}{T}\sum_{s=0}^{T-1} u_s^2 \xrightarrow[T\to+\infty]{P} \gamma_x(0) + \mu_x^2.$$

This proves the convergence in probability of $\hat{b}(\mathcal{D}_t^*)$ when T increases to infinity because we have the convergence in probability of each summation $\frac{1}{T}\sum_{s=0}^{T-1} u_s v_s$, $\frac{1}{T}\sum_{s=0}^{T-1} u_s^2$, $\bar{x}^*$ and $\bar{y}^*$ and $\hat{b}(\mathcal{D}_t^*)$ is simply a continuous function of these summations. Hence, we conclude that

$$\hat{b}(\mathcal{D}_t^*) = \frac{\frac{1}{T}\sum_{s=0}^{T-1} u_s v_s - \bar{x}^*\bar{y}^*}{\frac{1}{T}\sum_{s=0}^{T-1} u_s^2 - (\bar{x}^*)^2} \xrightarrow[T\to+\infty]{P} \frac{\xi(0)}{\gamma_x(0)} = \frac{\mathrm{cov}(x_t, y_{t+k})}{\mathrm{var}(x_t)} = b_o.$$

Electronic copy available at: https://ssrn.com/abstract=3709412

Appendix H. Theorem 5 from Magnus (1990)

If the following statements are true

1. $\boldsymbol{z}$ is a Gaussian vector with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma} = \boldsymbol{L}\boldsymbol{L}'$;
2. $\boldsymbol{A}$ and $\boldsymbol{C}$ are symmetric matrices, $\boldsymbol{a}$ any vector and $\boldsymbol{B}$ a nontrivial positive semi-definite matrix;
3. $\boldsymbol{P}$ is an orthogonal matrix $\boldsymbol{P}'\boldsymbol{P} = \boldsymbol{I}$ and $\boldsymbol{\Lambda}$ a diagonal matrix such that

$$\boldsymbol{P}'\boldsymbol{L}'\boldsymbol{B}\boldsymbol{L}\boldsymbol{P} = \boldsymbol{\Lambda},$$

then we have, provided the expectations exist, for $s = 1, 2, \cdots$,

$$\mathbb{E}\left[(\boldsymbol{a}'\boldsymbol{z})\left(\frac{\boldsymbol{z}'\boldsymbol{A}\boldsymbol{z}}{\boldsymbol{z}'\boldsymbol{B}\boldsymbol{z}}\right)^s\right] = \frac{e^{-\frac{1}{2}\boldsymbol{\mu}^{*\prime}\boldsymbol{\mu}^*}}{(s-1)!}\sum_{\boldsymbol{\nu}}\gamma_s(\boldsymbol{\nu})\int_0^\infty t^{s-1}|\boldsymbol{\Delta}|e^{\frac{1}{2}\boldsymbol{\xi}'\boldsymbol{\xi}}f_s(t,\boldsymbol{\nu})dt, \tag{53}$$

$$\mathbb{E}\left[(\boldsymbol{z}'\boldsymbol{C}\boldsymbol{z})\left(\frac{\boldsymbol{z}'\boldsymbol{A}\boldsymbol{z}}{\boldsymbol{z}'\boldsymbol{B}\boldsymbol{z}}\right)^s\right] = \frac{e^{-\frac{1}{2}\boldsymbol{\mu}^{*\prime}\boldsymbol{\mu}^*}}{(s-1)!}\sum_{\boldsymbol{\nu}}\gamma_s(\boldsymbol{\nu})\int_0^\infty t^{s-1}|\boldsymbol{\Delta}|e^{\frac{1}{2}\boldsymbol{\xi}'\boldsymbol{\xi}}g_s(t,\boldsymbol{\nu})dt, \tag{54}$$

where the summation runs over the vectors $\boldsymbol{\nu} = (n_1, \ldots, n_s)$ the elements of which are nonnegative integers satisfying $\sum_{j=1}^s jn_j = s$.[21] Moreover, the functions γ_s, f_s and g_s are defined by

$$\gamma_s(\boldsymbol{\nu}) = s!2^s\prod_{j=1}^s(n_j!(2j)^{n_j})^{-1},$$

$$f_s(t,\boldsymbol{\nu}) = \omega\boldsymbol{d}'\boldsymbol{\xi} + 2\sum_{j=1}^s jn_jr_j\omega_j\boldsymbol{d}'\boldsymbol{R}^j\boldsymbol{\xi},$$

$$g_s(t,\boldsymbol{\nu}) = \omega(\mathrm{tr}(\boldsymbol{\Gamma}) + \boldsymbol{\xi}'\boldsymbol{\Gamma}\boldsymbol{\xi}) + 2\sum_{j=1}^s jn_jr_j\omega_j(\mathrm{tr}(\boldsymbol{R}^j\boldsymbol{\Gamma}) + 2\boldsymbol{\xi}'\boldsymbol{R}^j\boldsymbol{\Gamma}\boldsymbol{\xi})$$
$$+ 4\sum_{j=1}^s j^2n_j(n_j-1)r_j^*\omega_j\boldsymbol{\xi}'\boldsymbol{R}^j\boldsymbol{\Gamma}\boldsymbol{R}^j\boldsymbol{\xi} + 8\sum_{i<j}\omega_{i,j}(in_ir_i)(jn_jr_j)\boldsymbol{\xi}'\boldsymbol{R}^i\boldsymbol{\Gamma}\boldsymbol{R}^j\boldsymbol{\xi},$$

and the matrices by

$$\boldsymbol{A}^* = \boldsymbol{P}'\boldsymbol{L}'\boldsymbol{A}\boldsymbol{L}\boldsymbol{P}, \quad \boldsymbol{C}^* = \boldsymbol{P}'\boldsymbol{L}'\boldsymbol{C}\boldsymbol{L}\boldsymbol{P}, \quad \boldsymbol{a}^* = \boldsymbol{P}'\boldsymbol{L}'\boldsymbol{a}, \quad \boldsymbol{\mu}^* = \boldsymbol{P}'\boldsymbol{L}^{-1}\boldsymbol{\mu}$$

$$\boldsymbol{\Delta} = (\boldsymbol{I} + 2t\boldsymbol{\Lambda})^{-1/2}, \quad \boldsymbol{R} = \boldsymbol{\Delta}\boldsymbol{A}^*\boldsymbol{\Delta}, \quad \boldsymbol{\Gamma} = \boldsymbol{\Delta}\boldsymbol{C}^*\boldsymbol{\Delta}, \quad \boldsymbol{d} = \boldsymbol{\Delta}\boldsymbol{a}^*, \quad \boldsymbol{\xi} = \boldsymbol{\Delta}\boldsymbol{\mu}^*.$$

[21] We warn the reader that a typo exists in Theorem 5 of Magnus (1990) as the vectors $\boldsymbol{\nu} = (n_1, \ldots, n_s)$ are defined by $\sum_{j=1}^s n_j = s$ instead of $\sum_{j=1}^s jn_j = s$.

Electronic copy available at: https://ssrn.com/abstract=3709412

Finally, the weights in some sums are given by

$$\omega = \prod_{j=1}^{s} \phi_j^{n_j}, \quad \phi_j = \mathrm{tr}(\boldsymbol{R}^j) + j\boldsymbol{\xi}'\boldsymbol{R}^j\boldsymbol{\xi}$$

$$\omega_j = \begin{cases} 1 & \text{if} \quad s = 1 \\ \prod_{i \neq j} \phi_i^{n_i} & \text{if} \quad s \geq 2 \end{cases}$$

$$\omega_{i,j} = \begin{cases} 1 & \text{if} \quad s = 1 \\ \prod_{h \notin \{i,j\}} \phi_h^{n_h} & \text{if} \quad s \geq 2 \end{cases}$$

$$r_j = \begin{cases} 1 & \text{if} \quad n_j = 0, 1 \\ \phi_j^{n_j - 1} & \text{if} \quad n_j \geq 2 \end{cases}$$

$$r_j^* = \begin{cases} 1 & \text{if} \quad n_j = 0, 1, 2 \\ \phi_j^{n_j - 2} & \text{if} \quad n_j \geq 3 \end{cases}$$

Electronic copy available at: https://ssrn.com/abstract=3709412

References

Alwosheel, A., S. van Cranenburgh, and C. G. Chorus (2018). Is your dataset big enough? Sample size requirements when using artificial neural networks for discrete choice analysis. *Journal of Choice Modelling 28*, 167–182.

Amihud, Y. and C. M. Hurvich (2004). Predictive regressions: A reduced-bias estimation method. *Journal of Financial and Quantitative Analysis 39*(4), 813–841.

Andersen, T. G. and R. T. Varneskov (2020). Consistent inference for predictive regressions in persistent economic systems. *Journal of Econometrics Forthcoming*.

Ang, A. and G. Bekaert (2007). Stock return predictability: Is it there? *Review of Financial Studies 20*(3), 651–707.

Bandi, F. M., B. Perron, A. Tamoni, and C. Tebaldi (2019). The scale of predictability. *Journal of Econometrics 208*(1), 120–140.

Bao, Y. (2007). Finite-sample properties of forecasts from the stationary first-order autoregressive model under a general error distribution. *Econometric Theory 23*(4), 767–773.

Bao, Y. and R. Kan (2013). On the moments of ratios of quadratic forms in normal random variables. *Journal of Multivariate Analysis 117*, 229–245.

Bossaerts, P. and P. Hillion (1999). Implementing statistical criteria to select return forecasting models: what do we learn? *Review of Financial Studies 12*(2), 405–428.

Boudoukh, J., R. Israel, and M. P. Richardson (2020). Biases in long-horizon predictive regressions. *SSRN Working Paper 3612600*.

Boudoukh, J., M. Richardson, and R. F. Whitelaw (2008). The myth of long-horizon predictability. *Review of Financial Studies 21*(4), 1577–1605.

Brandt, M. W., P. Santa-Clara, and R. Valkanov (2009). Parametric portfolio policies: Exploiting characteristics in the cross-section of equity returns. *Review of Financial Studies 22*(9), 3411–3447.

Bryzgalova, S., M. Pelger, and J. Zhu (2020). Forest through the trees: Building cross-sections of stock returns. *SSRN Working Paper 3493458*.

Campbell, J. Y. (2001). Why long horizons? A study of power against persistent alternatives. *Journal of Empirical Finance 8*(5), 459–491.

Campbell, J. Y. and S. B. Thompson (2008). Predicting excess stock returns out of sample: Can anything beat the historical average? *Review of Financial Studies 21*(4), 1509–1531.

Campbell, J. Y. and M. Yogo (2006). Efficient tests of stock return predictability. *Journal of Financial Economics 81*(1), 27–60.

Chen, L., M. Pelger, and J. Zhu (2020). Deep learning in asset pricing. *SSRN Working Paper 3350138*.

Electronic copy available at: https://ssrn.com/abstract=3709412

Chevillon, G. (2007). Direct multi-step estimation and forecasting. *Journal of Economic Surveys 21*(4), 746–785.

Chevillon, G. (2017). Robustness of multistep forecasts and predictive regressions at intermediate and long horizons. *SSRN Working Paper 3019901*.

Cochrane, J. H. (2008). The dog that did not bark: A defense of return predictability. *Review of Financial Studies 21*(4), 1533–1575.

Dangl, T. and M. Halling (2012). Predictive regressions with time-varying coefficients. *Journal of Financial Economics 106*(1), 157–181.

Deng, A. (2014). Understanding spurious regression in financial economics. *Journal of Financial Econometrics 12*(1), 122–150.

Fama, E. F. and K. R. French (1988a). Dividend yields and expected stock returns. *Journal of Financial Economics 22*(1), 3–25.

Fama, E. F. and K. R. French (1988b). Permanent and temporary components of stock prices. *Journal of Political Economy 96*(2), 246–273.

Farmer, L., L. Schmidt, and A. Timmermann (2019). Pockets of predictability. *SSRN Working Paper 3152386*.

Ferson, W. E., S. Sarkissian, and T. T. Simin (2003). Spurious regressions in financial economics? *Journal of Finance 58*(4), 1393–1413.

Fuller, W. A. and D. P. Hasza (1980). Predictors for the first-order autoregressive process. *Journal of Econometrics 13*(2), 139–157.

Goyal, A. and I. Welch (2003). Predicting the equity premium with dividend ratios. *Management Science 49*(5), 639–654.

Gu, S., B. Kelly, and D. Xiu (2020a). Empirical asset pricing via machine learning. *Review of Financial Studies 33*(5), 2223–2273.

Gu, S., B. T. Kelly, and D. Xiu (2020b). Autoencoder asset pricing models. *Journal of Econometrics Forthcoming*.

Hansen, P. R. (2005). A test for superior predictive ability. *Journal of Business & Economic Statistics 23*(4), 365–380.

Hillier, G., R. Kan, and X. Wang (2009). Computationally efficient recursions for top-order invariant polynomials with applications. *Econometric Theory*, 211–242.

Hillier, G., R. Kan, and X. Wang (2014). Generating functions and short recursions, with applications to the moments of quadratic forms in noncentral normal vectors. *Econometric Theory*, 436–473.

Hjalmarsson, E. (2008a). Interpreting long-horizon estimates in predictive regressions. *Finance Research Letters 5*(2), 104–117.

Electronic copy available at: https://ssrn.com/abstract=3709412

Hjalmarsson, E. (2008b). The Stambaugh bias in panel predictive regressions. *Finance Research Letters 5*(1), 47–58.

Hjalmarsson, E. (2011). New methods for inference in long-horizon regressions. *Journal of Financial and Quantitative Analysis 46*(3), 815–839.

Hjalmarsson, E. (2012). Some curious power properties of long-horizon tests. *Finance Research Letters 9*(2), 81–91.

Hoque, A., J. R. Magnus, and B. Pesaran (1988). The exact multi-period mean-square forecast error for the first-order autoregressive model. *Journal of Econometrics 39*(3), 327–346.

Ibragimov, R., J. Kim, and A. Skrobotov (2020). New robust inference for predictive regressions. *SSRN Working Paper 3580937.*

Ing, C.-K., J.-L. Lin, S.-H. Yu, et al. (2009). Toward optimal multistep forecasts in non-stationary autoregressions. *Bernoulli 15*(2), 402–437.

Inoue, A., L. Jin, and B. Rossi (2017). Rolling window selection for out-of-sample forecasting with time-varying parameters. *Journal of Econometrics 196*(1), 55–67.

Jansson, M. and M. J. Moreira (2006). Optimal inference in regression models with nearly integrated regressors. *Econometrica 74*(3), 681–714.

Johnson, T. L. (2019). A fresh look at return predictability using a more efficient estimator. *Review of Asset Pricing Studies 9*(1), 1–46.

Kan, R. and X. Wang (2010). On the distribution of the sample autocorrelation coefficients. *Journal of Econometrics 154*(2), 101–121.

Lanne, M. (2002). Testing the predictability of stock returns. *Review of Economics and Statistics 84*(3), 407–415.

Lewellen, J. (2004). Predicting returns with financial ratios. *Journal of Financial Economics 74*(2), 209–235.

Li, J., Z. Liao, and R. Quaedvlieg (2020). Conditional superior predictive ability. *SSRN Working Paper 3536461.*

Magnus, J. R. (1986). The exact moments of a ratio of quadratic forms in normal variables. *Annales d'Economie et de Statistique* (4), 95–109.

Magnus, J. R. (1990). On certain moments relating to ratios of quadratic forms in normal variables: further results. *Sankhyā: The Indian Journal of Statistics, Series B 52*, 1–13.

Magnus, J. R. and B. Pesaran (1989). The exact multi-period mean-square forecast error for the first-order autoregressive model with an intercept. *Journal of Econometrics 42*(2), 157–179.

Magnus, J. R. and B. Pesaran (1991). The bias of forecasts from a first-order autoregression. *Econometric Theory 7*(2), 222–235.

Novy-Marx, R. (2014). Predicting anomaly performance with politics, the weather, global warming, sunspots, and the stars. *Journal of Financial Economics 112*(2), 137–146.

Electronic copy available at: https://ssrn.com/abstract=3709412

Paolella, M. S. (2018). *Linear Models and Time-series Analysis: Regression, ANOVA, ARMA and GARCH*. John Wiley & Sons.

Pástor, L. and R. F. Stambaugh (2009). Predictive systems: Living with imperfect predictors. *Journal of Finance 64*(4), 1583–1628.

Pettenuzzo, D., A. Timmermann, and R. Valkanov (2014). Forecasting stock returns under economic constraints. *Journal of Financial Economics 114*(3), 517–553.

Phillips, P. C. (2015). Halbert White Jr. memorial JFEC lecture: Pitfalls and possibilities in predictive regression. *Journal of Financial Econometrics 13*(3), 521–555.

Phillips, P. C. and J. H. Lee (2013). Predictive regression under various degrees of persistence and robust long-horizon regression. *Journal of Econometrics 177*(2), 250–264.

Piatti, I. and F. Trojani (2020). Dividend growth predictability and the price–dividend ratio. *Management Science 66*(1), 130–158.

Rossi, B. (2005). Testing long-horizon predictive ability with high persistence, and the meese–rogoff puzzle. *International Economic Review 46*(1), 61–92.

Rossi, B. and A. Inoue (2012). Out-of-sample forecast tests robust to the choice of window size. *Journal of Business & Economic Statistics 30*(3), 432–453.

Sawa, T. (1972). Finite-sample properties of the k-class estimators. *Econometrica*, 653–680.

Shephard, N. (2020). An estimator for predictive regression: reliable inference for financial economics. *arXiv Preprint* (2008.06130).

Sizova, N. (2014). A frequency-domain alternative to long-horizon regressions with application to return predictability. *Journal of Empirical Finance 28*, 261–272.

Smith, M. D. (1989). On the expectation of a ratio of quadratic forms in normal variables. *Journal of Multivariate Analysis 31*(2), 244–257.

Stambaugh, R. F. (1999). Predictive regressions. *Journal of Financial Economics 54*(3), 375–421.

Turmon, M. J. and T. L. Fine (1995). Sample size requirements for feedforward neural networks. In *Advances in Neural Information Processing Systems*, pp. 327–334.

Valkanov, R. (2003). Long-horizon regressions: theoretical results and applications. *Journal of Financial Economics 68*(2), 201–232.

Van Binsbergen, J. H. and R. S. Koijen (2010). Predictive regressions: A present-value approach. *Journal of Finance 65*(4), 1439–1471.

Wecker, W. E. (1978). A note on the time series which is the product of two stationary time series. *Stochastic Processes and their Applications 8*(2), 153–157.

Welch, I. and A. Goyal (2008). A comprehensive look at the empirical performance of equity premium prediction. *Review of Financial Studies 21*(4), 1455–1508.

West, K. D. (1996). Asymptotic inference about predictive ability. *Econometrica 64*(5), 1067–1084.

Electronic copy available at: https://ssrn.com/abstract=3709412

Xu, K.-L. (2020). Testing for multiple-horizon predictability: Direct regression based versus implication based. *Review of Financial Studies Forthcoming*.

Zhu, X. (2015). Tug-of-war: Time-varying predictability of stock returns and dividend growth. *Review of Finance 19*(6), 2317–2358.

Zhu, Y. and A. Timmermann (2020). Can two forecasts have the same conditional expected accuracy? *arXiv Preprint* (2006.03238).

Electronic copy available at: https://ssrn.com/abstract=3709412